

MAN DOES NOT THINK, HE IS MOVED BY EMOTIONS

EMOTIONAL BRAIN & EVOLUTIONARY PSYCHOLOGY

J. C. FRAGOMENI

(Left Flap)

We aren't as rational as we believe. Our decision making process has a natural sequence: survival, emotions and then reason.

The understanding of the evolutionary process as a whole leads us to self-knowledge and the understanding that we are an integral part of Life, which is always in charge of maintaining and reestablishing the balance.

I'm not the one who lives: it is Life that lives within me!

This book is dedicated to those who believe; they are represented here by Ana and Darisa, who believed in it even before it existed.

Believing is the mortar of hope and faith, which are the constructions where man finds shelter to strengthen his will.

Man's will, whose origin or dwelling is unknown to me, is the force that moves this world.

Introduction

As I verified, through my own experience, a great duality between my emotional and rational side, and observed how I used reasoning to justify decisions already made emotionally, I became interested in the study of the human mind and the evolutionary process as a whole.

I came to the conclusion that man's capacity to think is an evolutionary artifice created as a competitive advantage for the mammalian proto-man's survival that already had an emotional brain.

I've structured this book into three parts, demonstrating in the first part that there is something wrong with the way we are and think. The famous phrase, "I think, therefore I am" is not an expression of truth. We've ventured into the Third Millennium in a state of perplexity and doubt.

The second part presents man as the fruit of the process of Life. It explains the evolutionary sequence from the first expression of Life up to man: I'm not the one who lives: it is Life that lives within me.

The third part rethinks man and demystifies current scientific and religious concepts, unifying them and rescuing the power of gnosis as complete knowledge.

Man does not think; he is moved by emotions.

We aren't as rational as we believe we are. Our decision making process has a natural sequence: survival, emotions and only afterwards, reason.

The understanding of the evolutionary process as a whole leads us to self-knowledge and the understanding that we are an integral part of Life, which is always in charge of maintaining and reestablishing the balance.

PART 1

THERE IS SOMETHING WRONG

1

The Two Selves

I had already been a medical doctor for years, practicing urology, when I became interested in the study of the human mind, especially interested in its emotional aspect.

At first, I became very interested in psychiatry. I had worked in a mental hospital in Rio de Janeiro, but my experience in such a place, which was more of a warehouse for human beings who had been outcast from society, made me lose interest in the specialty. All of the patients there had different personal realities from the societies they lived in, but society itself can only move forward when someone is capable of looking at what has been established as truth through different eyes. In a way, society is blind to the things it's not prepared to see.

However, a dose of insanity is necessary to see what we're used to accepting as reality from a different prism.

Working as a medical doctor, seeing patients in my office and performing surgeries, I have acquired an experience in the human condition, which no other profession allows. To me, the emotional aspect has always seemed very alive and present in the doctor-patient relationship.

This book presents a completely view of the conventional: a human mind outside of the standing concepts and accepted by society.

I dared to present my point of view on what I call the emotional brain after I read the book "Descartes mistake" by Antonio Damásio, professor at the University of Princeton, where the author presents a anatomical and physiological substratum, which in a way, justifies what I am about to expose.

The first observation I made in this sense is that man thinks that he thinks, but he doesn't. Man almost always reacts emotionally to an external stimulus, taking an emotional stand and, only later, uses reasoning and logic to justify and explain an initial attitude.

As an urologist, I live a life in my office and in O.R.'s, highlighted by what I always felt were scientific opinions and actions, purposely excluding my emotional being in this activity, living in an environment of challenges and competitiveness.

I live another life of spirituality or religious, roaming through religions where I had the opportunity of experiencing extraordinary things; experiences, which were basically emotional or magical that, if analyzed within my scientific parameters, would undoubtedly have to be filed under hallucinations or madness. Experiences in an environment, which was almost always filled with harmony, cooperation, mutual help and especially, of charity towards fellow men.

Two selves co-existed: the doctor self, rational and the religious self, emotional. Both survived in a certain balance for quite some time.

Over the years, my spiritual (emotional) experiences began to clash with my medical (rational) background. The incident that I'm about to talk about next gave me a practical knowledge of the illusion that I'd always called reasoning and logic, which made me see that the medical self wasn't so scientific and rational. This experience was the first step towards linking my "two selves".

Experience with the Vena Cava

Right Kidney	Left Kidney
Vena Cava	Short Right Renal Vein
	Aorta

(Illustration)

Dealing with the Vena Cava is what is difficult in surgery. Vena Cava is the main vein in the body and is connected directly to the kidney through the renal veins. Since it is fragile, it requires a delicate touch. It is situated more towards the right side of the abdomen, which makes the right kidney be close to it. On the other hand, the left kidney is further away from it, which makes the left renal vein to be long, and the right one very short as you can see in the illustration above.

Surgical accidents that wound the vena cava are hard to fix. Due to the low pressure of abundant bleeding, which floods the surgical field, the location of the wound is hard to find and if it's not corrected quickly, the patient might die in a few minutes.

Right kidney surgeries are more susceptible to this kind of complication because, since the renal vein is shorter, it is very close to the vena cava. Every surgeon who has seen a wounded vena cava is afraid of such a complication.

When I was still a resident, I operated on a poly-traumatized child with a right kidney trauma, the pincer that I placed on the right renal vein hand torn the vena cava causing frightening profuse bleeding. With both hands, I held the vena cava and aorta against the vertebrae, which partially controlled the bleeding, which allowed my assistant to suture the vein. This lasted for about half an hour but it seemed like an eternity. My hand went into tetany and became fixed in one position, which made it hard to let go. Everything ended well and the boy was saved but from that moment on, I became afraid of operating on the right kidney.

Since I had to keep doing it, I controlled my fear and continued to perform my duties as a surgeon. What was interesting about this fear was that I was unaware of it for I had buried it in my unconscious. Thus, I continued to perform surgeries.

About twenty years ago, we urologists, had difficulty in choosing a clinical procedure over another, or surgery in the cases of kidney tumors, in cases of remote metastases. Malignant tumors in the kidneys aren't very sensitive to other types of treatments such as, radiotherapy and chemotherapy, which makes surgery the great weapon when treating this type of tumor.

Those who opted for surgery claimed, and even presented case studies, that after the removal of the primary tumor, that the regression of the metastases was evident. The surgical option was also justified by the possibility of removing the metastatic tumor in another surgery. Another group of urologists believed that surgery was unnecessary and didn't justify the suffering that patients went through, not really improving their life expectancy or quality. There were studies demonstrating both approaches.

I was always up to date in the literature of articles in the North American and British urology journals and was familiar with several studies that explained both procedures.

When there was a tumor on the right side (close to the vena cava) I always opted for palliative treatments, emphasizing that surgery would not be beneficial for the patient by quoting several studies that corroborated my conduct.

When it was a tumor on the left side, I always opted for radical surgery and also used several studies that demonstrated the improvement of the metastases and the possibility of removing them with another surgery.

Young and bold, I defended my position with sufficient assuredness, always defending a point of view distinguished by and supported on current scientific papers. Since such cases appeared in discussion sessions of clinical cases, which were far between, I wasn't aware of my own discrepant conduct.

In one of the sessions discussing of cases, a kidney tumor with pulmonary metastasis. I defended the surgical procedure, demonstrating through several scientific papers, that there could be a regression of the metastasis and that the patient could be submitted to pulmonary surgery at a later time. An older urologist, who defended a clinical procedure, said that he found my position to be strange because I had manifested a different opinion in a similar case. I had been more incisive, more emphatic and more secure defending my point of view based on American and British scientific papers.

Once again I justified my conduct, feeling magnanimous for not saying that my colleague wasn't up-to-date.

Some time later, I was operating on a kidney stone in the right kidney along with my resident when he inadvertently injured the vena cava. As the more experienced surgeon, I was able to solve the problem, despite the stress, by suturing the vein.

At the first session of discussions about cases of kidney tumors with pulmonary metastasis, I was terribly confused and insecure about which procedure to follow. I had lost my security and even the petulance and magnanimity of the previous session. All the studies that I knew became jumbled in my head, which made me unable to make a decision.

Only then did I realize that all my previous security had been originated from my fear and ignorance; even my benevolence towards the colleague that had called my attention, was fruit of ignorance and emotional involvement. The more afraid I felt, the more secure I felt in indicating the surgical procedure on the left kidney and not indicating surgery for the right one.

I realized that all my knowledge, logic, reasoning and all my conscious brilliance served a fear, which was buried within my unconscious. A fear that became more unconscious the more I needed to overcome it to become the surgeon that I had set out to be. All my knowledge, my "culture", solely served as justification for a decision made previously by an "emotional state".

Thus, I thought that I thought, but I simply served with my reasoning, an unconscious "being" that lives within me.

Reason and Emotion

From the observation of the behavior of thought in this vena cava case, I started to observe our way of thinking, especially our behavior in the face of the problems that the world constantly presents us with and demands an answer.

My first impression was of great disillusion with myself! Man does not behave the way he imagines he does. Man receives the information, analyzes them and then makes a decision. But we aren't as rational as we imagine!

We receive information and an "emotional filter" selects the information that is favorable to an action that this very filter has already decided on. We analyze the selected information to justify a decision that has already been made.

This may be shocking for the man who pretentiously thinks he is moved by reasoning, the Cartesian man who identifies his existence with the adage “I think, therefore I am”. On the other hand, it might seem ridiculous for a large number of thinkers, traditionalists and academia that exclude the problems that don’t fit their point of view.

There is no better profession than medicine to confront and expose this duality between reason and emotion.

Over the last few years, researchers such as, MacLean, Ledoux, Pinker, Goleman, Damásio and others, have demonstrated the importance of the Limbic System or emotional brain.

More and more, the importance of emotions has been demonstrated. The emotional aspects are interconnected with cognition, with memory and with the sequence of our thoughts, which is very interesting.

Analyzing what determines the sequence of our thoughts, we find the first point that joins science and religion. In the gospels, Jesus many times said, “Why are you thinking evil in your hearts?” making it clear that he understood that the sequence of thoughts are originated from emotional impressions.

In the 19th century, Dr Claude Bernard, whose name is revered in the world of medical symptomatology with the sign that his name carries and refers to ocular protrusion found in hyperthyroidism, used to say that the day would come when scientists and religion would speak the same language.

I believe that day is coming soon. There cannot be two men, one who is driven by logic and reason, and another who gives in to the comfort of a faith beyond logic and reason. It’s even worse when these two men are one: at work, rational and at church, giving in to irrational ecstasies.

In this period of deep transformations that humanity is going through, certain wounds should be exposed so that they can be healed in the same fashion that a surgeon makes a large incision to remove a kidney tumor.

Observations from Experience

The harsh experience from my own behavior in my surgical decision was like lifting a veil, which exposed the fragility of our way of thinking. I began to observe that this happens more frequently than is imagined.

I’d like to highlight certain important observations that occurred from my experience with the vena cava.

1. In the first place, the fact that the decision had already been made before the information had been analyzed. Exposing this type of behavior demonstrates an aspect of human nature that we are not used to observing. Our behavior is much more derived from ancestral and unconscious associations from the mind than rational associations. Our decision making process is not as logical, rational and Cartesian as we pretentiously imagine. In a way, we can observe that modern, civilized man who has the knowledge and information to put a device on one of Saturn’s moons, is not completely free in his individual decision making process.
2. The more information I had, the more “dangerous” was my response and the more biased was my decision. There was a paradox in my “deformed” response; it was more logical and rational the more information I had. The more I read, the more I studied, the more memory I had to store the studies, the more rational and logical was that “deformed” response. Excessive information can be a double-edged sword. A large quantity of information can become dangerous when manipulated unconsciously in a biased fashion.

3. It's interesting to observe the "comfort" that this anticipated decision gives us. I used to be secure, aggressive, magnanimous and convincing before I lost my fear of the vena cava. When I lost my fear, I became insecure and indecisive. I also lost my arrogant self-esteem, realizing that my colleague had been right in challenging me. When we give in to pre-fixed emotional decision, we feel a great "comfort", and I will analyze this later on when I talk about religion.
4. I was much more convincing at the anatomy-clinical meeting when I was afraid of the vena cava. I challenged arguments with logic and perhaps gave the impression that I was a brilliant professional. On the day that I lost my fear, I started to demonstrate insecurity.
In this world of extreme competitiveness nowadays, security, determination and tenacity are qualities that are required more and more from those who make decisions. Lifting the veil, one can see that many times these qualities derive from insecurities and fears that are hidden within the depths of the human brain.
5. And what about the patient with a kidney tumor? Wasn't he taken into consideration in my decision making process? Yes he was. I analyzed the studies that came to my consciousness, searching for the one that would be better for him but, actually, my decision was made within a closed circuit inside of me; as if I were a closed circuit: me and the environment. As if I were enclosed within an enclosure like the Operational Closure of Maturana and Varela for living beings, which is also painfully applied to man even though man's enclosure is much broader than the other living beings.

Each one of these observations will be analyzed throughout this book. For now, I'd like to demonstrate that what we call "our" logic, "our" reasoning, which we assume is Cartesian reasoning, is nothing more than a biased decision where information has been previously selected by an immense unconscious structure, which, at least for now, will remain nameless.

2

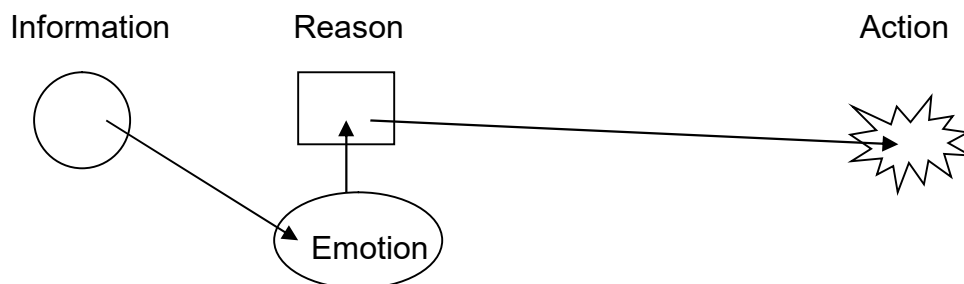
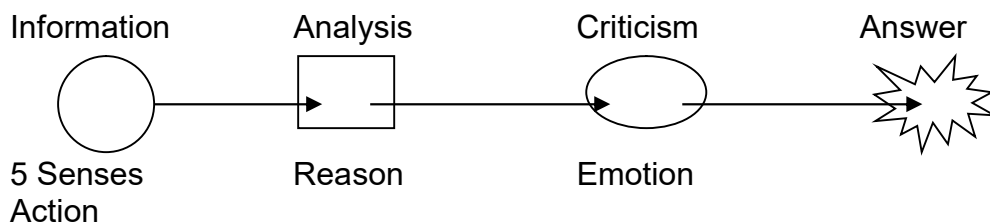
Emotional Decision

Since my experience with the vena cava, I observed that we act this way much more frequently than we imagine.

When facing any problem, we immediately assume that nothing is logical or rational, and only then do we use logic and reasoning to "justify" the position that has already been assumed in an "emotional" manner. "One never has a second chance to cause a good impression", a quite popular quote in marketing, illustrates what I am demonstrating.

During post-operative recovery from major surgery, thrombosis (the forming of clots) in the veins of the legs is a rare, but serious complication. The clots might dislocate to the lungs, causing sudden death. To avoid this complication, besides moving the patient quickly, small doses of heparin, an anticoagulant, can be used.

The doctor who chooses to use heparin in post-operative recovery to avoid this type of thrombosis is always one who had experienced such a complication – and not, as one might think, a doctor who has analyzed the statistics, weighed cost-benefit and has, logically and rationally, made this decision.



In the illustration above, we have the mechanism that we think we use to make decisions, since we “think that we think”. We can see, schematically, in the second illustration how decisions are really made.

This type of behavior is observed very clearly in people who root for a certain football team. Any information they receive about football immediately becomes distorted, justifying the preference for the team. In these cases, reason and logical thinking are “slaves” of a position previously taken in an emotional manner.

When was the initial decision made? The mechanism of such decision making process is lost in the meanders of the mind, but certainly relates to emotional experiences.

In politics and the economy, this type of behavior is very clear. Statistics and citations serve as a way to justify positions already taken. Even taking the honesty of a politician’s purposes into consideration, all of his reasoning, culture, memory, in other words, his whole mind, move towards justifying a position already taken previously. No socialist is a socialist because he or she has analyzed in a rational and logical manner, or because he or she has weighed the cost-benefit of a socialist structure. At some point in his or her existence, his or her emotions have grouped together in a special way and a position was taken, which made him or her become a socialist. Therefore, the greater the number of positions to be weighed, the stronger and clearer the initial emotional position is. This initial position is made by a brain that doesn’t reason or think, because we don’t have the logic that our pretentiousness imagines we have.

We have an emotional brain, which was abandoned, grew and was formed without education and disorganized.

Points to be emphasized

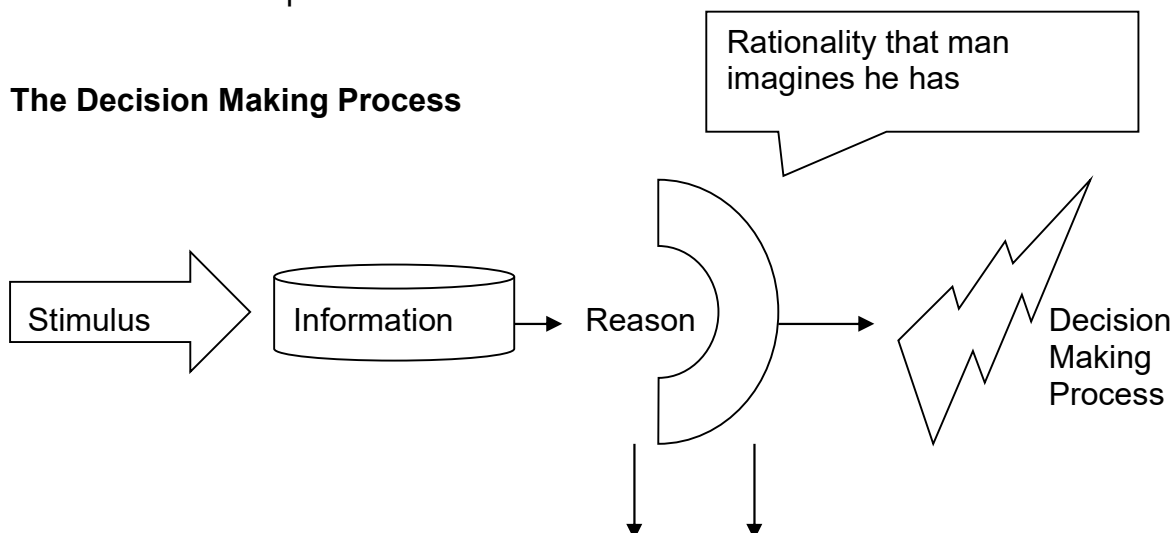
The observation of my own behavior in making decisions about kidney tumors made me realize that we aren’t as rational as we think.

“I think, therefore I am” is a statement that must be revised. I not only think, but I am also moved by emotions! My rational and Cartesian reasoning doesn’t seem to be as rational and Cartesian as I imagine.

I would also like to point out a few important aspects in that vena cava episode:

1. Anticipation: the decision had already been made after a first impression as if another sense, a third eye, had analyzed the fact, knowing that the vena cava represented danger. Even if I knew all the actions to be taken in the face of the complication, even if I exaggerated my security in dealing with it, even not admitting rationally that I was afraid.
2. Emergence: the initial impression made the studies that justified the decision to be made emerge in my consciousness. The fear “written” in my emotional brain selected the information, which I could weigh or think about. The emotional component of my decision making process overlapped my reasoning. I imagined that I was making a rational decision.
3. Emotional support: the emotional support of my decision making process strengthened a series of “competitive” qualities, which are valued nowadays. I became aggressive in order to justify my decision. My aggressiveness excited my memory so that I impressed others with the precision of my citations – there were times when I was able to cite the pages of a journal. I had the extreme need to convince others of my decision.
4. Information: the more I read, the more studies I became familiar with, and the better my memory got, the more dangerous I became in convincing others of the decision I had made. My medical culture could be dangerous.
5. Reality distortion: the fact that I didn’t admit that I was afraid made me study this type of complication profusely. No one knew about this type of complication better than I did. No one knew the measures to be taken in such a situation better than I did and no one doubted my security when I operated in the areas around the vena cava. I demonstrated exactly what I wasn’t. Even my kindness towards an “outdated” colleague was a distortion of reality.
6. Comfort: I felt very comfortable with the emotional decision. The moment I realized this fact, I became insecure and felt extremely uncomfortable. The emotional decision was comfortable.
7. The correctness of the decision: the emotional decision was right for me because I weighed (thought) the ‘self’ as a whole and I knew that there was danger in the proximity of the vena cava. Meanwhile, my rationality was unaware or insisted on being unaware of the danger. Thus, the ‘self’ as a whole was forced to distort reasoning so that the decision would fit reality. Rationally, I was convinced of my decision because I, as a whole, a living being as a being that exists, manipulated rationality.

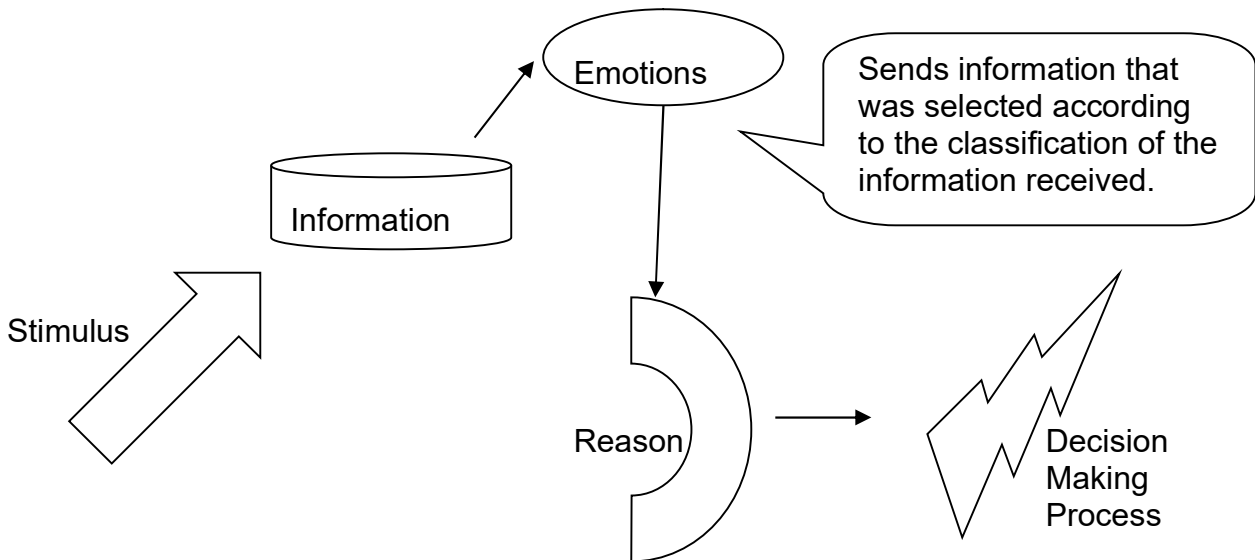
This experience of mine can be transposed to a many of our life situations; situations where we don’t see how we are led by emotional experiences. In these cases, all of our logic serves a decision previously made, resulting from an emotional experience.



Emotions

Generally speaking, we imagine that we possess logical and rational thinking. When information or an external stimulus is received, our brain weighs the information, crossing it with the information stored in memory and makes a rational decision.

When facing a problem, we analyze our files and make a decision? Wrong!



The files that are presented to us are selected within an initial position taken and only the files that support or justify a position already taken reaches our consciousness.

When influenced by a stimulus that requires the making of a decision, my emotional brain decides if I should fight or flee. When facing this decision, it opens specific files. If I must fight, I'm interested in the files that would give me a competitive advantage in the fight. The files that showed me how to run faster, or where to hide, are useless at this point. I am simplifying the example to demonstrate how the emotional brain selects the information that we think are available for what we call our logical and rational thinking.

Man isn't as rational as he imagines he is. Especially, when making a decision that requires urgency, it is clear that there is an emotional brain that functions as a third eye, which makes immediate decisions, making thought judge the data that were tendentiously selected.

It is easy to understand how the decision-making process works in cases of emergency. But, in cases that do not require urgency, it works in the same manner. My fear of the vena cava selected the information for a decision, which had already been made. A state of plenty peace and harmony, plus a deep knowledge of oneself is necessary, in order to identify this exchange of information offered to be weighed. These are situations that are increasingly removed from our current reality.

All of our science and technology haven't made us happier – it has, caused more concern and stress. If we could quantify the satisfaction my father felt when he listened to the radio, it would be no different than what I feel when I watch TV, as it won't be different for my great-grandson whenever he is being tele-transported.

All the great mystics foresaw what they called "the end of times" towards the end of the 20th century. During that time, Lucifer would be loose dominating the world. If we don't disregard the magical thinking, which has been done all these years, we will verify an amazing prediction of current times. "Lucifer" means "bearer of light". Science and

technology are nothing more than bearers of light. They illuminate, but they aren't light; they offer comfort but don't bring happiness, they add nothing to true human satisfaction.

We are living a period of great transformation and change. The creative revolution altered the axis that has guided human thought over the last five centuries.

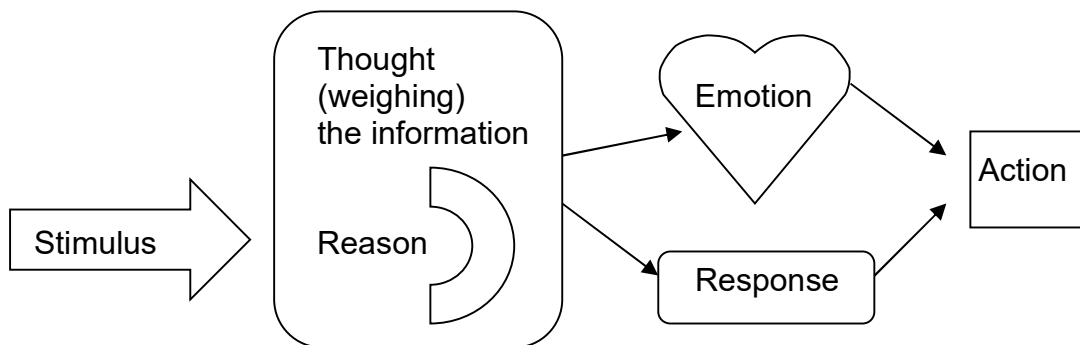
Emotional Decision and Comfort

I come back to the two main observations, caused by my experience with the vena cava.

In the first place the fact that the decision, which had been made before the information was analyzed, made it no longer important. The more information I had, the more "dangerous" and tendentious was my response. The paradox in my "deformed" response was to be so much more logical and rational with the more information I had.

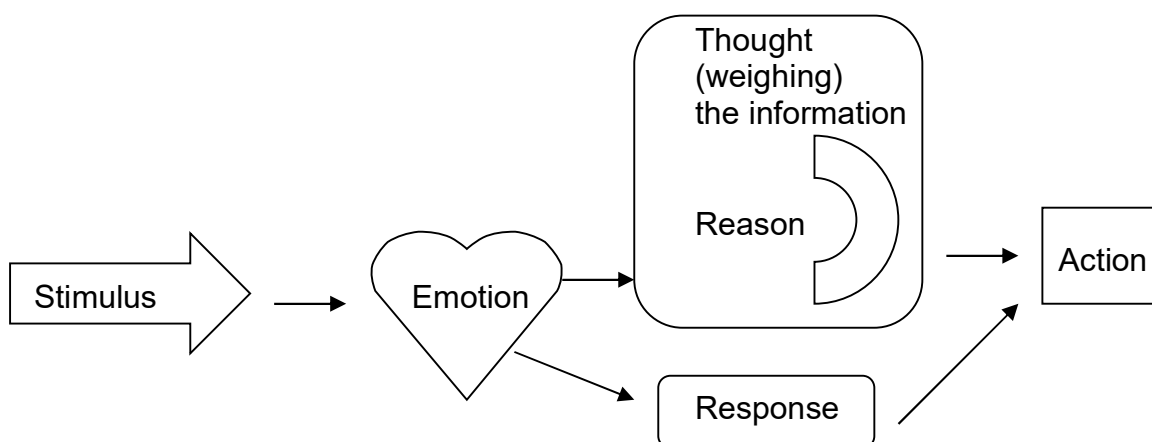
Secondly, it is interesting to note the "comfort" in anticipated emotional decisions. I was secure, aggressive, magnanimous, convincing, before I lost my fear of the vena cava. When I lost my fear, I became insecure, unsure and realized that my colleague was entirely right in contested me.

The sequence of our thinking goes first through a "selector" that will decide how reason will be used.



Logical and rational (Cartesian) thought only happens after it has gone through this "selector", which has a lot to do with our emotions, representing our life experiences, in other words, our knowledge.

It must be stated that man's "knowledge" has nothing to do with external reality.



Living beings are closed circuits that react to external stimuli in order to perpetuate and perfect life. In man, this "closed circuit" has multiple amplitudes.

Living beings suffer from “operational cloister”; a concept magnificently expounded by Maturana and Varela in their book “The tree of Knowledge”. We will analyze this concept in more detail further on.

Another aspect that deserves an explanation is the concept of “emotions”. I will cover this in a specific chapter. I am not tied down to academic conceptualizations, and also will not copy dictionary definitions. My work is based on an observation different from the emotional, where I need to shatter a few paradigms and develop observations outside of the conventional.

For now, I will leave the emotional concept open, adopting Plutchik’s basic emotions: sadness and joy, fear and anger, acceptance and aversion, surprise and expectation.

Amount of Information

An interesting aspect that can be observed in the decision making process has to do with the amount of information or data to be weighed (thought).

In the proportion that greatly increases the number of measures to be weighed (thought), man accepts and understands that he makes an emotional decision, and afterwards, uses reason and logic to justify his position.

In case we decide if two and two is four, the facts to be weighed are few; they are part of mathematical knowledge. The factors weighed are few and the emotional factor is small.

It might seem strange but emotional decisions also exist here. If I have taken positions of vanguard regarding the mathematics of chaos or of the infinite, my answer is that two and two tend to be four. If I don’t hold these social and emotional positions, I can affirm that two and two is four. This is a simple example with few factors weighed.

In a very complex decision, such as the choice of a partner, the factors to be weighed are so many that we openly admit that our decision is simply emotional.

The greater the number of positions to be weighed, the stronger and clearer is the initial emotional position. This initial position is taken by a brain that does not reason or think in a Cartesian manner and doesn’t have the logic that we think it does.

The emotional brain has its own reason and logic. An emotional decision is made through thinking that is usually identified as “magical thinking”. Magical thinking is rational and logical; the difference is that the magical decision uses different parameters. The emotional brain thinks with life as a whole and not as an individualized being. It thinks of survival and not the small-minded human arrogance. Symbolic language has a thinking broadness that transcends the small-mindedness of human beings, and seems to embrace humanity as a whole.

Another aspect of the decision making process is that, the more serious the decision – serious being understood as a threat to the being – the more frequently the decision is made in an emotional manner. If life is being threatened, the decision is basically emotional. This characteristic of the decision making process assumes great importance when we analyze the evolutionary process of the human brain.

Decisions in the Modern World

In the modern world, we are constantly bombarded with a large variety of information. With each bit of information received, we must make a decision.

As soon as we get up in the morning, we receive radio or TV news, and when we get to work, this news will be part of our work relationships. We must position ourselves in terms of each piece of information; and nowadays there are thousands of them. The great

quantity of data to be weighed makes modern man react more in an emotional and less in a rational fashion. This has been exploited in an absurd way by the media.

Modern man is gradually stopping being rational to gradually making more emotional decisions. Such decisions are being made by the emotional brain, known as the limbic system or intermediate brain. It is amazing how we are commanded by this emotional brain, and how little attention is dedicated to it. How could this brain be relegated to a secondary position in our upbringing! Furthermore, emotions are usually despised as if feelings were a weakness in relation to logic and thinking that we judge dominates our mental and social structure.

Selecting an Entrance

Another important emotional function is selecting the information that must be catalogued or stored.

When we receive a large amount of information, parts of it are stored while other parts are ignored. Thus, even the information we possess are biased, because we only keep the information that interests us.

Man practically only sees what he wants to and not what exists outside of his operational enclosure. This characteristic of the emotional unconscious can be observed in the reading of books written through emotional ecstasies such as certain biblical passages. Each time a religious person reads the Bible, he or she finds an explanation for his or her current problem.

Our reasoning (rational brain or brain cortex) behaves as if it were subjugated by emotions (emotional or limbic brain). It seems that man's emotional essence has a breadth greater than its individual form. The emotional brain thinks in a less individualistic way, it thinks about the group, the species and perhaps, about humanity before thinking about the individual. Art and religion are full of evidences of the breadth of knowledge, which can be accessed through emotions. During states of ecstasy, artists and the religious are able to express knowledge that they had no means of possessing rationally.

The manifestation of this knowledge will be analyzed later on when I deal with the theory of the emotional catapult.

The Mayan Brotherhood

I am intentionally not naming or diagnosing structures and behaviors. I am simply analyzing facts and observing the illusion we have that we think with reason and logic. We delude ourselves that we think, but we actually react to everything in a much more emotional manner than one can imagine.

Another interesting experience happened to a group of friends and I who studied the Mayan civilization.

I had been to Mexico where I visited several Mayan ruins and came back amazed with this civilization's knowledge.

I've always searched for an explanation for how ancient civilizations possessed knowledge and information that we, nowadays, only possess if we use very advanced technologies that they were nowhere near having.

I wished to enter the "brotherhood" for Mayan civilization studies. My friends gave me a book on the subject so that we could discuss it at the meeting. I read the book, criticizing it from beginning to end. I used countless cunning skills to dismount the theories presented by the author. My critique was so destructive and astute that I had to ask myself why I was behaving in such a way. I read the book again, trying to be as open-minded as possible with the author's sequence of reasoning. Nevertheless, many critical observations would enter my mind.

The second time I read the book, searching for what could have caused that initial antagonistic position, I discovered that the author had dedicated the book to his mother-in-law. I observed that this first impression had made all the all of my reason and logic “batteries” turn against the author.

I believe that this was the reason for my initial emotional position against the book.

At the next meeting, I exposed my self-criticism and faced the fact that all of my destructive aggression was due to the dedication of the book to the author’s mother-in-law. But, despite several of my reasons having been accepted by the group, I was not accepted.

I’m telling this story to exemplify how we are commanded by this emotional “something”, which is born with us, grows and develops in a total random way within our mental and social upbringing. This emotional positioning of ours is so hidden within our unconscious that when it is exposed it seems absurd, not to mention ridiculous.

I’d like to demonstrate with this how we live immersed and commanded by the emotional being that lives in each of us, and how we have little perception of it.

Man “thinks that he thinks”. He deludes himself that he thinks and is logical. However, he uses all the information available to him in his brain to serve an existence, which he has almost no control over and that is emotional.

Part 1

3

There’s Something Wrong

We have never known as much, been as comfortable, and dominated the world as much as now. We are leaving our planet and heading into the universe, and with all of this, we are not happier. There is a general dissatisfaction. Those who have almost nothing are as dissatisfied as those who have too much. Depression is the illness of the moment. Drugs have become the path that youth has found to face this chronic dissatisfaction. Violence is taking the proportions similar to the importance it had during the Iron Age, when man and humanity were defined by the law of the strongest.

Medicine

Everyday, medicine discovers new diseases and fewer solutions. The average lifespan has increased fabulously, but only for a small part of humanity. The vast majority has a short lifespan and in miserable conditions.

Even the life that we are able to prolong is many times, a life of poor quality, which wouldn’t be worth living. Other times, we restrict quality of life so that it will be prolonged and the person suffers more.

Medicine, in certain aspects, is still far behind. The prevention of cardiologic diseases is an example. For years, a person’s quality of life is limited, not allowing him or her to drink, smoke, forcing him or her to a methodic life, keeping him or her from being exposed to adventures, using preventive medication, all of this so that his or her coronaries don’t get obstructed and he or she doesn’t die from a heart attack at the age of 78. Then, this patient dies at the age of 79, consumed by pancreatic cancer, which is also called cancerous cachexia and suffering with terrible pain. We have done something terrible to the patient, despite all of our scientific knowledge and good intentions.

Very little is known about what can happen to a patient. Medicine must have more foresight. It should take into account the patient's will, his/her desires, his/her wishes, in other words, his/her emotions.

Genome

With the study of the human genome, we have lately begun to see some light in terms of being able to predict a little what is programmed for each one of us. The day will come when we'll be able to tell a patient: you can smoke, because you won't get lung cancer; and to another one: remove your prostate before cancer attacks it. We are still very far from this.

Medicine, like every science, treads a tortuous path where knowledge, erudition and good intentions don't always result in good solutions.

Love

Man doesn't understand himself. A small portion of men concentrates the entire wealth of humanity, much more than they would be able to consume, when any animal knows that having more than necessary can be its downfall.

Over two thousand years ago, men talked and discussed about love, and are still incapable of any love that is beyond their personal interests, his operational cloister.

Aging

Medicine is always inventing methods so that people won't show signs of aging, so that they can lose weight, to be thin and beautiful, when it was man himself who created this situation, prolonging life and degrade natural selection. If man hadn't destroyed his predators, all of humanity would be thin and beautiful. Lions would eat the elderly, the overweight and the disabled. Man doesn't know what to do or how to live with his achievement.

Man himself has created the problems he faces, behaving as who doesn't have "seeing eyes".

The Blind Man and the Dog

If I want to teach a blind man the difference between a German shepherd dog and a wolf, I can give the blind man the statistics of the animals' size, their tail size, the shape of their muzzles, all the characteristics of both animals. The blind man might read more than a thousand books in Braille and have a vast knowledge, but he will never be able to distinguish one from the other. If he has to put his hand in the animal's mouth, he will always do so feeling insecure, he won't know if he will be bitten or not. All of his knowledge is useless if he lacks "seeing eyes".

Seeing Eyes

Thus, it seems that we are treading along with our science. It seems that we lack a direction, an eye. This "eye" has to do with our relationship with other men, with our feelings, with our art, in other words, with our emotions.

We live as if we were blind, we have compiled thousands of data, we have drawn marvelous conclusions that fill us with pride and that materialize themselves in academic theses. With the advent of computers, we cross-reference data to the infinite, we discuss about an order for chaos, we contest God.

I believe that the "seeing eyes" that we lack are related to the emotional brain. If we don't understand, if we don't know how to use our emotional brain, we will be blind, without ever being able to differentiate the value facts.

Gaia

This is a clear matter of facts and values, which demonstrate how we don't have "seeing eyes". Nowadays, there are movements that lead towards the opening of these new eyes. There are movements linked to ecology and the concept that the Earth behaves like a living being, for instance, the Gaia hypothesis. Ecologists, however, are so disoriented that they engage in movements to save seals in Alaska and forget that children die of hunger in Africa.

As to the Gaia hypothesis that the Earth behaves like a living being, if it behaves like an animal, then man is its offspring who sucks at her teat, has grown teeth and is now biting on the breast that feeds him. At any moment, Gaia will give him a shove or a slap. It seems like the mystical end-of-times prophets who have been very active lately, somehow know about the Gaia hypothesis.

Bioethics

Within medicine, bioethics is imposing itself as a new scientific concept that tries to find the lost path. One of the greatest centers for bioethics studies in the world is the Kennedy Institute of Ethics, which is maintained by the Kennedy family, who had one of their daughters submitted to a lobotomy, at the time a surgery hailed by science. It was so acclaimed that it awarded Egas Muniz Nobel Prize for Medicine. Nowadays this surgery is considered a serious mutilation. At the time there was clearly a lack of sense (an eye) to evaluate the breadth of such an option.

Points of View

A few years ago, I operated on a Muslim patient. She lived in her house with two other wives and their children. During her recovery, my colleague was changing her bandages and talking to her, asking her what it was like to live with two other women in a house. When she realized that he found it strange, she was truly astonished. She said she found our customs very strange, where men had children and abandoned their women to raise them! In her culture, a man could have three wives; support them and their children, taking charge of their upbringing. She was the one who found our behavior very strange.

An Indian Young Woman

I read an interview with a young woman from a province in India, where she commented on the absurdity of the western custom where a young woman like herself having to choose a husband on her own. She didn't think it was possible to choose a husband without a recommendation from the parents because she had no experience about what marriage was.

Communism

My brother visited Russia during the communist regime, and he met a highly qualified engineer there who made very little money. My brother suggested that he move west where he could be a successful professional. He was surprised when the engineer

turned him down and said that he'd never work at a place where he'd have to please his boss on a daily basis, because if he didn't, he could get fired.

Justification

The observation of my behavior in the vena cava episode changed the way I looked at psychism and the world. I started to pay attention not only to my behavior, but also to all behaviors. I became aware of the importance and the way, which the emotional factors influence and decide behaviors that I had judged to be logical and rational.

Most of the decisions that seem to be rational, which one believes that the pros and cons have been weighed, have nothing logical and rational about them. The decision had already been made by an unknown emotional component.

By observing my own behavior, I was impressed with the security, confidence, and even the petulance caused by fear. When I lost it, I became insecure. The security resulting from my ignorance was, in a way, extremely comfortable. My discomfort was intense when I lost my fear.

This comfort caused by a pre-established emotional decision was the aspect that impressed me the most and opened a broad horizon for my studies of the emotional brain, because I realized that many of my religious beliefs lay within this "comfort" zone.

Homo Sapiens

I began to understand that the *Homo sapiens-sapiens* that we know, is not so much of the *sapiens* genus, perhaps belonging more to another genus, the *Homo emotionalis sapiens*, for instance. This metaphor serves to exemplify my work proposal.

In the latest studies of the emotional brain or limbic system, one can observe that the importance of this part of the brain is much greater than one had imagined. Despite the advances of neuroscience, we find ourselves far away from an understanding of the workings of the human brain. The theory of evolution does not explain how man does not use the totality of his brain. The appearance of man happened very quickly, almost suddenly, if we consider the evolutionary process. In this aspect, religion, when explaining the creation of man, is much closer to reality.

Emotional Brain

I believe that the emotional brain has a much more important role than is imagined. I will enumerate the main points to be developed in this study. They are all extremely controversial, concepts that are in a way aggressive to the Cartesian rational thought. They are new thoughts, for we are living at a time of search for new paths. If at moments there is a lack of "language" to express myself, it is because emotional language is symbolic and not verbal.

1. During the decision making process, it seems that emotional thought precedes rational thought. In the vena cava case, my "medical culture" solely served as justification for a decision I had already made.
2. In human behavior, emotional thought has a foundation role, not only influencing the decision, but also offering a comfortable structure, which we have gotten used to thinking about.
3. As to the dichotomy that our culture has displayed between science and religion, between rational and logical thought, the emotional brain is surely the link capable of joining science and religion. Apparently, it is in the emotional brain where the comfort provided by religion can be identified.

4. The emotional brain is not irrational. There is reasoning, which is unknown to our individual reasoning. Your rationality is not restricted to the individual, having greater amplitude, which can affect an entire species and not only an individual.
5. Our rationality, which served as the “mold” in the composition of the neo-cortex during the evolutionary process, was created through the emotional brain. This is basically the theory that I will defend: The Emotional Catapult.

The emotional brain would've served to transpose all “written” knowledge in the brain to the man's neo-cortex. The model for our way of thinking is the emotional brain.

Proposal

As you can notice, I am observing man, his mind and culture from a total unconventional angle. I'll have to break with many paradigms, for my mind has also been “built” and molded by rational, logical and Cartesian thinking. I believe that it is time to “turn the tables” and try to develop or at least open new paths to face the new world that is dawning at the beginning of the 21st century, a world that man is not prepared to face. Reestablish emotional factors within the understanding of the human mind. Give an explanation and a meaning for this New Man that is facing a transformation unprecedented in the history of humanity.

Part 1

4

Man Thinks He Thinks

The giraffe's neck, like man and all other mammals', has seven vertebrae. Nowadays, it is believed that the growth of its neck was the result of an adaptation of the species to be able to reach the highest leaves in the trees. The proto-giraffe that was unable to reach the highest leaves wouldn't be able to eat and would die. Thus, only those with longer necks would survive more.

Natural Selection

This is how natural selection happens. Within this same principal, because man was able to walk on two feet, he began to see farther, which gave him the possibility of more productive hunting. Those who were firmer on their feet were able to hunt more, while those who were closer to the ground found less game, dying more and reproducing less. This was how the natural selection of the “*Homo erectus*” occurred.

The polar bear is white because the dark ones died more. The leopard has a fur pattern that better adapts to its camouflage. This is what happened to all living animals and vegetables. Each one has developed a special ability that distinguishes and enables life.

Useless Organ

Man's brain represents a paradox within the evolutionary process. Man is born with a structure (the brain) that throughout his entire life, only a tenth part is used. Within the evolutionary principle, an organ that has no use within the process of qualification for life is not possible.

We have organs that we don't use, but they are atrophies of organs; at a certain evolutionary moment, they were of vital importance and, with the unfolding of the process, they stopped being important. It's the case of the intestinal appendix, which at a certain time, was important but atrophied because it was no longer useful.

The brain does not fit in this category, because without doubt, it is one of the main organs that qualifies man for life. It is very strange that man possesses an organ he doesn't use. How could this structure have been formed? What had the function of this structure been in man's evolutionary process?

Recent studies of the brain haven't help much in the solution of these questions. One can even say that they have confused us even more.

With the recent studies on emotions, we are even further from understanding this organ. It seems that emotions are not at a certain place in the brain, but are distributed throughout countless structures.

Paradoxes

With the brain that we have now, *Homo sapiens* suddenly appears on the evolutionary scale about 10 thousand years ago. The neo-cortex appears very suddenly, as if it were the result of few mutations, not having had transformations in the brain's structure over these last 10 thousand years. It is as if the neo-cortex had suddenly been created without a defined function.

The neo-cortex is, in itself, nonsense in an evolutionary sense: an organ that only 10% is used does not justify the existence of an organ. One cannot explain, within the evolutionary process, the sudden appearance of an organ that has solely usage potentialities, an organ that is not used!

Ancient Civilizations

Another paradoxical aspect of the human mind is substantiated in the history of a few ancient civilizations, especially the Egyptian civilization. Nowadays it seems impossible to build a big pyramid, moving blocks of stone in a process we no longer know, orienting these blocks in a way that small orifices in the stone allowed the light of a certain star to pass through on a certain day of the year. Someone who knew the world well chose the location of the Great Pyramid.

The first Egyptian dynasties possessed a knowledge that got lost over time. The only explanations, with a reasonable amount of logic for this fact is the existence of a more advanced civilizations (Atlantis), that had existed and disappeared in catastrophe, or the existence of alien contacts that had transmitted such knowledge to the Egyptians. Both explanations lack confirmation. No evidence of the existence of the Atlantis civilization has ever been found nor has any sign of a catastrophe that might have submerged this civilization ever been identified.

The most concrete evidence we have comes from Plato's *Timaeus* dialogues. These dialogues could very well be an allegoric report of another civilization that he knew about. As to contacts with aliens, albeit possible, it doesn't seem logical that such beings have been here and then disappeared without ever contacting us again.

This same phenomenon observed in the Egyptian culture can also be observed in other great civilizations. In the Americas, at a much later time, in this millennium, we have the example of the Mayas, which had an incredible knowledge of astronomy, mathematics

and philosophy; knowledge that has also disappeared without a trace. In Chichen Itza, there's an observatory with two constructions at the top, one pointing at the magnetic north and the other, at the top, which points to the geographic north, as if the first builders identified both points. On the other hand, the Aztecs in Mexico created an extraordinary calendar.

Greek mythology shows a deep knowledge of man. The legends were elaborated at the beginning of Greek civilization. The knowledge expressed in mythology goes beyond the understanding we have of the human being in our days, but was also lost during the evolutionary process of Greece.

These civilizations have a common characteristic. They have all acquired great knowledge, no one knows where. They were unable to develop it and lost it little by little. Indubitably, there is a common source where all these civilizations obtained this knowledge.

This source is not present at a determined time in history; these civilizations existed at different times. Such civilizations had no contact between them. They existed in different and faraway places. Despite their cultural differences, of course they had some origin within man himself, in his brain. It seems that the explanation of this paradox is linked to religion, which are part of the emotional brain.

Way of Thinking

The way man thinks is structured like a tree. At each choice he makes, he abandons others and starts to think within that sector, becoming incapable of understanding thinking done through other paths.

An example of this is in the choice of numeric structure that Native Americans used, which didn't allow them to understand the number two, but on the other hand, allowed them to understand the infinite.

When man thought that the world was flat, he couldn't understand what was under the Earth and how oceans ended. In reality, he didn't think that the Earth could be round; he was sure of his thesis because there could be no other way! This example demonstrates what I mean, comparing man's thinking with the branches of a tree.

(illustration)
Restricted Knowledge

New Branch of Knowledge

The moment that man admits that the Earth is flat, his reasoning becomes limited to a single sector of ideas or thoughts. So that he can proceed with his reasoning, he must "unlearn" that the Earth is flat, which is very difficult, because all the evidence of his senses shows him that the Earth is flat.

To be able to think beyond the limitation that has restricted him, man must break with his paradigms; he must renege his reasoning structure; renege in a way that is contrary even to his senses and observations. It is easier for someone to understand that the Earth is round for those who never knew it was flat.

Such a difficulty in understanding, when one follows a structure of thought, is so impregnated in man that extends itself to his action in an unconscious manner.

Urethroscope

When I was doing my residency, we had a great respect for the bladder valve, which does not allow urine back into the kidney when it contracts itself (ureter terminal). We feared the infections (chronic pyelonephritis) that could destroy the kidney and, since there were no transplants at the time, these infections were fatal.

Over time, we began to understand these infections from a different perspective. When the ureter endoscopes came around, we had to pass the device, the ureteroscope, through this valve. It was very hard for me to force the ureter terminal. I had an enormous difficulty. My resident passed the device with more ease than I, who was much more experienced, could. This happened because I had a limiting knowledge that he didn't.

This observation of my medical practice extends itself to the entire area of knowledge. When we acquire knowledge, we are limited to it, making access to other knowledge to be outside of our thought structure.

The Option of Thinking

Man made an option for a thought structure that has modified his entire way of thinking and that, if on one hand has lead him to technological wonders, on the other hand, it has created unsolvable problems within his thought structure. It is very difficult to express this idea because it is rooted in the way we are, the same way that we were sure that the Earth was flat.

In the 19th century, it was proved that the Earth didn't turn, demonstrating that a balloon didn't move from the place on Earth where it had gone up. If the Earth turned, it would have to be moving at an enormous speed in relation to the ground.

What I'm trying to introduce here, is a rupture from a paradigm that we must unlearn, renege, to follow along another path.

Renaissance

Soon after the renaissance, man adopted a thought structure where he was the center of knowledge. Knowledge began to be valued to the same extent that man could use it.

The human brain was placed in the center of the Universe. Man adopted an egocentric thinking. This egocentrism is wonderfully expressed in Descartes' phrase, "I think, therefore I am". Contesting this statement is strange, but before, man existed due to the will of the gods, because nature allowed it, because he was part of a group or a society, or because he should serve a king or a cause.

Man started studying the workings of nature, of the groups and of society.

Thought Structure

In the 16th century, a single man, Francis Bacon, organized a structure so that man could think within it. In his book, *Novum Organum*, Bacon guided the way of our thinking, defining what we started to call "science".

Francis Bacon practically tied and gagged our way of thinking, using a structure where the human brain was the center of knowledge, an egocentric way of thinking.

The whole world embraced this structure proposed by Bacon. However, he as a person wasn't deserving of the least credit. He betrayed his benefactors and had no scruples in using the wealth of others. When he was banned from the court of Queen Elizabeth I, who was disgusted with his lack of character, he started to fawn over the courtiers.

It seems incredible that a man of such caliber could have influenced the structure of human thought so much over the last centuries. I believe that this atmosphere, which was

favorable to the acceptance of Bacon's scientific rules, happened because of a revolt against God, due to the abuse committed in his name, due to the madness of monarchs and the lack of structure of the groups, of the nation and of the family.

Science

In this unfavorable environment, man had become more selfish, thinking more about him than about the whole. He searched for an understanding of nature, within a "rule" where his rational and logical brain was the center of the universe. "I think, therefore I am!" If I exist because I think, it is because my brain is the center of existence! Nature was no longer the center of the world and started to be analyzed and equated into this "rule" that was called science.

Before this, knowledge formed a whole with philosophy, magic and even with religion. God was gradually diminished as natural, social and psychological phenomena fit into this "rule". H. G. Wells begins his renowned *Natural History* with the affirmation that God gradually diminishes as science advances.

The abuse of material power demoralized the emotional and magical aspects of man, creating an egocentric man, centered in his brain's cortex. The association of religion with power forced man to think more about him; the God presented to him didn't care about him.

Magical Thinking

There could be no difference between "scientific" thought (Cartesian rational) and "religious" or "magical" (emotional), since both happen within the same structure that is the human brain. On the other hand, we understand very little about what the human brain is and we have no idea about the essential structure of its workings.

There must be something in common, capable of joining religious thought to philosophy and science, and this something, is unquestionably in the emotional (limbic) brain, which is responsible for our emotions.

Our thinking is structured in such a way that it is difficult to find a way to unify the entire workings of the brain. To find such a way, it is necessary to go back, to start over again, to follow a new path to reach an understanding.

Therapeutic Plant

Emotional thought is a form of thinking that involves parameters that are different than what we call scientific thinking. An interesting example of this difference can be seen in the studies made by anthropologists with the use of medicinal plants.

Transporting from the Americas to the Philippines, the local natives began to use them for the same medicinal purposes as the American native. This use was not fruit of logical thinking, but of a magical thinking, where the Philippine native associated the plant's taste with the curative power of another plant that they knew about. We, the so-called civilized people, so as to discover the same therapeutic use, we should experiment, over and over, the utilization of the plant to discover its use; or if we used science, discover the plant's essential and active substance so that we could use it as medication.

Religious Thinking

Magical thinking, which is also religious thinking (emotional), is not, as what is imagined, a random function. This emotional thinking weighs, or thinks, data that hasn't been coded in what we call scientific thinking (Cartesian). There is, without doubt, an

immense potential in emotional thinking – known and accepted as religious and magical. Within our current culture, we understand that it is separate from scientific, rational and logical thought.

By what I'm trying to demonstrate, what we call scientific, rational and logical thinking, is almost always based on emotional (magical and religious) thinking, which precedes it in the decision making process. There is an emotional way of thinking. This form of emotional thinking possesses reason and logic, which are different from the reason and logic form of scientific and Cartesian thinking.

We believe that our way of scientific thinking is "all mighty", however, it is nothing more than an appendix, a structure, "astuteness", used to justify what has already been decided.

Labyrinth

The Middle Ages was a period in the history of humanity where emotional decisions predominated in detriment of reason and logic. It was a time dominated by the emotional brain with an abandonment of the rational and logical brain (brain's cortex).

This predominance of emotional thinking is not, as one might imagine, a step backwards regarding human evolution as a whole. Emotional thinking also has an important component that is rational.

In religious architecture, especially in gothic cathedrals, we find examples of this reason and logic hidden to the eyes of those who don't have the capacity of understanding. The labyrinths that were drawn on the floor of the main cathedrals, and that were destroyed in their majority, are an extraordinary example of how a purely emotional idea can be expressed in a graphical manner without going through reason (rational brain).

The labyrinths were drawn on the church floors so that the congregation could follow their meanders. They all consisted of many sinuous turns and had common characteristics: during the first turns one comes closer to what the end should be, then the meanders move away and follow a sequence of approximation and distancing. In the end, they all have an almost straight line that leads from the periphery to the center. The labyrinth was designed to be followed by those who have an emotional certainty that there is an end to the existential path, a finality, and that the meanders of this path always offers a way out, regardless of being close or far from the end. The labyrinth is a graphical expression and materialized from a purely emotional idea (decision): faith!

Cathedral constructors, who designed them, did not start from rational Cartesian reasoning to get to the labyrinth. They designed labyrinths by intuition (emotional). Holy architecture is extremely rich in ideas expressed graphically and that are not understood by Cartesian rational thinking. There is a symbology, an emotional language that, as strange as it might seem, is much richer than our phonetic language.

Part 1

5

Fragmented Man

Modern man is experiencing a unique moment in his existence. He is experiencing an extraordinary progress in science, which hasn't been followed by advances in other areas that compose his entire being. Religion has been the same for two thousand years.

Psychological disorders began to occur with greater frequency, which are now being discussed what is normal and what is pathological. With the advance of psychology and science, man has not improved his quality of life and is becoming unhappier.

To escape this unhappiness determined by the fragmentation that science has caused to his being, man abandons himself to consumerism, desperately gives in to a religion or consumes himself within science, remaining forever fragmented.

He, who abandons himself to consumerism, soon realizes that his physical satisfaction has no limits. He, who gives in to religion, finds answers that confront science and with his physical needs.

Fragmentation

The common man suffers terribly with this fragmentation and throws himself at one of the three muses, satisfying his physical, emotional (religion) or his reasoning (science) needs, without realizing that none of them could lead him to being balanced. At first they give him satisfaction, then they toss him into the emptiness of anguish and despair.

Man must find a way to live with and balance these three muses, because only then, will he find balance. To do so, his physical needs, his emotions and his reasoning must head in the same direction. In the history of the great men who were extraordinarily developed in some of these areas, we find the recount of this titanic struggle.

Saints

The great saints, who through their dedication to a religion, extraordinarily developed their emotional being, they fought a painful fight against their physical or material needs. The terrible struggles that Saint Francis of Assisi, Saint Theresa of Avila, Saint Anthony and many other saints fought against their physical needs are renowned.

These saints, who projected themselves as emotional beings, left purely emotional paths. Among them there are a few who developed their emotional being with some harmony with their physical and rational beings. That it the case of Saint Augustine, who before dedicating himself to religion, lived a material existence, which enabled him to develop, if not a science, at least some of the scientific models applied by the Church to this day.

Science

Those who have developed themselves only scientifically, found at the end of their paths, the emptiness of their minds.

The beings that developed the satisfaction of their physical and material needs are generally found throughout the world, don't have other stories except for their own and are always unhappy.

Einstein was one of those who extraordinarily developed reason and science. However, when we analyze his life, we come face to face with a profoundly emotional being and, perhaps even, religious.

Einstein

There is a story that while Einstein was walking along a beach one day, he observed a young man gathering starfish from the sand and throwing them into the ocean. Intrigued with the young man's work, Einstein asked him why he was doing that. The young man replied that he was returning the starfish to the sea where they belonged.

With his rational thinking, the scientist explained to him that the few starfish that he threw back, if compared to the size of the sea and the amount of starfish, made his work have no purpose. When he got home, Einstein couldn't sleep that night, thinking about the young man's work. The following morning, he went to the beach to gather starfish and return them to the ocean.

"There is no path that can lead from the world of facts to the world of values" (Einstein).

Saint Francis of Assisi

On the other hand, a man like Francesco Bernardone, ignored any satisfaction for his body, going hungry and cold, living a life dedicated to the poor, the animals, to every living being, in other words, to starfish too. He finds immense realization and even satisfaction in this.

"For it is giving that we receive; it is by dying that we live an eternal life" (Saint Francis of Assisi).

These two men have the same brain, they belong to the same race, they lived in the same world, receiving information from outside, metabolizing this information and reacting to better adapt to the exterior world. There is a link between them. They are all human beings, with identical structures. This structure has much to do with emotion! The integration between these two men certainly goes through their limbic system or their emotional brain, that was separated from reason since Descartes' time, with his phrase, "I think, therefore I am". I believe that we are entering a new era where Descartes' phrase will be changed to, "I am moved by emotions, therefore I am!"

Ecstasy

Emotional aspects are interlinked with cognition, with memory, and what is really interesting, with the sequence of our thoughts.

We have desperately searched for a unity between logic and emotions, faith and reason, to join religiosity with science, the mystic being with the rational being.

As to the dichotomy that our culture has made evident, between emotions and rational thought is certainly the link capable of joining scientists with the religious. Apparently it is in the emotional brain where the comfort offered by religion can be identified, and that has made man to develop religiosity for as long as we have historical records.

The sequence of our thoughts is determined by emotional aspects. We can find this statement in religious texts in an apparently figurative manner. These books weren't written by neuroscientists, but by religious men in a state of ecstasy. It seems like emotional ecstasy give access to a broader knowledge than can be acquired rationally.

Carmel

There are a great number of Carmelite convents spread around the world. They began at Mount Carmel in Palestine near Haifa. In this place, the prophet Elijah fought against the heresy of Baal and decapitated hundreds of priests.

After the crusades, a convent was created there. Later, a catholic priest received the scapular from Our Lady and the cult to the Scapular of Mount Carmel was created.

In the 16th century, there was a deep reform led by Theresa of Ávila, a woman who was a member of Spanish nobility. An intelligent woman and a great religious leader, Theresa restructured the monasteries by giving them greater rigor of the rules and tendency to an accentuated mysticism. She was a practical woman who knew the art of politics, which helped her raise huge sums for the institution.

The restructuring with a strong mystical connotation was implemented by her assistant; father João da Cruz, who wanted greater rigor in the Institution's male units. He was a great mystic, having written a series of poems where a profound repressed sexuality can be noticed, in dealing with divinity. He was a man who probably repressed his sexuality by sublimating it in mystical devotion.

Since Theresa of Ávila's reform, what we now call Carmelite orders began to appear, convents where the nuns are called Barefoot Carmelites. In the Carmel orders, the nuns dedicate themselves to a deep mystical introspection; they practically lead a life of praying and praising God.

The association between sexuality and the figure of Jesus was inevitable for these women who isolated themselves from the world in an attempt to sublimate their physiological needs. Sexuality grows in a mythical explosion that makes them forget about their bodies and practically live within an emotional world. Many of them die at an early age, in states of ecstasy that generate an extraordinary emotional field, which transcends themselves and the Carmel order, triggering paranormal phenomena that are later, seen as miracles and are sanctified by the Church.

The greatest example of this ecstatic state is Thérèse de Lisieux, perhaps one of the greatest expressions of the manifestation of the limbic brain ever recorded. Her texts demonstrate a high understanding that transcends her sparse culture; her projection demonstrates the connection that exists between all the limbic brains, this probably comes from the large number of miracles attributed to the Saint.

There are several similar cases in the Carmel orders. In Chile, there is another saint with the same characteristics: Santa Rosa of the Andes. The value that the Church has placed on this type of mystical ecstasy was much greater during John Paul II's papacy. Several of them were proclaimed saints and beatified.

Michelangelo

A fabulous example of emotional power is expressed on the ceiling of the Sistine Chapel in the Vatican.

The painting that represents man's creation was painted by Michelangelo during an emotional crisis caused by pope Julius II who managed, by exerting a lot of pressure, that the artist conclude the job.

Certain knowledge transcends reason; they weren't expressed by logical thinking, but came from Michelangelo's deepest emotional being. In the Creation of Adam, the outline surrounding the figure of God has a perfect coincidence with an anatomical intersection of the human brain. Experiments have been done by superimposing the images and identifying, with great precision, several of the brain's structures. One of the most interesting is the identification of the brain's amigdalae. Another important point is

the fact that God is extending his arm through the frontal lobe, which is the brain structure that distinguishes man from other beings.

In ancient Hebrew, letters were represented by images that expressed their meaning:

IOD, represented by a pointing finger, meant individualization.

HE, represents a southern wind, a blow that comes from within one's being. In the same way it is pronounced. It is the blow of life that comes from a sigh.

VAU means duality.

We can see in the painting how the idea of creation of man is written and expressed: IOD, the finger of God pointing at man, identifying him; HE, the breath of life that is being transmitted to this man; VAU, the duality that joins God to man; and a final HE, which is the sigh of man identifying himself with God. Iod-He-Vau-He. Man is being created in His image!

The knowledge demonstrated in this painting does not seem to be fruit of a rational and logical study in Michelangelo's Cartesian mind. In an emotional crisis, an artist becomes unbalanced, almost loses his mind, and reaches a state of ecstasy where he has a perception far beyond the possibilities of his rational mind. Knowledge that was held within his brain flourishes, knowledge that he had no access to through conventional means.

There are several episodes such as these, described and expressed in works of art. Michelangelo was a man who had a brain just like any one of us. He might have had a small variation in the size of his brain lobe, like there are variations in the size of a nose or an ear. The fact that this knowledge "flourishes" from inside his brain leads to the idea that we all have a wealth of knowledge far beyond than our reason is capable of imagining.

Man has extraordinary knowledge, which he is unable to manifest and that is related to his emotional essence. The access to this knowledge "dormant" in the brain only seems possible in the proximities of loss of reasoning. In the case of the Sistine Chapel, if it weren't for Pope Julius II's excessive, even harsh, rationality, Michelangelo would've gone mad and the knowledge wouldn't have manifested itself. The association between these two men demonstrates the need for unity between them so that the knowledge can manifest itself.

Beethoven

Beethoven was deaf when he composed the Ninth Symphony. He went through a deep existential crisis that came near to madness.

The symphony wasn't composed by Cartesian and logical thoughts, but by emotional thoughts that gave humanity the foundations for romantic music in the 19th century, as if emotions acted towards humanity instead of the person. Rationally, no one would accept that Beethoven suffered to create his symphonies.

Poor Beethoven's reasoning, which thought of the part, was subjugated by his emotions, which thought of the whole, mistreating the man with a broader intent.

Part 1

6

The Revolutions

During the second half of the 20th century, man began to face an exponential acceleration of scientific progress, which caused radical transformations in society, in man's way of life and thinking. The transformations were characterized by revolutions that motivated changes in the life of man, in proportions never seen before in our entire evolutionary process.

It might seem to be an exaggeration, but the transformation that man is going through nowadays can only be paralleled with the beginning of humanity's evolutionary process, when man stopped being nomadic and began living in society.

Social Revolution

We still haven't been able to evaluate the entire extension and depth of the changes these revolutions brought humanity. On the other hand, the speed of the transformations allows the processing by nature of the necessary alterations for these metamorphoses.

During the Middle Ages, a man lived the same way as his father and grandfather. Nowadays, my children, with a five-year age difference between them, belong to different generations.

There's a social revolution underway, which I will talk about first, and there are thought revolutions, which are the revolutions of information, communication, knowledge and creativity.

The first of these revolutions is a social revolution that is reflected in man's educational and learning process, for it involves his molding within the family. I will present this revolution in a very unusual way, showing the family as an important part in the evolutionary process of mammals. Thus, we'll have a dimension of the greatness and complexity of the familial revolution in the 21st century. The manner with which I will present the family revolution will seem very strange and even grotesque. I intentionally expose this revolution firstly. Throughout this book, I will deal with man in his behavior within his psyche and his society as fruit of the evolutionary process.

Mammals

The appearance of mammals represented a great evolutionary advance in the expression of life on this planet. Life became specialized to such a point and reaching such a great complexity, that it was no longer possible for the living being to produce another being completely similar to itself.

It is among the mammals that we will find the acceleration of the development process that occurs "outside" of the living being. At birth, the mammal needs to continue its development outside the maternal organism, and this development, besides being physical is especially educational.

The education that needs to be transmitted by the mammal to her offspring is supplied not only through maternal contact but also in a broader way, is transmitted by the family.

Among mammals, this education is constituted by the male, the female, the offspring and the community. Among current mammals we find different options for organizing the education of their offspring. All these structural options of familial structure are based on the fact that the female has the burden of reproduction while the male, in

some ways, helps his partner with a consistent contribution or with a selection of species improvement.

Families

There are three options made by mammals in order to transmit the education necessary for their offspring's survival: the one where the female is in charge of her offspring, the one where the responsibility falls upon the male and the one that falls upon the entire group.

There are therefore, families where the offspring are reared solely by the female, such as the case of the bear. The male has no involvement in the offspring's education, falling upon him, reserved by the evolutionary process, solely the selection of its qualities as a reproducer.

Another option in the family structure is the one where mammals of both genders participate in raising their offspring. Within this organization there are various forms. The most common is of a dominant male with several females, each one taking care of her offspring, forming a family. This model can be observed among lions and is more common among mammals that hunt or defend themselves as a group. In the case of a family of lions, there is one lion with various females. When he is repelled by a stronger lion, the new leader kills all of his predecessor's offspring. In another model, the male and female participate in raising their offspring forming a monogamous family, this occurs among the mammals that hunt and defend themselves alone, such as the tiger.

There are also those mammals that opted for an educational process provided by the community. This model is well represented by certain groups of apes, where the offspring are raised by the entire group.

The species adapted according to genetic modifications and behavioral changes. Each species has developed their own way of selecting a mate and the way to give continuity to the offspring's development outside of the mother's womb. The selection for the choice of mate seeks the strongest and fittest for survival within the conditions and environment that the species lives in.

Within the vast majority of species there are mating dances where the potential for survival can be evaluated. Also through bouts, the males almost always establish a natural selection process through the strongest.

Each species adopts a pact where the male, female and the group participate, somehow, in the development of the offspring. This pact is nothing more than what we denominate a family.

The Human Family

By weighing our entire intellectuality, logic and reason intention, we cannot escape from ourselves. Despite man's social and intellectual veneer, he is still a mammal that evolved through the process of natural selection, holding this reality within his core. In order to demonstrate my theory of the Emotional Catapult later on, I will take evolutionary psychology to its last consequences.

The manner with which the human species equated this need of continuing the newborn's physical development, of transmitting an education to him, was very peculiar, also generating a much different physical shape between males and females, which almost never occurs in other mammal species. The male has a larger structure, a more muscular shape that makes him more apt for hunting. The female is smaller, with rounder shapes and a brain that works differently than the male's.

Human Adaptation

The new imaging through magnetic resonance (MRI) techniques have shown that the woman's brain functions differently than the man's brain. This difference has been observed for a long time regarding cerebral vascular accidents, which present different symptoms in both genders. Aphasia, the loss of speech, rarely occurs in women. Nowadays we know that this is due to the woman's ability to speak using both hemispheres of the brain, while the man only uses one of them to speak. It is as if the woman spoke and at the same time had a perception of what she's saying with the other hemisphere.

The male became adapted and became stronger to hunt, while the female, who became smaller physically, began to develop behavioral mechanisms to involve the male and make him to hunt more for her and her offspring.

Proto-Man

So that we can have a better understanding of this process, I will call this being that preceded us in the evolutionary process a proto-man. The male hunted and brought the hunt to the female who was probably sheltered in the caves, breastfeeding her offspring. At this point in the evolutionary process, she started to feel the need to stay more at the cave to take care of her offspring, since the evolutionary process was forming a more complex being. The female proto-man had to develop a mechanism to involve the male. This involvement occurred through the heat, which is essentially the basis for the reproductive process. In almost all mammal species, the male and female have similar physical shapes and the desire to mate only occurs when the female is in heat. Beyond this period, both have indifferent behavior.

The proto-man female "swallowed her heat" and began to manage it so that through the heat, she could exercise a certain control over the male. This happened through the development of psychological processes, which now reflects in the difference we now find between the women and men's brain.

Female Sensitivity

Women have developed a perception mechanism of reality, which is very different from men's mechanisms, and women have a much superior sensitivity than men. As a consequence of this evolutionary process, modern women have a mental capacity different than men. She has more sensitivity and an extraordinary perception of everything around her than men do. On the other hand, man has a stronger physical structure; he is larger in size and best equipped for hunting and fighting.

With these characteristics, over many centuries, man completed his physical development, being taught within a family structure. The male would hunt and, later on in the evolutionary process, provided food for the group. The female breastfed her child, so that he could continue his physical development, transmitting the first impressions and teachings to him, while the male, as we've seen previously, he participated in this family unity in a different way than the female, with a participation of the group in this complex education. From the beginning of his evolution, man evolved as a hunter in a group.

Education

This development and educational model has accompanied man from the remotest ages, since the most primitive societies, since the cavemen.

Man has been developing and being taught in the same way over a period of many centuries, but this development and education way is being radically modified lately.

I was raised 50 years ago, with a family structure similar to the cavemen. The male provided for his family group and the female took care of her offspring, breastfed and, in a way, ran the household, involving the entire group in a unit, which depended on her in an emotional way, the same way that the means of survival depended on the male. Such a transmission structure of the complement of man's molding remained the same for many centuries, even millennia. Any change in the structure of transmission of molding can impact us in a totally unpredictable manner.

Women's Work

Two important alterations modified this structure. In the first place, the industrial revolution and the consequent consumption revolution, creating a great complexity for man's complete molding apt for survival in his environment.

In the second place, the most serious, was the discovery of the birth control pill, which freed women from the onus of pregnancy. The evolutionary process of man's (mammal) molding is founded in this fact. The woman begins to choose when to have her offspring. Reason and logic begin to control reproduction. Is there is no longer the animal (mammal) dependency where for millennia the complement of man the mammal's molding is based upon.

The woman enters the work market. And enters with qualifications that man does not have. It was as if the cave woman began to hunt and did it better than the cave men. The stronger being nowadays makes no difference in providing the means for survival (hunting).

Man is losing his role in the structure that was set up and that lasted for millennia to complete his molding as a mammal. The woman, with her sharper sensitivity, developed over millennia in order to structure the family, and who would complete the raising of her offspring, is entering into the work market, in a new society where the physical force of the hunter gave place to the flexibility and to adaptability. A new society, which is in an exponential progression, where unpredictability is the only certainty; where creativity is the gold standard.

Calf

By observing a calf, we can have an idea of the alterations that the change in this "educational" program has determined in the mammal. When a newborn calf is separated from its mother, for only an hour, this calf will develop a different attitude, no longer participating in "games" with the other calves.

When the mammal is born, it is still very unprepared for survival. It needs to be fed in order to continue with its physical development and upbringing to that the neural circuits proper to its species can be connected. Being away from its mother for an hour prevents the calf's mother from licking it and thus determining that those circuits not be completed.

When man is born, he has to complete his physical development while he completes a great number of his neurons' connections, which will work throughout his adult life. If such a small alteration in the calf determines an evident transfiguration in its behavior, imagine the modifications that an alteration in this "upbringing" will determine in an adult man.

The repercussion of this fact will manifest itself in an unpredictable way in the man of the 21st century. By analyzing the family as a complementary process in man's

formation, we can speculate that the tendency for a certain “craftiness” and the increase of egocentrism can be a repercussion of these alterations that the family is going through.

The Revolutions of the End of the Millennium

The final years of the 20th century brought radical changes, which still haven’t been fully evaluated, perceived and considered in their entire extension and depth.

Information Revolution
Communication Revolution
Knowledge Revolution
Creative Revolution

These four revolutions occurred almost simultaneously and involved the entire civilized world. Never before, in the entire history of humanity, has there been a revolution of such proportions or breadth.

Science has capacitated man to manipulate data and information in quantity as well as intensity never imagined before. It was the information revolution. We began to understand processes that hadn’t been accessible to us before due to the vast quantity of data involved.

An interesting example of this advance is in the identification of climate variations such as what happened in the identification of the “El Niño” phenomenon, which cyclically alters the weather in South America. The immense amount of information, which can now be processed, has led us to a new approach as to what infinite is, expressed in chaos mathematics.

Would chaos have an order when the information crossed was close to infinite? We’d end up back at the same previous situation of the mind and soul.

Is there a great order or does everything happen by chance? Einstein said that “God does not throw dice”, to show that chance was not possible. But it is as if God played chess. There is a rule in the game that not even God can change.

Man’s thought process is in a vicious circle that can only be solved if we break from great paradigms. The questions brought by the information revolution do not allow any advance in the understanding of the world or of man, with little improvement to his quality of life.

The information revolution occurred along with the revolutions of knowledge and communications.

Knowledge is the ability to manipulate information and gain some profit from it. The capacity to manipulate information, when applied, is immediately treated as new information.

With the advances in communications, the whole world became connected. With the dissemination of the Internet, the revolution of communications made information and knowledge available to everyone – it has transformed information and knowledge into commodities. Nowadays, a patient with prostate cancer comes to a consultation will all the most recent information about his disease. I can, at any moment, find out about the best way to fabricate a certain type of motor.

Man began to have information and knowledge available almost as if all the information and knowledge were universal assets. Information or new knowledge is harder to hide: all information needs is to be applied so that it becomes a commodity.

Within this perspective, the 20th century ended with the last of the great revolutions, creativity. With information and knowledge at the mercy of anyone, the competitive

differential became the creation of new knowledge, which, being created, degenerate into information. Creating constantly became the emphasis of the 21st century.

We are experiencing the Revolution of Creativity! Creativity is not ruled by logical and rational thinking but basically through emotion. The creativity revolution has disarticulated many years of one type of rational and logical thinking, initiated by Bacon and Descartes.

It is true that all the scientific progress over the last five centuries was the product of standardization of the way of thinking and reasoning propagated by Descartes and coded by Francis Bacon. Science presented a spiral development that, by mid 19th century, began to change the world, and did so in such a manner, that man himself can no longer recognize it. It changed the world but not man.

Man cannot change through information and knowledge, but by experiences that are more identified with emotions. The revolution of creativity decrees the end of an era of rational thought to begin a new emotional time.

Part 1

7

The Three Brains

The brain is an organ that is hard to be studied. The difficulty exists especially because we don't have an order or a definition of its structural functions.

When we study the digestive tube, we know that what the pancreas or the liver does. When we study the central nervous system, we don't know what the role of each structure is in relation to the whole. This lack of definition is added to the great anatomical distortions presented by the brain. The brain is a very compressed organ, as if it were smashed so that it could fit into the cranium, especially the brain's cortex is as if it were a crumpled sheet of paper.

MacLean

Neo-cortex
Limbic system
Primitive brain

This difficulty in studying the brain became easier by MacLean's studies when he identified the triple brain. The brain is composed of 3 distinct neural units that act with a certain independence and as a whole; units that represent different modes of behavior and act through several chemical means. These three units identified by MacLean are as if they were three superposed computers.

The three brains were created at different moments of the evolutionary process, they are like stratified layers that show us the evolutionary sequence that the brain went through.

The anatomical definition of these areas has been done based on the anatomy. It is not possible that this division be considered anatomically due to the fact that the brain is an extremely compressed organ within the cranium.

Evolution

The triple brain described by MacLean, is the key that identifies the steps that the organ went through during the evolutionary process.

The primitive Brain corresponds to the manifestation of the reptile. Not that it is similar to the brain of reptiles that we know but corresponds to an evolutionary period where the living being reproduced and transmitted its experience to its descendents, only through a genetic code. The alligator is already born as an alligator; its brain circuits are born ready, not needing any more learning than the environment provides it.

The Intermediary Brain appears in mammals, resulting from the need of knowledge that will still be transmitted by the parents.

The Superior Brain appears almost suddenly in man. The neo-cortex is a structure with which the Pulse of Life has endued a living being, which was the proto-man and proto-simian, in a crucial moment for its survival. When the forests disappeared, this totally unprepared living being came down from the trees to live on the African savannah.

The Primitive Brain

In a simplified way, the human brain is arranged in three layers that represent three evolutionary phases as if they were three superposed computers. These three layers are not totally sealed and were not formed independently, but they are clearly defined.

The Primitive or Reptilian Brain is formed by the brain stem, medulla oblongata, pons and mesencephalon, globus pallidus and olfactory bulb.

This brain is capable of maintaining life in a vegetative state. It is capable of adapting to the cold, heat, hunger and thirst. It is capable of commanding the greater or lesser intensity of multiplication of the parts and the whole, able to correct countless deviations in the process of forming the being. It is capable of identifying danger and commanding the response. It is the greater expression of the initial pulse of life that one day began in the evolutionary chain.

Pulse of Life

This is a term used by Freud as a “pulse” of life in a psychical manifestation. I’ll give a broader connotation of this term. What I mean by Pulse of Life is life itself, which manifested itself at a certain moment on Earth and through transformations and adaptations formed all living beings: the force or pulse that expresses itself as Life.

Considering the Primitive Brain as an experience of the manifestation of life, it is in this part of the brain that the most important information in man’s evolutionary experience is found; from when he was a single cell to nowadays. In a way, this entire evolutionary epic, all of man’s experiences, are written in the primitive or archipallium brain’s synapses.

The primitive brain communicates with the body, integrated to the sympathetic and parasympathetic nervous systems. It communicates through the internal secretion glands. It is the greatest experience of the evolutionary processes that began with the Pulse of Life.

As a whole, the primitive brain and the body are written, as if in a book, the steps taken by the Pulse of Life until it reaches its current state. In it secrets are kept, which little by little are revealed. In the sympathetic connections of the solar plexus we find neurons with the capacity of receiving, analyzing and responding to stimuli from the digestive tube, which are practically capable of thought. This type of formation tells us that during a

certain evolutionary moment, Life had an option to develop the brain, an option that was abandoned but if it had continued, would have developed a thinking being very different from man.

This is the most powerful of the three computers: it basically holds the information of the entire evolutionary process and is capable of maintaining and perpetuating life. The primitive brain is the Pulse of Life, which during a certain evolutionary process opted for a consciousness experience of all the information written in its synapses.

Intermediary or Emotional Brain

The limbic system appears with the mammals. This brain is situated immediately above and covering the archipallium. In it we find the limbic structure, which has as its main function coordinate the emotions. A lot has been studied and discussed as to the location and the paths that emotions travel, ever since the establishment of the Papez circuit up to the new areas attributed to it by MacLean.

It has been observed that there is no precise delimitation of the area where the emotions are. One can, however, certainly affirm that the area that intermediates the emotions is located between the primitive brain and the neo-cortex.

If we make a comparison with other brains, we can observe that the limbic system behaves very differently from the other two as if one computer that operates with a different system such as analogical, while the others would operate digitally.

Superior or Rational Brain

The Neo-cortex appears in man. This brain covers the encephalon. It is composed by two cerebral hemispheres and a few sets of sub-cortical neurons; formed by a type of cortical tissue of recent formation, called the neo-cortex. Above all, it is the neo-cortex that operates as the headquarters of our consciousness.

By observing the evolutionary scale and the evolution of the human embryo, we verify that it was the last to be formed.

This brain is the great wonder that neurosciences face.

Man, through the neo-cortex, becomes aware of himself and takes on a dominant role in the planet's evolutionary scale. Man is now the first being of the creation.

Through the neo-cortex, man has projected himself into the world of science seeking knowledge. The knowledge generated through this structure, in a way, became independent of its origin, the Pulse of Life.

Science

Clearly, from the Renaissance and rapidly over the last centuries, a Science that is totally independent from the manifestation of Life has appeared. Methodologies were created to solve problems that were generated by this structure (neo-cortex).

A total separation between Science and the Pulse of Life occurred.

Life

Life, in a type of adaptation experience, created an awareness structure, which is taking on the evolutionary command along a path that has separated from its origin and has distanced itself from its original purpose.

A perfect man could draw a perfect pentagon with a single line, because this information, deriving from multiple experiences adaptation to life; the same way that a flower can form a geometric figure. Science has created a mathematical, geometric and technological methodology, in order to enable an imperfect man to draw a perfect pentagon.

However, a perfect man wouldn't be able to pass his SAT's!

The dichotomy between Science and Religion, which has been emphasized over the last 500 years derives from this; fruit of the superiority taken on by the neo-cortex.

I Am

Man has identified with this new organ that gave him the awareness of himself as being the "I am", the "self" – when in reality man is the Pulse of Life that lives within him.

Consciousness was nothing more than an evolutionary experience, which might have been successful, in case man uses this consciousness for the perpetuation of the Pulse of Life. It might have been a disastrous experience if man definitely destroys the possibility of Life on the planet, as he is doing.

Virus

If, at a certain instant, the Pulse of Life realizes that it has been a disastrous experience, it will certainly destroy the human race. We can see that this is no fantasy: we have faced, on a daily basis, the mutations of viruses appearing faster than man is capable of finding or creating defenses.

Computer

The Neo-cortex is the main star in the evolutionary process of the Pulse of Life, having an immense potential, the same way that our modern computers have resources that we are unable to use in their totality. The neo-cortex is very similar to the latest type of computer.

As MacLean compared, the three brains are like three integrated computers in a functional unit. Each one of them has a different mode of behavior regarding the information and the response. It seems like the middle computer (limbic system) works differently, resembling an analogical computer.

The Superior and Primitive Brain work through a binary system, where the countless synapses, through which the neurons communicate, act as if they were chips in our modern computers. The Intermediary brain acts differently; it creates "neuronal assemblies", which can serve as a great diversity of situations. For example, the escape and aggression commands can be used in different types of situations.

A Different Approach

Up to here, I have tried to present a different approach to our brain. I want to demystify the brain and show it simply as an organ; and experience of the pulse of life to perpetuate and reproduce itself, because it is ephemeral.

The question of how such an organ was formed remains. What would be the motivation behind this life experience?

This is a palpitating subject, which holds innumerable discussions. The sequence serves to demonstrate didactically what I call the Emotional Catapult.

Catapult

The neo-cortex was created to give the living being the possibility to use all of the evolutionary experience that is written in the primitive brain, which was structured through the intermediary or emotional brain. The emotional brain served as a catapult to pass to the neo-cortex the possibility of usage of the evolutionary experience written in the primitive brain.

The hypothesis that I present here can and must be contested, because it certainly requires further studies.

Explanation

For didactical purposes, I call the being that precedes *homo sapiens* as *proto-man*. When the proto-man left the top of the trees and stood erect, he created a special circumstance for the adaptation of life.

The fact that he stood gave him the possibility to foresee the problems and situations that would present themselves next, before they happened. Life needed some adaptation to be able to use all of his evolutionary experience, stored in the archipallium (primitive brain), so that the proto-man could make decisions. Thus, the necessity that life has the “knowledge of this experience” was created.

Consciousness

The confronting of the brain's two hemispheres, which certainly appeared as a defense mechanism, it gave the proto-man a dichotomy in the impressions of himself and the possibility of seeing himself in two ways. These two ways of seeing himself, in confrontation with the external world, seem to have been one of the important causes of the evolutionary moment where the proto-man identified himself as “I am”.

When he was still in the trees, the proto-man also had the possibility of foreseeing the unfolding of certain events on the ground. From the top of the trees these events had no vital importance.

From the moment he was on the ground, participating in the threat, foreseeing became vitally important, creating in man a new type of need, which would be the vital involvement with a situation about to happen. These factors created a unique condition in the evolutionary moment, which was solved by transporting all the evolutionary knowledge from the archipallium to a new structure, a new organ, the neo-cortex.

New Organ

This new organ would have the possibility, but not yet the capacity, of being aware of all the accumulated information in the primitive brain for millennia, originated from the evolutionary process. This transposition was not of information, but of the possibility of making it conscious so that, during a moment of aggression, the proto-man could basically decide between fleeing and aggression.

The neo-cortex would have been formed, a set of new connections of a different type, with the potential of expressing all the knowledge and consciously selecting it when making decisions.

Sudden

This transposition happened very quickly, if we follow the evolutionary sequence. It seems to have been a moment of genetic mutation, expressing itself very discretely in the chromosomes, because the chromosomes of a superior mammal are not very different than man's for such a huge difference between both of them.

The transposition happened more or less suddenly, as a catapult that projected this knowledge in a new organ with the possibility for consciousness. The catapult seems to have been exactly the intermediary brain, the limbic or emotional system that interposes itself between the other two brains.

Emotional Catapult

The need for conscious knowledge in the proto-man appeared along with his emotional involvement. He was emotionally involved with the foresight of the problem. He had to be prepared, not only by the awareness of the situation, but also with his entire body involved: prepared for the possibility of aggression or escape.

Therefore, I call the sudden transposition of information from the primitive brain to the creation of new connections with the possibility of blossoming of this information in the consciousness intermediated by the limbic system, the Emotional Catapult.

The intermediate or limbic brain, operating as an analogical computer, allowed a new structure to be formed over it, composed of countless connections with different functions, with the potential of transforming all the information accumulated by the Pulse of Life into conscious information.

The connections were created to be used in moments when life was at risk. All human knowledge, fruit of the experience of the evolution of life, from a single cell to man, was catapulted by the intermediary brain (emotions), creating a new organ that allowed the awareness of this knowledge.

The Dictatorship of the Neo-cortex

The neo-cortex is nothing more than a Pulse of Life experience. Man's consciousness is an adaptation for the survival of Life – ephemeral by nature. The neo-cortex is not a wonder: LIFE is wonderful.

The (I am) man turned his back on his origins, feeling that he was the center of the wonder. He forgot that he dies! He developed his own "science", separated from his origin and his purpose. Absurdly, he prolongs his own life, many times artificially, causing suffering to the living being. He systematically destroys the environment that sustains Life because he feels superior to his environment. He has developed the use of his new

connections in order to create greater comfort, without creating greater happiness for the living being.

As long as man remains disassociated from his origin and his purpose, he will not find happiness. Religion is always saying this in many different ways. Man has separated science from religion as if it were possible.

Part 1

8

New World

At the beginning of this millennium, a new world is being designed, which began to dawn in the last years of the last century, fruit of an exponential growth of human knowledge.

This exponential growth is changing the world, which will take on characteristics even harder to predict. We are realizing that the world will change, but we haven't realized that we are at the edge of a transformation beyond the paradigms that we are used to. Any prediction is temerarious. We can see that it will happen in terms of values and not in terms of facts.

The exponential growth of human knowledge occurred towards the end of the 20th century through the Information, Communication and Knowledge revolutions, and we entered the 21st century with the revolution of Creativity.

Through the revolution of information, man began to have access to information and the possibility of crossing data in a way that transcends his intellectual capacity. The communications revolution disseminated the information. The revolution of communication happened through the manipulation of information. The moment that a piece of knowledge is expressed, it also becomes information: information and knowledge have become commodities and man enters the 21st century challenged by creativity.

It was the revolution of communications that truly globalized the world. Globalization and the dissemination of information and knowledge perhaps represent the most important revolution that the world has suffered since the beginning of organized society.

The world became united and globalized in information and knowledge, no longer distinguishing anyone because of these factors. Neither information nor knowledge is competitive differentials at the beginning of this millennium.

The moment when I create something new and put it into practice, from this moment on, everyone has access to my idea and my knowledge, and I will always have to create new things to continue having a competitive advantage.

The competitive differential became creativity.

These transformations have occurred in an exponential rhythm, astonishing the world in the face of the unpredictability of any situation.

If, on one hand, these revolutions have given man a potential never before imagined, to generate goods and wealth, on the other hand, they have increased the chasm between the part of the world that obtained competitive advantages from these revolutions and the part of the world that remains without access to these revolutions.

Consumer goods were created, with which man wasn't used to dealing with, in other words, the goods deriving from knowledge and creativity.

The world has become much richer, but much more unfair.

Dot Com

Towards the end of the last decade, the world observed a euphoric moment of increase of prestige of knowledge and creativity goods, especially in the stock market.

Since there was no prior idea of how a society based on knowledge would develop, there was a super-valorization of these goods, generating an extraordinary distortion between the new and old market values, which still predominate in the world.

The world has been experiencing, at the beginning of this 21st century, such transformations, with changes in paradigms, which have led us from a world of facts to a world of values.

The companies that manipulated these new values began to have a market value much superior than traditional companies. GM was worth less than Microsoft! The “dot com” companies created an expectation of generation of goods that was soon put in its place. Man became removed from the expectations of this society of knowledge; he was not able to keep up with the market projection and towards the end of the last century, the “dot com” structure crumbled.

Capital

Since then, the world has taken on a better-balanced position in this transition of market capital to knowledge capital.

The speed with which everything has changed and the great accumulation of goods by the groups and countries that immersed themselves in these revolutions have determined important alterations in the way that governments face this new globalized world.

The speed of the transformation and creativity has generated serious distortions in the management of groups and governments.

The United States

Lately, we have observed in the world, a situation of domination, not only militarily but also culturally, never seen before in history. Since the crumbling of socialism in the Soviet Union, the United States became the superpower that has the military control of the world.

The destruction of socialism in the Soviet Union was the fruit of the new capitalism propelled by globalization, catapulted by the revolutions of information, knowledge and creativity.

The North American power is only threatened by the European Union and by China. There is no knowledge, in the history of a period, where one nation had so much power over the world. Neither Cesar nor Alexander had so much power. The rest of the world became, in a way, isolated within this new organization of power. This makes these abandoned countries seek for a way to maintain their cultural identity, which has been done through an exacerbation of their values.

We observe that the world as a whole has advanced economically but such an advance is not being followed by the respect to man's emotional values. The countries that are not integrated in these revolutions became oppressed, without any opportunity to participate in the new wealth that is being created. So as not to disappear, these countries, in a clearly emotional manner, exacerbated their cultural values, their ancestral

traditions, their dance, their music and especially, their religion, in a desperate attempt to remain above the surface.

Shock

The superpowers on the other hand, inebriated by the exponential generation of goods, tend to forget their values. Not only of the excluded populations but also their own cultural and human values. They believe that by imposing what they call democracy, they will be strengthening their values.

Values cannot be imposed; they are not generated through reason or logic. Values are the fruit of the emotional brain, while facts are created by man's superior brain or neo-cortex.

"The heart has reasons that reason does not understand."

The world is experiencing a shock between facts and values. The facts are represented by the economic power, by the possibility of satisfaction of material goods and by money. The values, through the ancestral cultural of each people, their religions, through their emotional values.

The shock between Facts and Values was tragically marked by the events of September 11 in New York, and the superpowers continue disconnected from the real situation.

Bladder Probe

I have a friend who is an urologist, atheist and a joker, who always asks a patient with urinary retention if true happiness consists in having a probe passed through the bladder. They all confirm his theory. It seems that the rich countries conducted a survey similar to my friend's. They want to pass a probe through the bladder to make the world happier!

The United States insist on combating militarily, the desperate yearning for cultural, religious and even economic recognition of the Arabic people, by trying to impose a democratic regime, which is good for them, the United States.

The same situation can be observed in Israel, who benefited from all these revolutions of capitalism, assumed an economic position that will inevitably destroy the culture of the Palestine people. They, in turn, emotionally exacerbate their cultural and religious values in a desperate attempt to simply survive.

The relationship between the United States and Mexico are also heading towards an impasse between the fact of North American wealth and Mexico's cultural values. The revolutions of capitalism tore down the Berlin wall, but they are building a new wall to separate the United States from Mexico. This wall is already a reality along most of the border.

Somehow, the rich countries will have to find a solution for such an impasse, an extremely difficult solution, because there is no path that leads the world of Facts to the world of Values. It will require all the ability of creativity developed over the last years so that a solution can be found.

If we consider that man reacts much more emotionally than with reason, it is probable that this impasse will be solved in the area of emotions. Even rich countries have found difficulty in adapting their own people to these revolutions that occur exponentially.

State

Capitalism took over the State. When Adam Smith exposed his theory, he never imagined that the State would be conducted by capitalism in such a way. The values of the State cannot be managed by merchant facts. Health, education, well being and culture cannot be considered as capital goods.

The State has gradually abandoned its position as manager of the people's values. Over the last few years, we have observed the bankruptcy of large North American companies, which basically benefited from the extraordinary revolution to manipulate their data and cheat their stockholders. The most extraordinary thing is that these companies were involved with senators, congressmen and public officials.

Capitalism defeated socialism but is defeating itself.

The incapacity of the democratic State is clearly outlined in this new world, despite having an average representation of its population, of managing the nation's values.

It is possible that this is a transitory moment, but it is extremely difficult to find a solution, since it is about a shock between the worlds of Facts with the world of Values.

The States incapacity of managing its values is becoming obvious. They are transferring this obligation and this onus to the companies. So, the companies became horizontal, no longer having an owner, a figure pulverized among its shareholders, for example, here in Brazil, almost all of our telephone technology belongs to the retired elderly in Spain. And these companies have become very similar to States.

Companies

The companies however, are experiencing difficult times derived from the speed that things are changing, the instability of world market facts. When the first companies were started, they were probably hunter associations in a same valley, or of the same region, that had a certain surplus but on the other hand, needed other goods.

These first companies would have the same characteristics if they lived in the same area, therefore having similar scarcities and abundance. This group that lived in a certain area, with a certain need, became what can be called as the first nations.

The nations were originated from the communion of several companies or even of several families with similar material needs. Similar needs made them organize themselves into groups that cultivated certain values. The function of these nations was to especially manage the group's cultural and emotional values, making them more cohesive, stronger and more apt for survival.

The nations probably managed the Values while the companies managed Facts. The companies' Facts were probably connected to the Values of the nation. The companies might have had a very mercantile concern, and the nation a cultural concern.

Lately, and especially after the capitalism revolutions, the nations are taking over the mercantile management of the people. They are more concerned with managing profits than with quality of life, the values and culture of their people.

Impersonal

The great competition and the small profits are making the companies resort to manipulation of values to obtain profit more and more.

When a large company used to be managed by one man, the company could occasionally act in defense of values in synch with its nation. A man like Rockefeller might have emotional attitudes in synch with the North American nation. Nowadays, the company managed by impersonal stockholders is only focused on profits, no longer acting as humans.

The ecological movements that intend to limit the companies are insufficient and incapable of imposing their position. A demonstration of this is in the fact that the United States hasn't signed the Kyoto Protocol.

An outsourced State is being outlined, where the companies and especially the large companies manage values. The management of values by the large companies will not be restricted to this or that country, but to the group of people involved with it.

Geography

All of this tends to redesign the world's political geography; the globalized world would have a very different political geography from the one we know in our maps. A nation will no longer fight another nation to defend its values. The companies will fight amongst themselves to create profit through the manipulation of the values of degraded nations.

This is the Great War fought in the world since the beginning of the millennium!

The rich nations cannot forget that they need consumers for their production and must therefore, stimulate the buying power of those who have not integrated into these revolutions of capitalism. They must be aware of their values and not only pass a probe in the bladder of these nations believing that it will bring them happiness.

The Emotional Factor in the Genesis of Socialism and Capitalism

The States, in their origins, differentiated themselves by their principles, which in a final analysis were an expression of their members.

The people that inhabited the forests (Saxons, Anglo-Scandinavian) became less religious, because early on, they learned that survival depended especially on their efforts. Nature presented harsh cycles, allowing them to program themselves for the winter.

The people who inhabited the desert (Arabs and Hebrews) based their social structures on religion. Survival depended on imponderable factors such as rain. Survival depended on something more imponderable than their effort. They were States with religious princes.

We can also say that the princes who had ethical concerns generated socialistic States and princes of religious States generated capitalism.

Religion and Ethical Flexibility

The States ruled with a predominance of ethical norms were run by princes or by revolutions with ethical concerns, and generated the socialistic States.

The States with a predominance of religion, dominated by princes who not always had ethical concerns, allowed the flexibility of ethics, which generated capitalism.

These standards of States haven't always revealed themselves in a pure manner because the Church, exercising a domain over the western world, imposed its religious principles and the flexibility of many ethical principles and in general, allowed the appearance of capitalism.

The States that developed with an ethical predominance and generated socialism were especially those beyond the borders of the Roman Empire that the Church took over.

These characteristics of man got mixed with the States ever since the Roman Empire was invaded by the barbarians and later dominated by the Church. The Church clearly imported this model of man who believes in religious values from the inhospitable areas.

Equality

At the end of the 18th century, these States of ethical and religious domination clashed with each other in a brutal manner during the French Revolution, where the principle of equality among men was born.

From this mixture of man's ways with the States' ways, a dichotomy in the management of State appeared in the last century: the so-called democratic State and the so-called socialistic State.

Both had the primary concern of promoting equality among men. The democratic State, as I've mentioned before, more flexible in its ethics, developed the capitalistic regime. The socialistic State, more committed to ethical principles, generated socialism. Both management ways were basically concerned with equality among men.

Inequality

This equality however, is a utopian State because men are not equal.

In the 1920's here in Brazil, Rui Barbosa lost an election to Hermes da Fonseca, especially because of his famous phrase, "Inequality means treating the unequal with equality". Rui Barbosa was a man ahead of his time and therefore, was not understood.

The socialistic regime clearly demonstrated that it did not promote equality among men. The State was created on one side, and the people with their union on the other. This division of socialism, making it clear that the State was not the people, caused the initial conflict of its defeat in Poland through the Solidarity union.

The End of Communism

The Catholic Church, in a crafty manner, enabled this possibility. When the Church elected Cardinal Wojtyla as the pope, the world did not understand the extension of this action, which represented, as in a chess match, a checkmate on Communism.

The people of Poland, who are very religious, found support in the Church when it made the dichotomy evident between the Solidarity Union and the State, thus the rupture.

As you can see, even nowadays, the emotional power associated to religious power is still one of the great forces that rule the world. The democratic regimes, with their capitalistic principles, did nothing more than accentuate the inequality among men, creating enormous abysses between the rich and the poor and generating a brutal inequality in the distribution of wealth.

From the shock between capitalism and socialism, the certainty was born that the principles of equality proposed by the French Revolution were not feasible. In the first place, because Freedom implies in responsibility, which not all men take on with Equality; and in the second place, because equality is only possible with the emotional component of Fraternity.

While man does not reach the emotional development that would lead him to Fraternity and to the ethical development that would make him take the responsibility of Freedom, Equality among men has to be replaced by a principle called Equity.

The principles of the French Revolution are completely utopian, if we don't take into account man's emotional development. This emotional development was strangled by religion for many years.

The Raft of the Medusa

In 1850, a ship named Medusa shipwrecked in Europe. When sixteen survivors were found days later on a raft, the fact caused a commotion throughout Europe and one of Delacroix's disciples painted the famous "*The raft of the Medusa*".

In the painting we find the survivors expressing all types of emotions, from fear to anger, from hope to despair. In it, even if it isn't very clear, there is the suggestion that the castaways, in order to survive, ate each other.

I chose this painting because it represents man and the world today; as if all of us, with all of our emotions, were adrift on a raft. It is with all these emotions, himself being the raft.

Salvation

The new world will have to find a new rule, a new regime, with a new model of State to manage men within the principles of Equity.

Men are unequal in their productive capacity, abilities and relationships. In this new world, it will be up to us to find a path that leads to true equality among men.

This path begins at the beginning of this millennium as a path that passes through man and society's emotional structures.

We are all in the same raft of the Medusa, adrift in an unknown ocean. The only thing that unites us at this moment is hope.

Those who are afraid, those who hate, those who have dreams, those who have strength, the strong and the weak, are all united in this raft, in the hope of finding salvation. Salvation could come from the ethical union of everyone, if everyone rowed in harmony, which is not possible because they are unequal. Salvation might come, or a belief that it might come, if everyone prayed together.

We now know that salvation will come because we know the tides, we know the ocean, we know the routes of savior ships; we have the communications that was able to pinpoint where the shipwreck occurred. We are sure that salvation will come. All that is left to know is how we will be in this raft the moment we get saved. How many of us will there be? Will be able to live with the trauma (anthropophagi, for example) that occurred in the raft while we were adrift?

What unites us at this moment is an emotional force, which can be translated in the hope of knowing how to manage equity until we are able to live in equality. This emotional force of man is totally out of control.

Man feels the need to find a path, and believes that this path is the defense of the environment we live in. Ecologists organize themselves and raise the flag of hope, but they also find themselves in a complete state of disorientation. We observe this when we see an enormous amount of people and a huge quantity of resources spent in the movements to save the whales, to prevent the killing of seals, totally forgetting about the men who die of hunger in Africa.

Infinite Mathematics

Man nowadays has a degree of information available to him, which has never been seen before. In order to make a decision, man has an immense amount of data and simulations that can repeat themselves to the infinite.

With the possibilities that the computer gave us to calculate experimenting actions close to infinite, we have entered a new world where we have precise information but we don't have the logical reasoning to analyze it.

If we repeat infinitely the spinning of a roulette, we will find a pattern that will repeat itself with a certain rhythm. Within this pattern, a storm that happened over a hundred years ago, where the tree used to make the roulette used to be, is expressed, because this storm caused a deformation in the tree's growth. We can find the pattern of repetition in the spinning of the roulette and the storm will be "written" within it. But how can we recognize it within the pattern? This will be the question of the new times.

By repeating an infinite experience, we will have a pattern that records an infinitude of facts that we did not realize that they were related. If we infinitely repeat the dripping of a faucet onto a plaque, we will have all the curves of the pipe recorded.

Man's thinking is used to cause an effect in a logical and rational sequence. With the advent of infinite mathematics, we approach chaos. Chaos would have an order involving an infinite quantity of parameters that we would be unable to identify.

Through this thought process, we realize that everything is inter-connected. EVERYTHING IS LINKED TO EVERYTHING. If this is so in the field of materiality, how could it not be with LIFE?

Meteorology

By applying this mathematics to the weather, it was observed that the entire Earth's weather is interconnected. The 'El Niño', responsible for important weather alterations in the southern hemisphere, was identified for the first time. This interconnection of the weather is dramatized in the story about the butterfly.

The butterfly that flaps its wings in China can have been the 'dx' of an established sequence that determines a hurricane in Florida three days later. The information that the butterfly flapped its wings seems to not have any significance, and in fact, does not. What is important is the sequence of phenomena that were as if interlinked so that the initial movement would trigger the sequence.

It's like a series of falling dominos: The first one to fall has no importance; what is important is the disposition of all the dominos, or the intent of the person who displayed them in those positions.

The dominoes could not be placed that way at random; someone or something put them there with a certain intent. Such intent could be conscious or subconscious but, in some way, there was an intent in the action of displaying the dominoes.

Going back to the butterfly example, flapping its wings is not important, but the sequential disposition of the other phenomena. Removing God from the sequence, there were several atmospheric phenomena, such as the monsoons in India, a sirocco in the Sahara, a snowstorm in the Alps, a large storm in the North Atlantic, or a great heat wave on the western coast of North America.

This is the sequence of dominos that were positioned to leverage the initial movement. This sequence cannot be foreseen only through linear reasoning. The factors would be countless coming close to what is called a chaotic system.

Ecologist

The butterfly that flapped its wings in China and caused a hurricane in Florida is linked to a chain of phenomena such as dominos, which are impossible to predict rationally. However, the distribution of these phenomena (dominos) can be related to irresponsible action of man, who has not respected the atmosphere with his emission of gases, has polluted the seas, deforested the Alps and warmed California.

The placement of the domino pieces, as we can see, can be representing a common action, whose unit has unconsciously determined such a sequence. All the information in the world wouldn't be capable to recompose the sequence that preceded the hurricane in Florida, but a dreaming ecologist, even a sort of crazy one, could point out the sequence of events.

Wisdom and Emotions

Such knowledge that approaches what we call Wisdom the most, comes from brain connections that do not follow the linearity of our common reasoning. It comes from a thought or from an assembly of neurons close to what we call Emotion. Through this type of access to a superior knowledge could Beethoven compose, albeit being deaf, the Ninth Symphony; could Michelangelo paint on the ceiling of the Sistine Chapel. This will be the new path to be trailed in the new millennium.

Marketing

Man will change in a way that has never been seen before. He is communicating with an infinite number of other people, with all the information he desires being available to him. The true power will be held by those who know how to dominate the usage of the information, those who, in some way, can pay attention to this or that data.

The determining factor of true power will be the valuing of information, the mode of divulging it and especially, the association of the information with some emotional charge.

Due to the unpredictability, globalization and small profit margins, there must be a radical change so that the business world can adapt to the new millennium.

This change will be related to the radical change that mankind is going through. Man has never experienced a period with such transformations. It is as if we were once again in the industrial revolution, or as if the time when man abandoned his nomadic life to live in fixed communities like cities.

The New Manager

We might have all the available information nowadays, but the real capital is in the way of using and dealing with it. In the near future, there will be individuals who are more or less equipped for this purpose: they will be managers with qualities that have very little to do with their knowledge of business management, but with his psychological configuration and his situation in the world.

Such individuals will be distinguished by the emotional Self, not in the sense commonly given to the term, but in the sense that the emotions give access to the greater world of knowledge management. This new manager does not have the same profile as the current manager, because he is involved in ecological processes and has a great concern with life as a whole.

Such a manager might seem hypothetical, but it will be happening soon. This manager does not have concerns of becoming wealthy as the yuppies of the 80's did. He

wants to live well. The way to entice him will not be through high salaries, but with the opportunity of building the Work!

Optical Entropy

Everyone can see a wall full of Van Gogh paintings. They will become lost in the analysis of the display of the paintings, in the form or in height of the wall and will not see any paintings if the wall is completely filled with paintings.

If a single painting is displayed, the observer can see Van Gogh's art and at some level understand all of his paintings. The amount of information is, in a way, harmful to knowledge.

How do we make an observer, standing before a wall full of paintings, opt to observe deeply a single painting? Those who hold this formula will have the power to lead the observer to a very special world with the mind of a Van Gogh.

The true power will be with those who are capable of manipulating the input of information in the mind of the observer. Of course that the observer won't do anything that isn't already his latent desire, but in most cases, he doesn't know what his own needs are.

Information has lost its value. Everyone can have access to the information of the sequence. Only a few people, through non-linear with emotional components, will be able to perceive the sequence of the paintings on the wall, and understand that the butterfly flapping its wings in China has triggered the hurricane in Florida.

Emotional Reasoning

This type of emotional reasoning, still unknown to man, has been mixed to religion and art for a long time. It is not accepted in our society, which still uses linear reasoning, always appearing to be biased, related with man's psychological and emotional structure, always contaminated with the innermost prejudice and with the innermost personality unbalances. Thus, it has no value.

We must find a way to use this work tool and, in some way, be able to clean it from these impurities, a way to purge them from the imperfections of the emotional thinker's personality. This is the great challenge of our time. We cannot underestimate this tools at all, especially when it becomes the only feasible way to imagine the sequence of the domino pieces or the planet's meteorological phenomena.

Part 1

9

New Man

We are experiencing a moment of serious reflection, a moment where man has reached his plenitude of manifestation on this planet, a moment that needs to be rethought.

At this beginning of the 21st century, man has been going through a huge transformation. Not long ago, man was struggling to survive. He gradually occupied the spaces that became habitable by using his thinking to adapt to the adverse conditions of certain areas. At this moment however, the scientific progress that man reached, allowed him to survive in all the areas of the planet and even, out of the planet.

Up to the last century, man had fought against the external environment in order to survive. Since the beginning of this millennium, he begins to have to face a fight against himself. Man has dominated his survival conditions in such a way that he is escaping from the rules that Life (life pulse) impose on living beings. He managed to survive beyond what life permits.

Survival

Life has made man to be born, grow, multiply and die; it equipped man with the means to reproduce and multiply the species. Science, in a definitive manner, has detached the need of reproduction for the multiplication of the species, where reproduction is no longer propelled by Life, as it was since the most remote antiquity, Life commanding the reproduction of the species as it does with all living beings.

In this beginning of the 21st century, an evolutionary moment where man has practically defeated the external world, he has definitively stopped fighting against the exterior environment to survive and perpetuate the species and started facing a fight against himself. This has happened because science has allowed such an alteration of the external environment, that the man became an important stabilizing factory in this environment. In this fight there will be no winner. Either all mankind comes together, understanding that we are a unit and must behave as such, or we will create an unbalance in the external environment that the planet's self-regulatory system will not be able to balance.

Life Correction

Another hypothesis which must be suggested is that Life (life pulse) can correct this harmony. Life has the experience in dealing with such a situation, when the living beings created by it become a threat to its need for perpetuation and survival.

These needs derive from the protein molecule, upon which life was structured. Because the protein molecules are ephemeral and unstable, Life itself is ephemeral and unstable.

If we consider that Life manifests itself by creating living organisms, tending to occupy all the space allowed by its metabolism, we will reach the conclusion that the pulse of life behaves in an intelligent manner.

With the episode of the disappearance of the great reptiles, we were able to observe how the pulse of life arranged for its manifestation, through the living beings, could create the mammals.

The mammals are capable of transmitting to their descendents a knowledge that was not transmitted to the reptiles, singly through genetics. The pulse of life certainly behaves intelligently in the determination of the forms with which the living beings manifest themselves in each evolutionary moment.

Even considering that the evolutionary process has occurred in the form of an Evolutionary Drift, the intelligence of the pulse of life cannot be denied.

Escape from Responsibility

With the advance of Astronautics at the end of the last century, we began to consider the possibility of man living outside of planet Earth. This possibility would be an incredibly clever way of a minority, holding the knowledge and science, to continue with the struggle that man has always fought against the external environment, abandoning the rest of humanity in an arid terrain improper for survival, escaping the destiny that the evolutionary process has placed: that man could understand himself as humanity.

The possibility of man living on Mars is a way of not facing the struggle that has been initiated in this new millennium, the struggle against himself. Man must abandon his selfishness and his egocentrism which makes him think: "I think, therefore I am", to say: "I live, therefore I think".

The millennium has begun with clear indications that man will escape this confrontation with himself. The possibility is that those who hold more knowledge and more science, continue to exploit the natural resources until their depletion, and then moving to other areas, leaving the living beings of that area doomed to disappearing.

Current Moment

This moment when a single power dominates the world is very symptomatic, and this single power is exactly the one that wastes millions of resources in the attempt to establish life out of this planet. The same power that refuses to sign environmental conservation treaties.

We are living a crucial moment in the evolution of the human species. As I've mentioned before, it is a serious moment, where man must rethink himself. He must initiate a fight against himself.

Religiosity

This expresses itself especially in the religiosity that gives man a broader sense than his individuality. It is manifested in the terrorist actions that are done because of religion, causing so many innocent victims. It is a moment when man has to stop fighting against the external environment and begins a new fight with himself.

As strange as it might seem, this is the same fight that exists within each man ever since he acquired a consciousness, and that is called Religion.

Man and Society

Just like the whole world is changing at the beginning of this century, man too is changing. Changing within himself, in the way he is and also in his relationship with this new world.

Man's relationship with the world occurs through the family, the society around him and his work.

The family has suffered a great transformation since the end of last century. The large families have disappeared, there are no longer families with many children. Nowadays, the families have one or two children, and the number of families who don't have or do not intend to have children cannot be ignored.

In this relationship between man and the family, it must also be noted an important transformation during this period, which results from a great number, of both men and women, especially women, that make the option of not having a family.

Woman

The repositioning of the woman in society became an important factor in the restructuring of the family. Since the pre historic ages, the woman had adapted herself to an inferior position inside the family structure, because she lacked means for her child's and her own nourishment. Since the discovery of the antriconceptive methods, specially the pill and its diffusion in the 60s, the woman could free herself from the obligation of pregnancy in the sexual relation and was able to position herself in an equal situation with the man in the social means, family and work. Such change of behavior of the woman was the most important transformation factor that occurred at the end of this century.

The female adaptation had been done in such an important way and during such a long time, that the woman developed a diverse way to be and to think.

The woman has a special way of thinking, using in a different way from the man of the left hemisphere of the brain. This, today, demonstrates itself when observing the brain activity of the woman and the man in the magnetic resonance. When the woman talks she uses, seemingly, both brain hemispheres; while the man uses predominantly one of the hemispheres.

The social repositioning of the woman brought a radical change in the family structure. The woman's aspirations to an equal social position made her vindicate a professional realization and caused her mass entrée in the labor market.

Provision

Before, in most of the families, only the man worked, and the woman dedicated herself to taking care of the house and children. This type of family became totally impracticable nowadays, in face of the material necessities that force both man and woman to work in order to survive. The entry of the woman to the labor market, restricted just to the man determined, at the beginning, a less gain for the job, as if the working labor had doubled in the market.

Due to the characteristics of the way of thinking and being of the woman, she became an inexpensive labor and with very special characteristics in certain situations – a differentiated important factor in the labor market.

The positioning of the woman inside the family again made her less dependent. The man, not being prepared to this new situation, started having a great number of marital misunderstandings, with a consequent bigger number of separations.

Modern Man

The families suffered significant modifications, with a big number of separations. All this made us have today a new man (man and woman). This New Man is totally different from the man of the last century.

The relationship between man and society also goes through a transformation never seen before. The man is lonelier than ever.

The means of communication and the consumption facilities have made man each time more trapped inside his computer – the new man is becoming increasingly more virtual!

Job

The concept that we had of the job during thirty years is gone. The existing jobs are in the industries that use physical work and, even before, are rapidly being substituted for the mechanization of the industry and by the informatics and electronic technology in the jobs.

In Brazil, the jobs offered by the government are still a glimmer of the old concept of the job. The government is also changing, outsourcing more and more its services and establishing temporary contracts for the execution of certain tasks. The delay in the changing is because of the ancient culture, impregnated in the Brazilian society, which relates security with public service.

Today one is contracted to do a determinate task and, when this is finished, the job ceases to have the necessity for the company. To continue in the same company, the employee needs to qualify to some new type of job, normally generated by his previous work.

In case one is not able to qualify oneself for a new job, this person will have to look for a new company that needs the work that he is apt for. This change of job will soon

wear out the working market for his kind of job, because all the companies have, on the other hand, been constantly changing.

Workability

Such a situation created a new type of concept: workability. The new man doesn't have a job any more. He has his workability. The security of his old job was substituted by the workability. His security originates of the necessity of the working market for his line of work.

The new concept determined a constant necessity of improvement, which tends to be the principal characteristic of the new man. The constant actualization inside a universe in constant change and instability is generating a man who will have to adapt to this through a radical change in his way of being, living and, mainly, of dreaming. Besides that, the continuous learning is the only way to have some form of work surviving security.

Transformations

Among the constant transformations of the world and confronting the unimaginable quantity of information in which they are exposed constantly, the new man has to know how to "abandon what he doesn't know" to be able to go ahead in his constant learning.

An example of this "abandonment of what he doesn't know" can be observed in the computer system programs. Even before knowing well the Windows 95, I had to learn the 98 and the XP.

Presented this way, it could seem that this is a simple adaptation. However, man has been raised during many centuries in a different way. He learns to add, so that then he can multiply and this way develops the understanding of mathematics. Today he has to resolve a square root without knowing how to add!

It is a more complex transformation than he can ever imagine, and it has to be accompanied of an alteration in his whole way of seeing the world.

With regards to this type of adaptation – abandon what he doesn't know – the woman is much more prepared than the man, exactly because she didn't have her mind structured as the man was during these many centuries of learning. If I suddenly change the purpose of my firm and start doing a diverse job, my women employees have a bigger capacity of adaptation than my men employees. The man wants to know well how Windows 98 functions to start using the XP, while the woman is more worried with the use of one or of the other.

Leisure

Twenty years ago, people imagined that the man from the beginning of century XXI would have his work load reduced and more time to dedicate to his leisure. There was the certainty that, with the revolution of the computer science and of the communication, the man would work about four hours a day, in his domestic environment, having much more time for his leisure. There was even an increase in investments in the leisure sector.

Today, with all the changes, the man works more than ten hours per day, and very little time is left for leisure. If, on one side, the changes facilitated the work, they also determined many dismissals. The banking sector was one that suffered more, with dismissals of employees caused by the mechanization.

For each job opening there came to be more candidates, and even more qualified. The integral dedication to the job started to be an indispensable condition.

Be a Team Player

In the past the big corporations, such as Bank of Brazil and Petrobrás, “would buy the soul” of their employees, moving forward to the demands and opening a social space for them. The dedication was not only full time, but the company was part of the employee’s family life. Today the employee has to really “deliver his soul” to the company, not being able to count on the compensations of before. “Live the company” and “deliver a soul” is nothing more than the indispensable condition to work.

To be able to stand this transformation, the new man had to adapt his way of being to the new situation; work had to be transformed in leisure.

To survive in the new world, man has to find in his job his leisure and, this way, be able to put up with the work load which would be unthinkable with the old concept of work. Such adaptation has to be done inside an emotional component: you can’t turn work into leisure if there is no *passion*. The emotional involvement of the new man is an indispensable condition to his work.

Structuralism

Saussure’s linguistic, transported to the anthropology, and came to form the bases of a thinking way which influenced all the thinkers of the second half of the twentieth century: the structuralism.

Who better expressed this chain of thought was Levi Strauss, who transposed to the anthropology the structuralism, delineated in the linguistic, studying the indigenous cultures of Brazil.

The cultures have a unity and a structure, which can be reduced and, somehow, it is possible to anticipate some of its reactions. Observing an Indian tribe in the interior of Brazil, they found Indians that were not capable of understanding the numeric concept. They could not admit that two rocks and two trees were the same number “two”. It was unthinkable for them that a tree and a rock were expressed in the same way; there’s no correlation between a rock and a tree.

This apparent ignorance was, however, inserted in his cultural context, very different from our European cultural. The numeric concept is, in some way, so rooted in our form (structure) of thinking, which makes it difficult for us to abstract from it. We could think of the Indian as an inferior being or that he processed an inferior culture.

This way of thinking or “logic” of thinking, in the past, generated a distorted concept of superiority of race and culture. Analyzing the structure of the Indians culture and ours, we can observe that our thinking is structured differently from his – neither better nor worse.

Numeric Concept

The moment our culture accepted the numeric concept, we made the option for a path, like a tree branch of the thought, abandoning other paths. Accepting the numeric concept, we opened a path to our technology and science, which is capable of launching a man to the Moon. However, at the moment of option, we limit our understanding; we can no longer understand what infinite is. The infinite does not have a place in the numeric concept.

Levels of understanding

The Indian, who didn’t understand how it was possible for two rocks to have some kind of identity with two trees, understood and experienced the infinite in his

understanding and in his concept of Everything and All.

The more we advance in the science and technology, new chains of thought appeared that put in doubt our option of culture.

We no longer despise that Indian or see him as an inferior or uncultured being. The Indian has a different structure of thinking. He climbed different branches of the thinking tree, as I expressed in the picture above. (Page 86)

The acceptance of the numeric concept corresponds to a dichotomy on the base of a culture. All our technology, resulting from the numerical option, has not given us true inner satisfaction and, in an objective way, has not made man happier.

For each option that we make in the construction of our culture, at the same time we abdicate of our options. Each choice implies in the giving up of numerous other paths. Each choice in the structure tree of thought makes all the problems that we begin to analyze after that dichotomization become "impregnated" by the anterior decision.

Limitation

When we choose one path, we become tendentious in our reasoning, not permitting that we have the freedom to analyze other aspects of the problem.

Within this form of seeing it, our culture is a limited factor of our thought. When I make the option because of a determined impression, I am limiting my understanding to this mode of impression, and abandoning numerous other options of interpretations. If I am seeing an image, I can imagine that it is a landscape, or that it is a painted picture on a canvas, or that there is or not a window pane between the landscape and the place where I find myself.

Art

Art can express this dichotomy of the thought, as in the surrealistic paintings of René Magritte.

Detach

I try here; again, show how the art can express concepts way beyond our logic and rational understanding. Art gives us access to a world of knowledge which our reason and logic are not able to access, because the pure intellectual culture is limitative!

The structure of my thoughts limits my understanding. Two people can receive the same information and interpret the same information in a totally different way.

Unlearn

and learn
again

To reach the same understanding they will have to return on the tree branches, so they can later follow another path.

It is necessary, then, to UNLEARN so The other can understand the other. One will have to reconsider one certain option of knowledge, so that he can have the same understanding of the other, even though they are at the same place receiving the same impression.

As can be seen, all our mental structure is mounted on bases that LIMIT our knowledge. We are not as “scientific” as we think, neither are we as free to think as we judge. We think though determined “paths”, that take us very far, but that, on the other hand, impede us to arrive to new places, simply because we don’t know the many other paths that exist.

Part 1

10

Thought

Knowledge
Concept options

For didactic reasons, I will continue here the structure comparison of our thought with the tree.

Whenever we accept determined information, we stop accepting many others; in every decision we are committed to the anterior decision, as if we followed the branches of the trees.

This way, each time we limit more our possibility of understanding, instead of, each time, understanding more. To have a different understanding, it is necessary that we return and follow another path.

This occurs more frequently than we imagine. The entire creative ideas move from a different way of seeing what was a convention named “reality.”

The Tree of Knowledge

Knowledge
Option for the concept of non numeric

The moment we accept the concept of numbers, immediately, we limit our thought in understanding the infinite. It is not possible to understand the infinite inside our cerebral structure.

This is an example of the modification at the structure bases of our thought.

This occurs with each decision we make, each time we make the option for a determined concept, we abandon many others. We proceed into unfolding our thoughts up the lines or branches, which become farther apart from the initial option that we made for the adoption of a concept.

Someone who has adopted another concept will develop his thought up the branches in a totally different way. It’s as if the cerebral machines (hardware) worked with different programs (soft wares).

New Path

Paradox
Bifurcation

When we contest a concept inside our thought structure, we open a totally new field, which not only lets us see a problem in a better way, but, find a new solution. This is more or less what we want to say with “break paradigms”.

If we do not break paradigms, if we do not change certain concepts that structure our thoughts, we will not be able to find solutions for what we so call “paradoxes”.

In the Indians case, the paradox of what is infinite is resolved, or better yet, does not exist for him. The paradox of what is infinite is like a “koan” generated in the way our brain functions.

The paradoxes that are presented to us in our culture are the results of the options for concepts that we abandoned when an option for another concept was made.

Hardware

The machine of the brain (hardware), in its potentiality, can be compared to a tree with a very big tree top. The branches of this tree, which are few at the beginning, divide, multiply and open themselves, forming at the end a wonderful tree top full of leaves, flowers and fruits, the expression of our thoughts of the material world in which we live.

Each time we choose a path, though a branch, though a concept, we abandon others. For each concept, each branch that we orientate our thought; we create a bigger limitation in the tree top area in which we will be imprisoned.

With this image of the tree as the potential and structural of our thought, I want to give the graphic idea of how the knowledge can be limiting.

Abandon Paths

When we make an option, we abandon others, and to turn back, many times it is necessary to unlearn. Only by unlearning, turning back in knowledge, climbing down the branches of this tree, is when we can break paradigms and initiate thoughts based in new concepts.

Following other branches, we can arrive at the tree top in a same region, between the same leaves, between the same flowers and, this way, understand paradoxes which were there expressed.

Our culture and our civilization are full of paradoxes, which can only be understood if we unlearn a little of our anterior knowledge and we turn back though our other concepts. This image of the tree makes me understand, in an interesting way, what occurs, nowadays, with the youngsters, who have a culture and a way to see the world totally different from my generation.

Old-fashion Upbringing

In the past we were brought up in how to climb the branches of the structural tree of the thought. We learned concepts and developed knowledge. Though basic concepts we could reach finally, conclusions and knowledge; knowledge very similar to what we call today information.

The teachers taught us to never memorize anything but, yes, reason and, though reasoning, reach knowledge and information. We created a tree of our knowledge, based on branches or options of concepts, and we got to the flowers, the fruits and the leaves. The quantity of information that existed and that we had at hand was very limited.

The doctor restricted himself to his medical knowledge and information. The architect restrained himself to his knowledge and information in the construction and art area.

Revolutions

The revolutions at the end of the twentieth century, brought by the information and communication, the development of the computers and its web intercommunication,

culminating with the global web and the Internet, made the world suffer a change which reflected in the way of thinking and structuring the thought.

The revolution of the information permitted that man cross and manipulates information with a capacity much superior than the human brain. He began principally to consider the form to manipulate or play with these pieces of information.

The form, the way in which the information is worked, is what we call Knowledge. We entered the Knowledge revolution, which also caused another structure change of the human thought.

We find, today, a new revolution in progress: the revolution of the Creativity.

The revolution of the Information gave place to the revolution of the Creativity which was almost concomitant with the revolution of the Communication, flowing today into the revolution of the Creativity.

Information Knowledge

The tree top became too big. We can no longer climb the branches.

All these revolutions alternated and modified the human thought and, more important than this, modified our way of thinking.

If we consider the human brain as a machine (hardware) capable of intercommunicate or to interlace nervous synapses as computer bits, we can say that we need to change the programs (soft wares) of our brain. We have to change the programs with a lot of speed.

Youth

Returning to the tree image as the human brain potentiality, we will have the image of the tree top as accessible to any one, though the machines of the Internet.

The youth of today now has access to all the information and to all the knowledge available in any area. It's as if he had access to the whole tree top of the human knowledge.

The man of today does not have the necessity of hierarchizing the formation of knowledge and information. What is important nowadays is to know how to abandon what is not known, to be able to go forward. It's different from the way of thinking that we had in the past, when we dedicated ourselves to one subject and tried in some way to exhaust it, before we went ahead.

Windows

The man of today has to adopt and use Windows 95, without yet knowing Windows 3.0, and go by Windows XP, without having more than 10% of Windows 98.

The man of today started to have the information of the whole tree top. When he finds the opportunity to create something, he should climb down though the branches and communicate himself, this way, with the concepts that originated this or that information.

Coming from bottom to top, it is much easier for a man of today to tear up paradigms to resolve paradoxes. Again, it is the image of the tree representing the structure of the human thought.

New Tree of Thought

Nowadays, a very important difference has been observed between the way of thinking of the youth and the way of thinking of the elder. The elders complain that the

youngsters don't know history or literature. The youngsters complain that the elders want to impose unnecessary knowledge.

There is, unquestionably, a very different way of thinking that, nowadays, dominates the youngsters. Something changed and, forcefully, changed the way of people thinking.

Knowledge and Information

Observing the youngsters of today, we notice that they dispose of a knowledge that comes very close to what we call information. Information is not knowledge. Knowledge is the ability to manipulate the information, use it, and make it productive and creative.

The old formation aimed, basically, the knowledge and though the knowledge we obtained the information. I had to know Napoleon's life, dates, what was important in his life so that, then, conclude though the information that Napoleon propagated though Europe the concepts of revolution and equality, liberty and fraternity. To understand that Napoleon propagated these concepts though Europe, I have to know all of Napoleon's campaign, since the period of the revolution until his second deposition and, even, his death in the island of Saint Helen.

Today all of this became unnecessary. It is known that the concept of revolution was propagated though Europe. That's sufficient. The historic dates and Napoleon's life is irrelevant.

Young Men

Knowledge
Information

It is said that young men don't know history or geography nowadays; don't have the slightest notion of the historical periods. That's true. But what's the use of this if today the information is accessible to all? The information becomes a commodity. We do not need to think to obtain the information; the information is always at our disposal.

In the past, it was necessary to know how to add so that we could multiply. To calculate a square root all the mathematical operations needed to be known. Today, a square root will not be wasted time with calculus: the result is accessible in the plainest calculators.

Before, our culture structured itself as the climbing of the knowledge tree. The cult man had a beautiful knowledge tree, which cost exhaustive years of studying the humanities. The revolution of the information, followed by the revolution of the communication, led us to the revolution of the knowledge. As soon as the revolution of the knowledge initiated, it was then substituted for the revolution of the creativity. Once the knowledge being exposed, it starts behaving as information.

The youth of today can no longer have the same thought structure of the elder. The quantity of information is so big that it is no longer possible to "climb" the knowledge tree. At each moment, such a quantity of information is presented, that it is impossible for the youngsters to have "humanistic" information like the elders did.

Climbing down the Tree

Knowledge
Specialization
Super specialist

Whoever tries to venture the “climb” of the knowledge tree, will have an extremely deformed tree, whose tree top covers only a determined sector. This one will be the super specialist.

It is no longer possible for the young to have a complete formation in all the areas. The labor market has become extremely competitive, only accepting the best, demanding that the man becomes a super specialist, which is no longer fixable because of the big number of information and of knowledge.

On the other hand, the creativity revolution has made the market even more volatile in relation to the specialization. A highly specialized man is needed, but it is not known in what!

Professions

The classical professions disappeared. Before one was a doctor, engineer or lawyer; today we look for a specialized in underwater welding, a doctor specialized in images and sound, a lawyer specialized in a determined tribute. This specialized is impossible to form because while an engineer specialized in welding is being formed, the doctor in images and sound and the lawyer in determined tribute, the market necessity has changed, and welding will no longer exist, the image of the sound was abandoned and that tribute disappeared.

Specialists

The man has to prepare himself to be a super specialist, but he does not know in what he should specialize. The market does not know its necessities of tomorrow. A super specialist is needed in a specialization that does not yet exist.

Inside this situation, the man is modifying his knowledge tree to adapt himself to the new age. He makes an adaptation in such a way that it calls the attention of the elders who insist in saying that the youth no longer knows humanities. The interesting fact is that this adaptation is done in an independent way from the school or university curriculums, which find themselves in an even more perception backwardness and in the assimilation of the transformations.

Transformations

The man adapts himself and life adapts itself in an independent mode from the reason of man himself. Everything behaves itself as if there were a determinism forcing man, facing the necessities, can find his own path, as if there were a same life for all the men, which makes them find the same path.

The same way, the natural selection acts: that's why, the proto giraffes began to have a bigger neck to reach the higher and better branches of the trees and survived creating the species of giraffe (which, like the other mammals, have the same seven cervical vertebrates).

Inverse Path

To enter the market, man, today, has to know everything in a superficial way. No longer is it possible to know everything profoundly. Knowing everything in a superficial way, he will apply for a super specialist position in the labor market which is needed in the labor market at that moment. One will become a super specialist “climbing down” the knowledge tree.

His competitive differential will be the capacity to, parting from a necessity of a specific knowledge, to find an inverse path which will give his knowledge solidification.

The lawyer will be specialized in a determined tribute today, the doctor will be specialized in images created from the ion K⁺, discovered this week, the engineer will no longer weld because yesterday an alloy was discovered that glues spontaneously.

Knowledge Information

Inverse path

Knowledge of the New Man The youth's tree

The adaptation capacity to the new pieces of knowledge of a productive chain, or to make the information usable though all the knowledge tree of his culture, is what will turn him into a super specialist.

The information and, the knowledge are no longer important in the professional qualification of the new man. He can acquire this at any moment.

Unprepared

As can be seen, the elders are right in saying that the youths are unprepared, that they no longer know the structures of their culture. What changed was the culture itself, and the elders no longer have a way to adapt to the new culture.

An important change is taking place in the thought of the man, and this determines the change in our culture. We have not yet perceived the great transformation in which we are going through.

Competitive Differential

The competitive differential in the market, nowadays, was transferred from the knowledge and information area to a new area: the emotional areas.

The adaptation capacity of the new man to the market will depend more of his psychological structure, of his way of being, of his capacity, finally, of his emotions. The competitive differential is being transferred from the world of facts (information and knowledge) to the world of values (emotional).

The schooling was defiantly put over the instruction.

Schooling and Instructions

Our general structure of schooling finds itself out of step in this respect. The schools still tend to offer a lot of information and little schooling.

I was a Medical College student at the University of Brazil, in Praia Vermelha, one of the oldest educational structures of our country.

Right in the 60's, the old structure of the medical formation began to dismount. The old professors of a life time, who had their serious defects, were substituted for young professors, full of new concepts and information.

Masters

The old structure, today dismantled, had qualities that were despised. The professors of the Medical College were Miguel Couto, Carlos Chagas and other names that were idols to us, Medical students.

The image that Miguel Couto spread was the one of a humanitarian doctor, who went up Rio de Janeiro's slums; from Carlos Chagas, the image of a man dedicated to the

research and his work and, from Oswaldo Cruz, the image of a recognized sanitarian in the whole world.

These images created in the mind of each student an ideal Master. This master was an image with an emotional form: a form that corresponded to a series of emotional values necessary for the execution of the medical profession. Even if these masters were nothing compared to the image that was made of them, they were offering the doctors a “schooling” which was there being formed.

So many times, not knowing how to conduct myself in certain cases, I would imagine how one of these Masters would behave and, almost always, I would make the right decision. This form of schooling was abandoned in detriment to a teaching based on the information.

The teaching structure came, somewhat, to reinforce the thought structuralism in the old form, like a tree in which we try to climb. This form became totally incompatible with the actual reality being, uniquely, capable of constructing deformed knowledge trees, as the form of the super specialist.

The Revolution of 1968

The teaching revolution was marked by the year of 1968, with the students of Paris's revolution. An important moment was defined, in which the students began to intervene in the definition of their own school curriculums, interfering in what had to be studied by them. The power of how to freely choose the youths formation from the masters was taken from them. This was, undoubtedly, a big advance in the education – certainly it is not conceivable that the elders determine what the youths should study.

The iconoclastic movement that followed this revolution destroyed values that should have been maintained and, today, lack. These values are all basically related with the emotional, with the formation of the image of the Master (emotional) to be followed. This type of mental creation has been exhaustively explored in self helping movements that try to adapt the new man in the new market system, reinforcing his importance in the professional formation.

Good Upbringing

The new man requires more a good upbringing than knowledge and information. The schools have not yet noticed the changes that occur in the labor market and insist in forming professionals full of knowledge and information, without at least caring for the education of these professionals.

The new firm does not seek a professional with more knowledge. They seek the one that has an upbringing (set of values) capable of integrating himself in the firm and transform the knowledge of the own firm, generating intellectual capital. No longer was it interesting for the firm the man that comes with a lot of his own knowledge. Many firms have programs of disinformation for the new employees.

As I said before: the knowledge can be limiting.

Formation

I began to notice that there was a new way of thinking following the studying of my children in the secondary course. The change was not only in the “memorize” as it was in my high school; the objective was not to teach the subject, but to prepare the student to

“do the SAT test”. They would try to develop a “sixth sense” in the student which permitted that a right choice in a question of multiple choices was made.

I observed that the formulation of the question was more important than the knowledge of the subject.

Medical Formation

At the University of Brasilia, as an Urologist professor, I offered two courses; Urology I, and Urology II, optional.

In the Urology II course, the students were selected by the general average. Since there were few openings, only the best students of the Medical College could do Urology II course. Observing these students, I learned to distinguish which ones would become good doctors and the ones that wouldn't be good professionals, although all of them were the best students. There was, even, a period in which there were two exceptionally good students; I noticed that one of them would be a good doctor and the other not. I wanted to differentiate them though their grades but I couldn't, because both had very good grades in the exams. After ten years, my observations confirmed themselves.

There is in the student of the health area “something” which makes us see the good doctor. This “something” has this thing with the instinctive way (emotional) of knowing how to identify what is important in the clinic area of the sickness, what is irrelevant and what is essential.

There is “the good sense” which was not transmitted to him though information, but though Upbringing. I'm using the term Upbringing here in the most ample sense, including not only the cultural, but the emotional formation which is structured since the earliest age.

Upbringing and Instruction

When I was a student in the old Medical College of Praia Vermelha, I decided to study physiology with a neighbor classmate, one of the worst students in the class. We almost flunked in physiology.

Analyzing what happened, I could observe that my friend studied and memorized the process sequences, but did not know its finality. This way, he memorized the chemistry reactions that the sugar suffered in the liver, but did not know that the liver, receiving all the blood of the intestine, prepared the sugar to be put in the circulation.

It was as if the information had nowhere to be put in his mind. Since there was nowhere to “put” it, the information was lost itself easily.

My friend wasn't stupid. He was in fact very intelligent. But he had a “flaw” in his educational structural.

Christmas Tree

Comparing the thought structural with a Christmas tree, I could say that, when some “decoration” was offered, he didn't know where to put it. Comparing the doctor's knowledge with a Christmas tree, I can understand: what makes a good doctor is having a good tree, a good thought structure which is given to him though education.

Those that have a good structure (thought tree) know where to put any information that he receives. Information received in the hallway, a conclusion glimpsed in a moment, in a class, in a reading, everything can generate “decorations” for his knowledge tree. The good doctor has a beautiful Christmas tree as a knowledge tree.

This comparison serves to identify why the professors that deal with medical students are capable of identifying that one who will be a good doctor.

Mutation Point

The concept of lineality was undone when Max Planck, studying the atom, developed the quintic theory. An atom that circulates in a determined orbit concentrates energy and, suddenly, leaps to another unpredictable orbit. This occurs when linearly it concentrated one "quantum" of energy. The orbit leap of the electron was given the name of quintic leap.

Discontinuity

This Quantum discontinuity occurs in many of nature's phenomena. There's an energy concentration that's not visible and can't be measured. In a certain unpredictable moment an unimaginable transformation occurs. The Quantum Theory concept has been transposed to infinity of physic occurrences, natural and psychological. Many of these transpositions could be explained in a linear way, if we knew where and how the system of energy concentrates itself. With the studying of mathematics of the infinite and the theory chaos, we can understand that many of these phenomenons could be, in reality, linear systems. We are the ones that do not dispose of conditions to know where, how and when the energy concentrates itself in the system which would be pseudo Quantum.

A quantum theory example applied to the psychology would be of a person that would irritate himself with a certain fact, concentrates anger energy and, in a determined moment explodes in unpredictable reaction acting in a totally different way. He was hit by concentrated quantum anger and a quantum leap occurs. The quantum energy is not pondered, or for being aleatory, or for its dimensionality being impossible.

It was conventionalized to call this moment of quantum leap the Mutation Point, extending a concept that already existed in the oriental philosophy.

Changes

The physicist Fritjof Capra transposed this concept to an actual world situation showing that we are close to reaching the Mutation Point. We are, undoubtedly, concentrating energies in our civilization and cultural lofts. There is a tension state that is difficult to identify or measure, but that materializes and manifests itself though the disharmony of our civilization and culture.

It looks like we are about to reach a mutation point in which our civilization and our society will take unpredictable paths. Like water being heated, in a determined moment reaches a boiling point. Like a butterfly larva that transforms itself in a cocoon. Inside this cocoon there's a transformation that we don't see and, at a determined time, a butterfly is born. The comparison with the chrysalides is very interesting, because the butterfly is anything unimaginable from the lava's view point.

Interesting is that all the big mystics and ecstatic (which come into ecstasy) make similar affirmations. The prediction varies according to the culture and hallucination. As for the time in which the transformation will occur, it is also consonant with the Quantum Theory: it can't be predicted.

Prophets

Such prediction of ecstatic makes it clear that the new order has something to do with the emotional being that manifests itself in its ecstasy. All agree that, or man transforms himself though understanding, or he will transform though pain and suffering.

The terrorist acts at the beginning of the millennium show that the equilibrium of the world can only be reached inside a new world order, in which we neither have a model nor

parameters. The new order will not be a new rule, or a new norm to rule the man. It will neither be a new capitalism nor a new socialism. Not even a new mercantile order.

We are very near a mutation point. Man will change and, this way, will establish a new order, where our actual values will no longer have a meaning.

The apocalypse prophets preach of a great destruction provoked by great geological disasters – showing statistics that show the proximity of this point. The falling of an asteroid can undoubtedly cause a similar transformation that occurred with the dinosaurs in the Mesozoic era. It is not a possibility to be discarded for ten years yet, when man will start controlling the proximity of meteorites and destroy them before the shock with the earth.

This type of cataclysm could leave a small number of survivors who would behave in a new order.

“God does not throw dices.”
said Einstein (algorithmic reasoning).
But it looks like he plays chess (heuristic reasoning).
Nature does not act by chance; it acts though rules.

In chess the rules are maintained. The knight moves, always, through three positions, the castle moves in one direction and the bishop in another. At each move, all the possibilities change. Cataclysms are not essential for great changes to happen; depending on the “move” made by life’s beat, all possibilities change.

Possibly the apocalyptic prophesies are not in nature’s plan.

A possibility of transformation from the mutation point would be a correction that life itself is capable of exerting over man, adapting him to a new situation that perpetuates life.

The most possible is that it will start producing beings with global mental capacities, that is, with ecological conscience, as has been happening with the new generation. This affirmation, which might sound absurd, will be better analyzed forward, in the developing of the Emotional Catapult theory.

Another Path

With the exponential growth of our technology there is, at every moment, the activation of processes before abandoned for not being factual.

The example of the television shows this reality. When a way to create images was discovered in a cathode tub, the discovery had no meaning. In a certain moment “climbing down” the knowledge tree was done and a completely different branch of knowledge was developed, almost as if it were another tree. With the television, the birth of a new culture was done which followed different paths (branches), giving us the possibility of new understandings, but limiting us to others. How different our “head” would be if there was no television!

All new knowledge comes “contaminated” by the previous decision. Each option we make brings in itself part of the anterior decision. This way, we go on creating a thought world of ours, in which we restrict ourselves. Every thought of ours is tendentious, and guards the tendency of homogenizing itself with our previous decisions.

Rationality

As can be seen though the example of the vena cava, there is something wrong in the way we think and very little do we notice. Our logical and rational thinking does not let us perceive that we are not logical and rational; it serves a predetermined decision by the limbic brain.

I develop here some thinking sequences not very conventional and, for this, I need to “climb down” the knowledge tree and follow though other paths in order to find a different approach for the human brain.

I might be tracking strange paths, but I believe that this is the only way to find a unity and unit Science and Religion. To go ahead, we have to go back.

Break Fences

In the first part of this book I tried to expose the fragility of our logic and rational thinking, and show the importance of the emotion in our decision making.

My intention is demystify the brain, especially, the neural cortex. I wish to show that the cerebral neural cortex is an evolutionary artifact, in which it capacitated Life, to confront a critical moment in the evolution process.

Though the theory that a present as follows, I will try to explain some paradoxes of our culture and knowledge, besides joining the fragmented parts of man’s rezoning and religion.

Rupture

This point of view meets a great deal of concepts and prejudices ingrained in our way of thinking, which makes it hard to justify such affirmation. The rupture of certain concepts needs to be made, in a certain way going back in our thought, to follow a new path. Because to climb the knowledge tree, we have to climb down and, after, climb though other branches.

PART 2

EVOLUTIONARY SEQUENCE

Part 2

1

Proposal

In the first part, I tried to expos the fragility of our rationality and of our logic. Man stubbornly thinks that he is capable to think with an independency from his physical structural. At the moment he is a living being as any other, committed with the necessity of surviving and reproducing. The capacity of thinking each time more is an artifact in which life’s pulse capacitated him to increase his competitiveness, in a more complex and difficult world at each time to perpetuate Life’s pulse.

The capacity of man’s thinking was “mounted” over his emotional brain and is strongly linked to this part of the brain that appeared with the first mammals. My approach is to show that the neural cortex is only an evolvable artifact, in which life’s pulsing capacitated a type of human being to better perpetuate Life.

In this second part, I will describe the evolvable process of Life’s pulsation, since the appearance of life until man; from the unicellular to the multi cellular and from the reptile to the mammal, reaching the man.

Since it’s not possible to separate the man from his structure of living being, I will approach the evolutionary process of the living being with the evolutionary process of the

knowledge. This approach requires an important rupture with some concepts and pre concepts ingrained in us, and an opening to a type of rationing which, although logic and rational, it is centered in life's pulsation and not in the man.

Pau-d'Água

In front of my house there is a tree which grows like a palm tree. I planted it myself when we moved to this house. Here in Brazil we call it pau-d'água, and it is said to protect the house.

My tree grew a lot, almost reaching 4 meters tall. The strong wind of autumn would curve it to the point of almost breaking it in half. In the morning, I would observe it in my matutinal meditations. I noticed that its trunk could break and the tree could die. I had already decided to cut it down and plant another to substitute it.

One day, suddenly, I saw that, exactly at the biggest inflection point of the stem new branches started to sprout, forming new tree tops, which fortified the tree. Because of these new branches, the stem of the center branch stopped growing, eliminating the threat of destruction of the tree. As an experiment, I hurt the tree bark in many points, to see if there too new buds would appear which didn't happen.

It seems as if the tree "knew" where it had to develop new buds. The solution for this survival impasse came with mathematical precision because, in the highest inflection point, the bark broke making it possible for new buds to appear. Surely there was an intelligence hidden inside this tree, capable of adapting it to the adverse conditions and make it survive the adversity. This "intelligence" found a better solution than mine, who considers himself at the tip of the evolution scale as Homo sapiens.

The pau-d'água tree, that doesn't have a nervosa system nor a brain like me, resolved the problem better than I would have, who was there meditating. We can understand that, though a genetic code, the tree had information of how to develop itself from the starting point of a seed – fact that for itself was wonderful – and, even yet, of how to behave when confronting adverse situations. We can consider that the genes nucleic molecules function as computer chips.

My tree was selected by nature, making it possible that the trees that react this way survive and the ones that didn't disappeared. The disposition of the nucleic acid chains in its genes keeps the information. The quantity of information inside these molecules is something unimaginable. It could have been said that the plant "learned" that, when its bark breaks, it should launch buds in that part and this would be coded in its genetic code. But if I hurt the tree bark in another point, it responds with a type of scare. How does it know how to respond to a similar aggression? There is some kind of tree rationing in this tree, like intelligence. This intelligence is Life!

The tree that, through a mechanism of phototropism, grows towards light, presents a type of intelligence written in its generic code. The complexity of this codification for this type of behavior is extraordinary. The light can present itself in different forms and intensities that vary into the infinite, but the plant always recognizes these variations.

When a new year season begins, the plant begins to behave in a different way, as if it were guessing the season sequences. This is written in its codification. An interesting example is the bamboo blossoming, which happens at each 100 years, in the whole planet, at the same time, showing that the bamboo codes are synchronized and influenced independently of each particular bamboo.

Inside the cells, identification has already been made, of particles that determine how many times the cell can multiply itself, making life end in a mathematical moment of duplication. This has been proven with the clones that "age" very soon. These particles (telomeres) practically make the clone have a life time equal to the being that was cloned.

It is clear that, although the man was able to reproduce a physic body, he is not capable of prolonging the life.

With the current development of the stem cell, the man will be capable of reconstructing a part of his body that, somehow, was damaged. It will be marvelous the moment in which we will be able to reconstruct the nervous system of patients that had there marrow damaged and remove them from their wheelchairs. This will not be done by man the same way that he constructs a machine or a computerized tomography appliance; it will not be a “constructed” piece of work done by man, but, a “discovery” of how to modify life’s pulse contained in the stem cell and how to conduct life’s pulse in a specific direction.

Life’s intelligence, or the pulsation of life, has been doing this during many millenniums. We can see its acting in the reconstruction of the tail of a lizard. If it were the interest of this pulsation of life she could redo the arm of a man.

As can be seen, what we do with our “intelligence” is nothing more than “to discover” how the great intelligence of life functions. Life’s pulsation acts according to a self desire, independent of the brain of the Homo sapiens.

Inside the pau-d’água tree of my garden an extraordinary intelligence exists, capable of finding a solution to its problem of better surviving to what I call “my intelligence”. What I call “my intelligence” seems infamous, if compared with the greatness of intelligence of the pau-d’água, which behaves as if it had a memory of all the paus-d’água that preceded it. This intelligence is also inside the seed of this tree, codified in spirals of nucleic acids capable of repeating this tree as many times as necessary. Each cell of this tree would also have all this information.

Genetic Code

We are even capable of understanding the tree’s genetic code, if we think that the nucleic acids function as computer chips. There is something beyond the genetic code, a force, a will that orients and give destination to this codified process in the molecular chips. The strength that moves the pau-d’água tree in its constant adaptation to the exterior world is in the chips that were structured by a great intelligence or a great force.

There is a pulse or an impulse that moves it, that determines this continuous adaptation to the external way. It’s a pulse of life that existed since the first life manifested itself, and continues to manifest itself, moving the pau-d’água tree in a continual adaptation process to the external way.

If we consider the time-line of the pau-d’água tree, we will observe that it takes, through its many diverse material expressions, a force, an impulse, a pulse of life that propagates itself from one to another living being. This life pulse has intelligence and a reasoning, which makes it realize its will of showing itself in the world, adapting itself continually. There is something that functions as a determination, as a will, which drives the pau-d’água tree to surviving. It is something that the genetic codification of the nuclear genes simply would not justify.

This is LIFE.

Intelligence

There’s intelligence in the living being which is independent of the genetic code. Every living being assumes surviving attitudes in response to the stimulus that were never presented to it before, but it’s as if it could make a comparison with previous experiences, as if it reasoned. This intelligence is in me too, in the regeneration of my tissues, in the unconscious adaptation that my organism makes when facing adverse conditions. It’s in

my dog that, likewise, or maybe even better than me, can adapt to new situations. It's in the growing plant, searching for light. It is in everything that lives!

I do not know how to define Life, and I have no intention in getting into a discussion that could take me off track in my attempt to be logic and rational. Life exists, is real and is present in our existence in a very clear way. It has nothing to do with a philosophical or abstract concept. It exists, although we don't know how it is built and we can't reproduce it with our brain's neural cortex rational science.

We are capable of analyzing it though the made up carbon molecules in which it is constituted of but, even if one day we can mount these same molecules, we won't reproduce it. We make countless considerations and emit many theories in respect to the way in which it could have appeared on the Earth's surface but, we don't even come close to understand it. I don't know what it is.

I don't imagine how it formed itself and I will not here say that it is divine. I don't want to get into religious or philosophical abstract discussions. I'm forced to affirm that it exists, that it's real, that it has an intelligence that many times surprises us Homo sapiens. Many times it perplexes us and makes us think that we are ignorant. It is a higher power that is given us to understand.

It is intelligence. There's no doubt about it. It's superior to our intelligence. This intelligence, this greater power, somehow, is also in me, because I have Life!

Part 2

2 Life

In The Outline of History, by H.G. Wells begins with the affirmation that God diminishes Himself as science develops. Primitive man saw God in a flash of lightning, in the rain and in the fire. As man started understanding nature, God diminished.

This simplistic affirmation does not take into account two important aspects.

In the first place, man will never be able to understand all the processes of his existence, because he will always bump into life. The more we try to understand the evolutionary process of human life we bump into the beginning of this process, where an initial "impulse" determines that the carbon molecules would form a living being.

Secondly, all the knowledge processes that man has been through "diminishing God", is being, constantly, being incorporated into this living being who is man. All this knowledge is constantly being "written" in man's mind.

I want to make a clear separation between Life and the living being.

Life is clearly defined in an elegant way by Maturana and Varela in The Knowledge Tree, as a self poietic system, capable of producing, in a continuous way, to itself.

Life would be the impulse that "determines the formation of a living being or of a self poietic system." Life is the initial impulse that generated the appearance of the first living beings. The Life that exists in man is the same that exists in a tree and animal cell. The same "impulse" of Life that generated the initial cell that drives man.

Life is the moving piece of evolution. To better characterize this impulsion, I will call it Life Impulse. Life Impulse manifests itself all over the world disguised in uncountable forms of life. The life impulse that drives a virus is the same that drives man. The life impulsion generated the appearance of the first carbon molecules and isolated them from the environment by a membrane, forming the first self poietic system, or the first living being.

Life on our planet has structured itself over the proteomic molecules. These molecules are structures that present great instability and, on the other hand, cannot

maintain themselves for a long time, therefore, are ephemeral and unstable. These qualities were transferred to Life. Being unstable, it goes through immense transformations – and being ephemeral, it has to replicate itself.

Life characterizes itself by its necessity to constantly replicate itself, of reproducing itself, because it's ephemeral, like the protein molecules which compose it. In case it doesn't reproduce, the tendency is for it to disappear. This same determination deriving from the proteomic constitution reaches the Homo sapiens who, with all his arrogant way of thinking, cannot escape from this imperious necessity of Life that lives within him. When the hypothalamus determines the sexual character appearance, man manifests the same tendency to replicate himself, which the proteomic molecule has that will one day disappear.

The instability of the proteomic molecule was given to the living beings. The instability of the living beings determined a plasticity of the forms, which resulted in the acceleration of the evolutionary process. If we analyze the lapse of time since Life's appearance, the speed in which Life adapted itself until it reached man is surprising. Such evolutionary acceleration is because of these two qualities of life we inherit from the proteomic molecules: ephemerality and instability.

Individualism

Life's Impulsion has the necessity of expressing itself. It always has to be transferred from one being to another and, if it weren't this way, it would disappear from the Earth's surface. This transference generates a more specialized body, more apt for surviving.

Man, as we know, is a specialist in this initial life impulse, which has as a finality to perpetuate and has as a necessity to exist! In his pretension and arrogance, man judges himself in the highest tip of the evolutionary chain, as the big reptiles did one day too. Because of his conscience, man has positioned himself as being the marvel of evolution. He's simply nothing more than an adaptation experience of Life on the planet.

Definitely, man is not the best expression of Life's Impulsion. At the beginning of the 21st century, we are confronting the possibility that man, enclosed in his individuality, will be capable of altering the condition of life in a way that will make his own survival impossible. Due to this individuality, man is destroying the planet's resources, leading it to a situation in which man's own life will no longer be possible.

Experiencing a crucial moment of his existence, he should start thinking about his species or he will destroy it. Life's impulsion has a prognostic for this occurrence. There are species capable of surviving the devastation man can create. In this particular occurrence, there are insects and bacteria that would be able to redo the evolutionary process and recreate a thinking being that could be physically very different from the modern man. Surely, the new creation would correct the possibility of this new man incarcerating himself in his individuality.

It is a Life that Lives

Our evolutionary process began the moment a membrane ended the first unity of life in a unicellular being. Life, in a way that we cannot yet comprehend, appeared in our planet and, from this initial moment, evolved until it reached man. Structured over the ephemeral and unstable proteomic molecules, it could not maintain itself for a long period of time. That's why it is, constantly, disintegrating and reconstructing itself; constantly adapting itself to the external means which, in its way, is also in constant transformation.

We are no more than an evolutionary moment of something that we don't understand: it's the first impulse of life! The same way, the insects are also Impulses of

Life. Life is capable of reproducing the entire evolutionary chain from a cell or insect, but will be incapable to do so if man uses his power to destroy it.

When we perceive that man is a manifestation of the impulse of life, which has been evolving through a chain of bodies that were inhabited by it, we can conclude that the human brain is a potentially dangerous experience in this evolvable chain.

The insect, through its adaptable power, may be the best form of life impulse manifestation nowadays.

The molecules that compose the human being are constantly reconstructing themselves. Consequently, the human being is constantly changing (adapting himself). All the cells of a man may change every ten years. The man of today and one from ten years ago only have Life in common!

Man has tried to explain, in a mechanical way, life's appearance, using radiations on top of molecules and showing that there is a tendency of certain molecules to group together. Some even show structures in which molecules group together as if they were going to form a membrane. However, nothing or nobody was able to show how life was formed on this planet. Surely there were very special and favorable conditions for its appearance, but even these conditions do not explain such an appearance.

Life is like a tendency, an impulse, a pulse, which finding favorable conditions, "materialized" itself and materialization is a good term that I want to use. Life made itself present through a physical structure. After the materialization, it went on adapting itself and evolving but always existing.

The same impulse which materialized the first membrane and isolated the first living being continues to exist. Everything that exists is part of the same initial impulse of life! The animal world, the vegetable world and the entire living world are, until today, the materialization of the same initial life impulse, which because of very special conditions, manifested and materialized itself in the first living being.

The impulse of life is therefore, the initial point of the phylogenic sequence and ontogenetic which I present in this book.

The Human Brain

All this philosophical digression aims at showing the point of view in which I want to present the human brain: a manifestation experience of life, which in a certain evolvable moment, experienced a new organ that had its own conscience. An organ which, due to the identification with individuality, withdrew itself completely from the initial impulse of life in which it was generated.

This organ is an experience that can be positive or negative in the continuous process of Life's adaptation. Positive, is the consciousness that manifests itself through the organ understanding that it is nothing more than the manifestation of the impulse of life that at some point materialized itself on the planet, and finds harmony with life and with the planet. But if it is negative, before it can self destruct, the impulse of life will not hesitate in destroying man with his brain, using the same impulse that can show itself through a simple virus. More viruses have been discovered than our capacity to create defenses against them.

Finally, it was not man who acquired a self-awareness: it was life that became conscious.

Part 2

3

Evolving Theories

Our entire evolutionary process initiated at the moment when a membrane concluded a first unity of life in a unicellular being. This Life, concluded in the first unicellular membrane, works like an Impulse which has conducted the evolving process.

The human body is the result of an evolving process that can be identified in many moments when we compare it with the many animal bodies that we know.

The links that unite the evolving chain, will surely all disappear, but from these links, which are contemporary to us, living beings that guard “memories” of the links will develop.

The human body has a segmentation similar to a centipede. The human embryo presents branchiae similar to the fish. Our cerebellum commands and articulates our movements in a similar way to all the mammals. Our brain is made up of layers that seem to be overlapped like archeological layers.

However, we we're not centipedes or fish, nor are we monkeys.

Unicellular

Life began through a unicellular being, that surely has nothing to do with the unicellular beings that we know nowadays. The unicellular was the first manifestation of life impulse and it happened at the moment in which one membrane detached on one hand, from the exterior world and the impulse of life on the other.

Trapped Fire

The living being characterizes itself through its capacity to produce itself, in a continuous way. This organizational capacity is defined by Maturama and Varela as “self-poietic”. The living being is, therefore, the self-poietic one (capable of reproducing itself). This capacity is granted by a cellular metabolism that is, in a final analyses, the manifestation of the impulse of life.

The impulse of life is like a flame, a trapped fire inside the cellular membrane, and it behaves as a burning flame in a restrained way, carrying out a destiny that seems to be the perpetuation of life. At first, the flame is capable of producing its own molecules (self-poietic) and, afterwards, of producing itself.

Because of its instability, this impulse or fire needs to initially produce its own molecules! Because it's ephemeral, it needs to produce itself. The instability determined by the multiform (protei) molecules, triggers an incredibly fast process, which we identify as the “evolutionary process”.

Speed

The instability of the protean (protei) molecule explains the speed in which life has expressed itself in our planet. For the effect of greatness, if we consider the existence of time on the planet as a day, the time in the course since the beginning of life until now would be measured in minutes.

Fiction

Life can be present in many other planets, structured on top of other types of molecules. If this restrained fire were to structure itself on top of another type of molecule, living beings could be very different. As can be perceived, life is capable of presenting itself with forms in which our reasoning does not have the capacity to imagine.

Man, through his neural cortex and his reasoning, has the pretension of being a unique being in the universe. However, in the last years, with the advances of astronomy,

especially due to the images of the universe taken outside of Earth's atmosphere, large quantities of planets have been identified.

The great variety of life forms present in the universe cannot be "fathomed" through our reasoning – but, strangely, it can be identified in man's imagination, through fiction. As can be seen, our brain, specially our neural cortex, is nothing more than a mere manifestation of the impulse of life. The Life that lives within man has a knowledge that transcends immensely our pretension of reasoning.

Olympus

The impulse of life progression from one life to another, in a continuous way, from the first living being until man, can be compared to the symbolism of the Olympic fire. The continuous fire, that one day came to Olympus, continues being transmitted from one man to another with the objective of reaching eternity.

Boat

I will summarize the main theories that try to explain the marvel that is a multiplicity and variety of life's manifestation on Earth, thus justifying the existence of man.

With a didactic and even romantic intention, I show life as a boat sailing in the ocean.

Life's constant flow and ebb follow an evolutionary drift, like a boat adrift being taken by the running waters and winds. Where running waters and winds behave like the constant variations of the exterior environment.

This boat, which separates the living being's external environment from the internal environment, is a Membrane, something extremely important in the entire process. Inside the boat, The impulse of life follows.

The impulse of life, inside the boat, is the commander. The fact that the boat is adrift doesn't mean it doesn't have a commander, and this one does everything in its reach not to allow it to be shipwrecked. The commander will always be aware of the latitude and the longitude where the boat is, always observing the ocean to defend it from the dangers. Undoubtedly the boat has a commander, which is The impulse of life.

Several Theories

Concepts have evolved a lot over the last centuries. I believe that all the theories have their truths when we understand the evolutionary process as a manifestation of the impulse of life.

Saint Thomas Aquinas believed that the impulse of life, sometimes, could put an oar into the external means and row in the direction of the "Island of Perfection".

Within this same concept, Lamarck proposed his theory of evolution, as if the living beings had adapted themselves to the external means by walking towards perfection.

Around the 19th century, with the definitive rupture of science from religion, Darwin proposed his Natural Selection theory. The boat would navigate fighting against the current and with the winds, confronting itself against the exterior world.

At the beginning of the 20th century, it was discovered that the boat had a cargo. Part of this cargo was the nucleic acids which formed the genes. It's as if the boat had a book aboard.

Based on this "book", The Genetic Theory was proposed.

Composing the cargo of the boat with its struggle with the exterior world, the so called Synthetic Theory was established which, practically, joined the Natural Selection to the Genetic Theory.

At the end of the 20th century, Maturana, exposing revolutionary knowledge concepts, shows that the boat defiantly doesn't have oars, and does not fight directly against the currents and winds. It simply molds itself to the influences that it notices from outside, registering in its log, the currents and the winds the way they are seen from inside the boat. This is the Natural Drift Theory.

At this transitional moment in which our culture is going through, at this very "mutation point" in which we find ourselves, the Natural Drift theory began a great controversy and a radical positioning that divides science and religion.

In certain North American states, it is prohibited to talk about the evolution of the species. This theory has generated revolt in the conservative circles, which position themselves against any theory that speaks of evolution.

Certain radical sciences (rational Cartesians) want to demonstrate that life was formed by chance and also developed itself by chance, through the natural drift. On the other hand, radical religious people and reactionaries say that God created man as his own image.

The skeptics cannot prove their ideas with experience, nor can the religious people rationalize their dogmatic concepts.

This confrontation has worsened the polarization between science and religion. Science has evolved greatly, to the point to be able to give phylogenetic and ontogenetic explanations. However, has not been able to penetrate into The impulse of life. On the other hand, religion has tried to bring science close to them in order to find reasons and logics to their dogmas. But at the moment that we accept the concept of Life Impulse, we may speculate with reason and logic the ontogeny and phylogeny.

We are now, at the beginning of the 21st century, at the verge of a great transformation of knowledge. At this moment we cannot have distinction between science and religion. Both currents have men with the same brain, evolving history, same necessities and deficiencies. We have to admit the fact that we do not know what The impulse of life is. On the other hand, we have to think and theorize as to how she has manifested herself.

"Then render to Caesar the things that are of Caesar's and to God the things that are God's."¹

The solution to this impasse is the "cargo of the boat". The boat definitely does not carry, only, DNA sacks (genes). The boat carries The impulse of life". We cannot know if an impulse orients the boat in some way, as if it had the intention of arriving at a "perfect island", or if it has, only, the intention of surviving.

The boat seems to flow adrift, because it has no oars. Apparently it cannot influence the exterior world to create its destiny. However, The impulse of life that follows inside the boat may have intentions, pretensions and even dreams of arriving somewhere.

The intention of the impulse of life is not accessible to our understanding, even less to our analysis. The impulse of life "is what it is"!

¹

Matthew 12:21; Mark 12:17; Luke 20:25.

The Importance of Membranes

The impulse of life, navigating adrift in this sea which is the outside world, is detached from it by a membrane. The membrane detaches the impulse of life from the outside, thus characterizing three units: the impulse of life from inside; the exterior world from the outside, and the membrane.

In the search for the understanding of this process, we have concentrated on the inside and the outside without giving the proper value to the membrane as we should. I believe that the broadening of the membrane's concept will enable a better understanding of man, joining his parts that are detached and uniting it as a whole.

The membrane functions as a frontier that separates the impulse of life to the outside. It can be the animal's skin, but it can also be the frontier that unites the beehive, or the social frontier that unites a baboon group. All of these frontiers are membranes that detach the inside core, where the impulse of life lies, from the outside, which is the world. I'll discuss the formation of the membranes when I talk about the multi cellular beings.

When we understand the initial membrane as a frontier, and understand the concept for all the living beings, we notice its importance in the whole process.

Part 2

4

The Origin of Knowledge

Until now, man has tried to understand the evolutionary process of the human species – Phylogeny – as a spectator who, using his reason, tries to assemble a picture, like a puzzle. In this puzzle, the pieces are not available, because all the forms of life in which man went through during the process are lost.

Phylogeny

The comprehension and understanding of the evolutionary process of the human species (Phylogeny) are not restricted to the assembling of a sequence of living beings, simply because these living beings no longer exist, and paleontology has helped very little.

The unicellular being of today is surely very different from the first unicellular being that at a certain moment initiated the evolutionary process. Man and apes had a common ancestor that looks nothing like today's apes. The first mammals that appeared after the great reptile era were very different from today's mammals.

Ontogeny

Another important instrument in the study of phylogeny has been the observation of the evolutionary stages which the human embryo goes through – Ontogeny. The embryo, during its formation, presents primitive structures that transforms and adapts itself as it grows. The embryo presents gills like fish, and tail like many mammals.

The human embryo's urinary apparatus, which initially presents itself with the primitive forms found in fish and reptiles, gradually, transforms itself into man's urinary apparatus that we know. Some of these primitive structures, which recede during the embryo's growth process, can leave cell residue that manifests itself, later, as tumors. They are known in urology as embryonic residues tumors of the ducts of Müller or Wolff.

The observation that the embryo growth process presents stages of the evolutionary process led Haeckel, in 1866, to formulate his famous law: "The ontogeny recapitulates the phylogeny". This law, initially taken literally, became a broader and more philosophical meaning with the discovery that the process of growth of the embryo is determined more specifically by a genetic code.

If the ontogeny does not recapitulate the phylogeny, at least it is an important factor in its understanding, proving that our physical body is a product of our evolutionary process.

Anatomy

If the archeology and the ontogeny can help in the understanding and in the comprehension of the phylogenic, the compared anatomy has shown the great similarity that exists between our organs and the organs of the other living beings.

The similarity that our body presents in relation to the body of mammals and primates is big.

Genetics

Genetics has also shown the great similarity that exists in the gene codification of the living beings. The chromosome difference between man and certain species of primates comes to the point of approximately 1%, but all these scientific advances have not yet been sufficient to be able to set up a rational sequence of living being that, through an evolutionary process, create man.

Science

The phylogeny, or the evolutionary process of comprehension of the human species, cannot be set up on a rational table, as if man were a separate observer to this process. When man wants to know something he cannot escape from the law that rules the knowledge: the knowledge depends on the one who knows.

Man's rationality, especially the scientific method inaugurated by Francis Bacon, gave man the illusion that he is capable of an independent rationality from his being as a whole. The scientific method teaches that by observing nature, man would be able to infer the laws that rule this nature. Observing the Sun that rises every morning, we could affirm that the Sun will rise tomorrow. Who first interposed this scientific method was David Hume, by saying that if the Sun will rise tomorrow is still a belief not a certainty.

Incapacity

Despite all the data that science has nowadays, it is not possible for us to have a rational and logical knowledge of phylogeny, excluding our logic and our rationality. The way science is structured is not, in any way, capable of making us understand the phylogeny.

It is not possible to disassociate man from his knowledge. Man is a living being like all living beings, like the monkey, the bear or the tiger. He differentiates himself from the other mammals because the impulse of life equipped him with a neural cortex for survival purposes – the same way as the giraffe was equipped with a long neck to eat the tallest leaves and the felines with teeth and claws.

The knowledge that the lion has of the savannah where it lives is totally different from the knowledge the gnu has of the same savannah. All the information the lion has is for the purpose of hunting the gnu. The information the gnu has is to run away from the lion. Knowledge depends on who knows what!

Presentation

The evolutionary process of the human species (phylogenic) cannot be disassociated from the evolutionary process of knowledge. A name not existing to identify the evolutionary process of knowledge, this study would have to be called the Origin of Knowledge.

Man is a living being that has the greatest knowledge, he is the greatest knower, the living structure that has more capacity of knowing among all the beings of the planet, even though he is not free of the fact that the knowledge depends on who knows.

The evolutionary process, in a totally diverse manner of what is given as conventional, consists of the following: in the beginning, a Life Impulse separated itself from the outside environment through the formation of a membrane, forming life's first unit. The initial unit became more and more complex and more capable of surviving.

The physical form of the living beings, changing and becoming more complex, began constituting itself of many units (multi-cellular), reptiles, mammals and man.

Knowledge Membrane

The surviving capacity (knowledge) also expanded itself, acquiring knowledge that was no longer restrained by the initial membrane, that detached the impulse of life from the external environment. The living beings had to inner-relate to survive (Plenitude Law). The initial units of life would not survive if information was not exchanged. The impulse of life began to involve more units of life through new membranes that were no longer physical (protean) and were now knowledge membranes.

When two living beings exchange surviving information to reproduce, its outside knowledge is no longer limited by its physical membranes. There is in these living beings a knowledge membrane that involves two beings. When an ant carries leaves to the anthill, there is a knowledge membrane that transcends the ant and involves the entire anthill.

The evolution of the knowledge occurred through the expansion of the knowledge membranes or power membranes. These membranes can be better understood when they assume the form of social membranes. The evolutionary knowledge is the process of the expansion of the membranes.

Here, the evolutionary knowledge process is compared to the evolutionary knowledge process.

As the evolutionary physical form is being shown (Phylogenic), the knowledge expansion as it was, is shown as the Origin of Knowledge.

Part 2

5

Membranes and Enclosures

In The Knowledge Tree, Maturana and Varela introduced two important concepts that revolutionize the understanding of Life and place the physiologic bases of the human comprehension: The Self-Poietic Organization of the Operational Enclosure.

The revolution which these concepts make in the human comprehension has not yet been totally digested by our science. Human comprehension cannot be disassociated from the comprehension of the living beings.

Self-poietic Organization

According to the first concept, the self-poietic organization, living beings are those capable of producing, continuously, themselves (Auto-poise). It's not exactly what is reproduced, but that one which can restore its own structure.

The self-poietic organization of many orders is composed of other self-poietic organizations, which behave in a similar way. The concept resolves a complex question as to the understanding of what a living being is. Man is a living being because he is a self-poietic system. However, his liver cell also has this characteristic.

The liver cell of man has, written in its nucleus cell, the code of the whole man. It's an organization inside another auto-poietic organization. Like a life inside another life, where both have the same code book. The impulse of life which is inside the liver's cell is the same in man.

Knowledge Membranes

These organizations are limited by a border, which in the unicellular structure is a membrane, and assumes other multi-cellular characteristics, called second order organizations.

The initial life impulse unit had, as a border, its membrane.

The multi cellular beings, like animals, have skin as a physical membrane. Other beings may not have a so well-defined membrane, which is the case of vegetables. Even more incredible are the auto-poietic organizational borders which reproduce as a group, like beehives and anthills. The ant or the bee does not survive alone, as well as the liver cell does not survive if it is not part of an auto-poietic organization of multiple orders.

This concept shows that the initial cellular membrane unit, that sheltered the impulse of life, can broaden in a way not yet analyzed. The border can be material, constituted by molecules, it may not have a defined limit, like the vegetable, and may, still, have a limit that is not physical, like in the case of the beehives. These membranes or borders are an important secret kept in the impulse of life.

In living beings, auto-poietic, the cellular membrane will expand as the living being's knowledge expands. It will stop being physical and start being a membrane that involves knowledge, as Maturana identifies well in what he calls social membranes.

A social membrane involves the anthill and a physical membrane involves the ant. A membrane involves the liver cell; another involves the liver, another one involves man and another involves the group in which man lives.

As the evolutionary processes advance, the membranes expand themselves.

They are nothing more than knowledge membranes. The social membrane that involves a group of gnus has more knowledge than each individual gnu. The expansion of these knowledge membranes is the expression of the evolution of the knowledge process or the Origin of Knowledge.

Operational Enclosure

Another interesting concept of the book *The Knowledge Tree* is the Operational Enclosure. The living being "operates" inside an enclosed system: it (the living being) and the other outside (outside environment).

In a first living being it is easy to understand the concept, because this one is separated from the world by its cellular membrane. As this membrane expands itself as a knowledge membrane, the operational enclosure also expands.

All the living beings are separated from the outside by a border, which limits the auto-poietic organization. These limits would be the knowledge membranes.

These borders can be identified in animals as a social frontier. As well as a social membrane separates the beehive from the outside world, a social membrane also makes the baboon group form a unity when it relocates and confronts the threat of a predator. When the baboon group sees itself trapped by a lion, the oldest of the group surrenders so that the whole group may escape.

Inside this knowledge border, the living being is a prisoner of the operational enclosure. The auto-poietic organization operates, functions, reacts, in short, exists as an enclosure, separated from the world by a Membrane.

Little does it matter to what is outside, if this outside does not interfere in its Enclosure. What is outside only matters for the organization, as long as it interferes in this organization.

Maya

The outside reality does not exist for the organization; what exists is a series of stimulus that reach them coming from outside. In possession of these series of stimuli, the living organization or auto-poietic forms an image on the outside that has nothing to do with the outside reality. The image that the auto-poietic organization makes of what would be the outside is very similar to what the Indians call Maya: an illusory reality.

If outside the border a tiger or a lion goes by, the wild animal will be registered only while it can interfere in the organization. The image the auto-poietic organization makes of the outside world is like a mirror: it reflects the image that comes to the interior; only that this image arrives in a deformed manner by the membrane, because it selects exclusively what interferes with the organization.

Maya

Operational Enclosure

Living Being
Membrane

What the living being sees of the outside world is a very particular image, uniquely his, that little has to do with the outside reality, but has a lot to do with itself.

Every living being suffers this operational enclosure process, from the cellular unity, passing through the multi cellular and reaching the animals, which will have a social border. The living being forms an outside image and acts as if what he saw in the mirror were the exterior. However, its mirror reflects its own membrane, and not the outside. The living being is locked within itself, sees itself and not the outside.

Conclusion

The living being is the one capable of reproducing itself. It operates locked within itself, in an extremely egotistic and egocentric way. The limit of the auto-poietic being expands itself with the progression of its knowledge, through what I call knowledge membranes.

This concept of knowledge membrane can be applied to all the multi cellular living beings and also to man.

Man can behave as an auto-poietic organization limited by the knowledge membrane of his skin or as an auto-poietic being limited by humanity's knowledge membrane. In a broader expression, man can behave as an auto-poietic being limited by the knowledge membrane of his own Life!

Part 2

6 Unity

The first manifestation of the impulse of life was a unity of life separated from the environment by a membrane: similar to a cell, a unity of life.

Starting from this first structure, the impulse of life began to multiply itself in an extraordinary way, like an explosion of life. Such an explosion occurred through great transformations, making the unit that initially sheltered the impulse of life transformed itself, adapting and joining itself to other units, originating what today are all living beings.

Unicellular

The impulse of life materialized itself, in its first unit, sustained in ephemeral and unstable protean molecules; this way life's unit also became ephemeral and unstable. Ephemeral because it "knows" that it can't last long, and unstable because it "knows" that at any moment it can be disassembled. Life's unit initiates with this "knowledge that made it reproduce and multiply itself.

Since the beginning, two big lines already defined that the impulse of life is present in its manifestation: in the first place, the necessity of eternally repeating itself and, in the second place, the necessity of constantly adapting itself. These two necessities were the ones that drove life's explosion immediately after its appearance, and has continued until our evolutionary moment.

Necessity

The process in which the impulse of life materialization has been through is determined by necessity. Man cannot be excluded from this process.

Man moves and thinks through a necessity of the impulse of life and not his own. Man has the same necessities as the first cell that sheltered the impulse of life. As a result of the fact of being built by protean molecules, he is also ephemeral and unstable.

Observing this materialization process of the impulse of life, we notice how bound we are, united, and how much we are part of the impulse of life. We notice how we have the pretension and the illusion of being a separate unit from the whole.

Drift

Let's go back to the image of the boat sailing adrift in the world's ocean, driven by the winds and by the tides: inside the boat is the impulse of life, separated from the ocean by the boat which is the membrane. The boat has, therefore, a forthwith necessity of being, constantly, adapting itself to each new current, to each new wind and to each new rock which it crashes into.

This is the situation of the manifestation of the impulse of life on Earth: a situation of extreme necessity and abandonment. A situation so dramatic that it could only have a continuity manifestation if the impulse of life had some "personality", some action. If it existed or had a will.

Will

The will of Life impulse was what enabled, after this beginning, reproduction to accelerate and the multiplication of their units, confronting the dramatic situation in which it found itself. Life's unit, at the initial moment, was the titanic fighting stage between the impulse of life, from the inside, and the constant transformations, from the outside.

Struggle

In its effort to reproduce itself, the living being had to face the outside changes. At each change it had to adapt, and at each division transmit to the "offspring" the information about the transformations that had happened within it.

Duplication

In the beginning, the unit duplicated itself, passing down to the other new beings, the transformations that it had gone through to adapt itself to the environment. This information was becoming greater and more complex at each moment, and the impulse of life began to use the protean molecules to register the information.

Genetic Code

The most logical and practical way in which the impulse of life found to transmit the great quantity of information to the new units, was to "write" onto the molecules being duplicated at each reproduction. This way, it set up a written form using the combination of four letters in one line, which if it were too big, could tangle in a helicoid form. The four letters were the four molecules that compose the DNA and the RNA: Adenine, Guanine, Cytosine and Thymine or uracil – AGCT. Initially, they could have used various types of molecules as letters, but in the end they were reduced to four.

The fact that all of life's code was written with four molecules is an important factor, that shows the unification of all of the planet's life.

Identification

We're all in the same boat! Man, insect and a wild beast that lives under the shade of an old tree, are all, part of the same impulse of life!

Coming back, again, to the image of the boat, a log was being written! All of this exposes the human fragility in the face of the outside uncertainties of the world, showing the necessity that drives Homo sapiens, not very different from the one that first unit that sheltered the impulse of life. The comprehension of our evolution confronts the human pretension of wisdom and knowledge, in an extreme reductionism which will pontificate the necessity.

The necessity also drives man nowadays, because of the fact of knowing that he is an ephemeral and unstable being, that has to reproduce because he will die, and that he needs to adapt constantly, because he knows that he can become unstructured at any moment.

Plenitude and Threat

The living being tends to manifest itself in plenitude.

This is a law that rules the living units and is present in life since the first cell appeared in man.

The living being tends to occupy all the space that the perception of its enclosure presents. The auto-poietic organization is closed within its own self, and has no

consideration for anything that has nothing to do with itself. It has no pity; pity is a human quality.

Plenitude certainly played an important part in the evolutionary process, and left its marks in today's beings. Life's first unit, through the laws of plenitude, certainly occupied all the available space, draining the nourishment and becoming a threat to the own manifestation of the impulse of life. The first living being was already a threat to the own manifestation of the impulse of life, the same way as modern man is.

Mars

To give you an idea of what went on during the initial moment of life, we can observe the programs that NASA has to make a planet like Mars inhabitable. If on Mar's surface, a bacteria that can metabolize a determined chemical element that exists there in great quantity is placed there, and that provides a sub-product, for example, oxygen, in one or two years we will have transformed Mar's atmosphere, because the bacteria will occupy the entire planet.

The same might have happened when life appeared on Earth. Life expands and occupies all the available spaces, not leaving any empty space. The moment, in which all the available space is occupied, the impulse of life can no longer multiply, because living beings have consumed all of the environment's nutrients.

Conscience

Like the organization that manifests itself, the impulse of life must constantly multiply itself because it is ephemeral, it is necessary to find a solution, or it disappears – “swallowing” itself.

The initial cells occupying all of the exterior space ended up altering the information that came from outside, because the environment had changed. The units occupying all the spaces will alter the exterior environment. At the moment in which plenitude threatens its own life, we find aware “embryos”, the manifestation of the impulse of life, like a “consciousness” of itself.

Oar

The operational enclosure of the initial unit, if not altered, will cause its own destruction, because of an external means without nutrients, which is promoted by itself. As it multiplies into infinity, the external environment begins to drain the nutrients.

This can be the “oar”, which the impulse of life casts away from the “boat's” living organization, in the sense of preserving itself.

In the case of the bacteria on Mars, in two years, all of the Martian soil would be occupied by them. If they metabolized themselves into iron, eliminating oxygen, in two years, there would no longer be iron and the atmosphere would be transformed because of the increase of oxygen. At this moment, the bacteria could no longer survive, and would die en masse!

Solution

The impulse of life notices a situation that transcends the living organization enclosure and arranges itself to resolve the problem. To solve the problem of plenitude, it started using the resources, in which it gathered certain individuals in a new membrane, making it possible for them to have another function, a new function that adapts itself to the exterior environment, transformed by it.

When the impulse of life launches a new membrane, it will do this in a way that may use the waste that the other organizations that are dying en masse leave in the external environment, or the metabolism leftovers that accumulate in the atmosphere. Some cell groups will unite to perform a new function: metabolize what is left in the external environment. The impulse of life will create a new membrane that, in the beginning, will be initially of power and later, physical.

Kingdoms

It is possible that at this first moment of manifestation of the first power membrane is when vegetables appeared. If we use the image of the bacteria on Mars we'd have, in two years, an environment full of dead bacteria, in an oxygen environment, where a group of cells could assume a new function, a new membrane.

There's a great division between the animal living being and vegetables. Some live from what is left from others – this could have been an initial membrane delivered by the impulse of life to confront the law of plenitude, which makes living beings extremely selfish. Anyway, the structure created by the impulse of life, so that one part of the organizations sustain the other, shows how the impulse of life has strength, a determination, and an orientation that creates a group of organizations.

Vegetables and Animals

There's a series of interesting observations resulting from this sequence. The vegetables, creatures that live from the decomposition of organic matter, don't move because they're always where there's an abundance of decomposed organic matter. The animals that search for their nutrients move to fetch them.

The impulse of life has a defined intelligence to structure itself in the place in which it manifests itself. Somehow, it "fights" against the living organizations, egocentric, in their operational enclosures.

Galapagos Islands

As we saw, the evolutionary process hardly happened by chance. The impulse of life is capable of throwing an "oar" outside of its membrane.

A couple of scientists from Princeton University, studying a type of bird from the Galapagos Islands, demonstrated that the length of the beak varied in size over a period of a few years. When the weather was dry for a long time, the birds were born with a short and thin beak, proper for digging the earth. But when the weather was rainy the offspring were born with a normal beak. It wasn't a natural selection that selected the ones with thin beaks for the dry period. Life's own impulse defined what would be better for the species.

Part 2

7

Multi-cellular

The first units to shelter the impulse of life, the cells, were then joined and formed multi-cellular creatures.

The concept that living systems are auto-poietic, capable of repeating themselves, continuously, finds in the multi-cellular case, a complex understanding. The auto-poietic

organization of many orders is composed of other auto-poietic organizations which behave in a similar way.

Maturana left the question open about the auto-poietic organizations from many orders in the definition of what would be the living creatures. In order to differentiate the organizations belonging to many orders, it is expressed by the quality of its membrane. In a second order, the organization acquires a new membrane. They are, therefore, the membranes which will define the organizations of many orders.

What separates the first unit from the external environment is a cellular membrane composed by glycoprotein molecules. The following membranes are not necessarily physical; they can be power membranes.

Group

Unicellular and the First Membrane

Multi-cellular and the Second Membrane

The appearance of the multi cellular living creatures represent an extremely important moment inside the evolutionary process in which the impulse of life went through, from the moment of the unicellular manifestation to man.

The multi-cellular is composed of cellular units which contain a code in their nucleus capable of reproducing the entire creature. If we could get only one dinosaur living cell, we would be able to recreate a dinosaur. Each cell of a multi-cellular creature has the capacity of restoring the original creature while the creature, as a whole, has the capacity of reproducing itself, creating a new creature different from the first. The new creature will have some competitive advantage in relation to the previous one.

Each unit does its job, but all are capable of reproducing the whole.

Membranes

New aptitude

Cellular Membrane

Two membrane	Three membrane	Four membrane
-------------------------	---------------------------	--------------------------

In the multi-cellular creature a similar phenomenon occurs as in the first manifestation of life: the creation of a membrane that has detached the outside from the inside. The creation of the multi-cell occurs when a group of cells unite, being enclosed into a new membrane and creating a new auto-poietic unit. It's a new manifestation of the impulse of life, or an expansion of the impulse through a new membrane involving a set of units to make them more capable for the survival.

The evolutionary process will become an expansion sequence of the membrane that involves the living creatures. At each membrane expansion, a new type of living unit (auto-poietic) will be enclosed in a new operational enclosure.

Power Membrane

The Moment of the Creation of Life

The impulse of life

Exterior Environment

Membrane

The impulse of life manifests itself

The first membrane forms

The first auto-poietic creature

**Ephemeral and unstable
because it's made of protei molecules**

**The impulse of life
becomes enclosed by the membrane**

As I've shown in the case of mammals, there are membranes, which form auto-poietic organizations and are social membranes. This type of membrane, which does not exist physically, may be called a power membrane. They are membranes that exist, manifest and limit an organization, but they are not physical. The limiting membrane in the auto-poietic organizations is always a power membrane that transforms itself into a physical and material membrane.

External Means

Membrane

The impulse of life

Before forming the first auto-poietic organization, before the initial "cell" had a physical membrane, there was a process which, guided by the impulse of life, organized what would be separated from the exterior by the physical membrane. In all the phylogenetic process, this fact seems to repeat itself. The impulse of life creates a power membrane, which expands beyond the existing membrane. The new power membrane, in time, becomes a physical membrane.

The first multi-cell would not have "skin". Probably the first multi-cell had been a group of cells that found a favorable environment. In this favorable environment, a new potential began, making a new membrane extension of the new life impulse to involve the new potential. This would have been done by a certain substance production, proper to the group, recognized by all the units. This same system still persists in man's auto-poietic organization as his endocrine system.

The membrane that limits and determines the frontier inside, which the auto-poietic organization is closed within its operational enclosure is not always a physical membrane as the one of a cell or as the skin of an animal. The membrane is, above all, a power membrane, limiting the extension of the impulse of life in the organization. The frontier limits the knowledge which the impulse of life manifests in that unit. Even before the first cellular unit existed, there was a power membrane that detached what was going to be enclosed.

Virus

This first form of power membrane exists in the virus. It's quite possible that the viruses are the "involution" of cellular units but also, it's unquestionable that even those, not having a physical membrane, have a power membrane. Although the viruses are complex molecules without a membrane, when enclosed inside a cell, they become auto-

poietic organizations. The limiting membrane isolates and encloses the virus in an operational enclosure, because it is a power membrane.

Frontier

External Environment

The impulse of life

Membrane

Second Membrane

The type of power membrane, which we call social, also creates a non-materialized frontier. The multi-cellular organizations are, therefore, auto-poietic organizations which expand their power membrane. We can say the impulse of life manifests itself through the expansion of the power membranes. At each new power membrane, the impulse of life transfers a greater biological knowledge to the closed system inside its operational enclosure.

A mammal can be physically limited by its fur; however its limit, its operational frontier, involves its partner, its offspring and its group. There is, therefore, a power membrane that limits the auto-poietic organization in which the impulse of life is manifesting itself.

In the case of bees and ants, the power membrane is in such a defined form, that it starts to generate different units inside the community. There are carrying ants, scouts and soldiers. In this case, it becomes difficult to identify what would be the living creature or an auto-poietic unit: would it be the ant or the anthill? That's why I prefer to characterize the auto-poietic units defined as power membranes.

Both the ants and the anthills are auto-poietic units, with diverse power membranes. The carrying ant has a limited operational enclosure for carrying food, while the anthill has a more ample operational enclosure.

My liver cells have the operational enclosure of promoting the transformation of the food that comes from the intestine. Yet, my liver has a much more ample enclosure. The frontier is limited by the cell membrane and by the liver's capsule. However, my liver participates in my digestive tube enclosure, which does not have a physical membrane, but a power membrane.

Life's unit manifests itself through the expansion of the power membrane, that continues to involve evermore cellular units, generating operational enclosures that stay locked within several levels. At each expansion, the impulse of life seems to transmit or pass a greater biological knowledge for the new auto-poietic organization from a new order.

This concept that the impulse of life manifests itself, continually, expanding its power membrane, will be very important when we deal with the brain's ontogeny.

(illustration)

A unit kept on multiplying and forming different units to face a new challenge. Later they joined into a multi-cellular creature,

Many units joined to confront the new challenge.

Multi-Cellular Origin

We can imagine that a multi-cellular form is an auto-poietic organization that was formed by the junction of different units, with different capacities, that came to form a multi-cellular group. This hypothesis seems unlikely, because the different units would not be able to modify their forms without a referring necessity to the group. The most probable is that various similar units joined, transformed or specialized themselves for some group task.

Nourishment

The initial impulse of life, enclosed in the membrane, had the necessity to fetch nourishment from the outside, to be able to reproduce. The impulse of life enclosed in the first living creature oriented it to feed and reproduce.

The necessity, as I said before, is the result of the unstable and ephemeral molecule structures in which life installed itself. The first unicellular unit, at a given moment, had difficulty to find nourishment.

Transformations

The impulse of life, enclosed in a membrane, continued a process of regular adaptation at each adverse condition that presented itself. It gradually modified the material structures in which it manifested itself, giving origin to an evolutionary process which undoubtedly denotes intelligence. This intelligence is present in life's first unicellular manifestation, and also, it is clearly manifested in the evolving process that has determined the constant adaptations of these structures (living creatures) to the environment.

Intelligence

One cannot deny that in the impulse of life there is, if not an intelligence, a desire of being strong, that leads it, through extreme difficulty, to constantly improve its modifications, reaching, at the actual evolutionary moment, the formation of the human body.

All the evolutionary processes initiate and go on by the necessity of finding nourishment in the exterior environment, to be able to reproduce another structure that shelters the impulse of life enclosed in its membrane.

Colonies

The first unicellular living creatures, probably, began multiplying in places where there was some abundance of nourishment. This started the formation of true colonies of cellular creatures in these abundant places. The cellular colony service began to feel the necessity to protect the place, avoiding that a great quantity of unicellular creatures be installed there, determining the death of all.

Preserving the Area

At a first evolutionary moment, the first cell multiplied in the proper places, where nourishment was. The impulse of life felt the necessity of protecting such a place, to preserve its manifestation. If such measures were not taken, which would protect the

place, there would then be an accumulation of units, which would consume all the food and all would disappear.

The impulse of life needed to find a way of reaching an equilibrium between the available food and the quantity of units at the place. For this, a group of units was defined which would be compatible with the rich and nourishing environment. If this intervention of the formation of a group or colony did not exist as independent units by the impulse of life, the unicellular creatures would destroy themselves.

The units that were part of the group or colony would have to find some way to protect the place from an external invasion, so that the colony could survive. So the colony could protect itself, there would have to be some communication code between the members that formed the initial colony. This communication started to exist through the secretion of substances which were common to the whole group, capable of identifying the members of the group and, also, capable of giving a new unity to this colony. Such substances, which at a first evolutionary moment made the creation of a colony of living creature possible then, until today, present in the human body, in the form of the endocrine system.

If the impulse of life that was enclosed in the first living creature did not have the intelligence or a very strong desire, it would not be able to transpose the initial cellular membrane to involve the colony as a new unit. If, on one hand, we have an evolutionary process that looks more or less delineated through the physical bodies that have continuously sheltered the impulse of life, on the other, there is also an evolutionary process of the knowledge.

This impulse of life that manifests itself initially in the primordial membrane cell possesses a knowledge in which it exteriorizes, at each necessary moment, to involve the other impulses of life. It gives a clear demonstration that all of these portions of life are one unit. There is, therefore, an evolutionary knowledge process, in which the impulse of life will expand its membranes, creating, continuously, new membranes which will no longer be composed by complex molecules, and will, continually, be creating living creature colonies.

Hypothesis about the Origin

There's a first moment, which is life's creation, the appearance of the first living creature, and we have no means of understanding this moment now. The evolutionary process which we present here starts at this moment.

The first cell capable of feeding and reproducing itself will surely occupy all the spaces where the nourishment will be available in the exterior environment. If the process of the impulse of life manifestation had ended at the first moment, this unicellular creature would multiply itself until it occupied all the spaces or use all the available food, and all the manifested living creatures would have disappeared.

Somehow, the impulse of life that manifested in the first living creature used its intelligence to create a state of harmony between the initial living creatures and the exterior environment.

A hypothesis would be that these living creatures organized in colonies, which made it possible for them to survive in that niche of nourishment, finding harmony with the niche. In this hypothesis, it becomes clear that life's production is manifesting a knowledge that goes beyond the cellular membrane.

Another hypothesis would be that, the living creatures would have first suffered some kind of transformation, which permitted them to use the protei material that begins to exist in great quantity with the cell's death. The impulse of life would have found a balancing point: the creation of new units with an initial metabolism and new creatures

that, somehow, used the proteins broken down by those that went on dying as nourishment. This is a very interesting hypothesis, when we imagine that the impulse of life sustains itself on the planet through the animal kingdoms and vegetables that, in a final analysis, are a manifestation of the impulse of life that complete each other.

A third hypothesis could be that the impulse of life had manifested itself, from the beginning, through a membrane's enclosed structure, with diverse metabolism processes. This hypothesis also makes it clear that the impulse of life manifested in the initial moment had a knowledge that transcended the cellular membrane.

It seems very clear that the impulse of life that one day manifested itself on our planet has a knowledge that has also been manifested in an evolutionary process; as such knowledge becomes necessary for the impulse of life survival. There's an evolutionary process of the material bodies or protei that shelter the impulse of life, and there, one or another evolutionary process through which it has continually manifested its knowledge.

We will show, initially, the evolutionary process of the protei structures which have manifested the impulse of life.

Part 2

8 Membranes

Phylogeny and ontogeny have, until now, studied the living creature as an auto-poietic unit that operates inside a closed system (operational enclosure), limited by a physical membrane or skin. This living creature, which is ephemeral and unstable, tends to be unstructured and will certainly disappear.

Timeline

The living creature must be understood as a dynamic process. It is a time line that starts at its arrival until its death. Each living creature's timeline is locked with all the other living creatures' timeline through which would be the evolutionary process.

The living creatures are structural connections, which move in space and time, impelled by the impulse of life held in each auto-poietic unit. The evolutionary process of the living creatures is, therefore, a constant manifestation of the impulse of life!

The impulse of life manifests itself through the emission or expansion of power membranes to the exterior environment. These power membranes acquire knowledge from the exterior environment and transform themselves into knowledge membranes. The knowledge is then transferred inside the living creature in diverse forms. When the new knowledge is transferred to the living creature, it expands its knowledge and is then involved by a new membrane, which can be called a social membrane.

The evolutionary process is not a sequence of living creatures that have manifested since life's beginning, as phylogeny has tried to demonstrate. The evolutionary process consists of the constant knowledge transference of the impulse of life to the living creatures. The evolutionary process can better be understood as the Origin of Knowledge rather than Phylogeny.

A passing aspect of this dynamic way of understanding the living creatures profoundly alters the concept that we, as Homo sapiens, have of ourselves. Man is a dynamic creature as are other living creatures. Each one of us is a time line belonging to the evolutionary time/space process, which is driven by the impulse of life manifestation – to which we still have no understanding or explanation.

Beginning

The impulse of life, in a way that is not possible for us to understand, generated a membrane that separated a group of protei molecules from the exterior environment, forming an auto-poietic unit, which would be the first manifestation of life. Such a fact occurred due to some extremely favorable conditions for this manifestation.

The initial auto-poietic unit that sheltered the impulse of life multiplied itself, probably through bipartition.

Through plenitude laws, life tends to occupy all the possible spaces. If a new impulse of life intervention had not happened at this evolutionary moment, the initial units would have occupied all the available space, consuming all the nourishment and would have died. The exterior environment would have become a great protei body mass in decomposition. If a new intervention or the impulse of life manifestation had not happened, it would no longer be possible for the impulse of life to move ahead. Life would have disappeared!

This is an important moment to understand that the impulse of life, which manifested itself in the first living creature, has the potential of expanding, in the form that is called the knowledge membrane, to resolve the impasses of its survival. Therefore, the impulse of life does not remain restricted to life's membrane!

There's a closed system (operational enclosure) that makes the initial unit be an auto-poietic being, capable of reorganizing itself. However, within it, is the impulse of life, which has the possibility of expanding its knowledge to the exterior environment and foresees its surviving impasses.

The living creature operates inside an enclosure; his knowledge is limited to the membrane. However, the impulse of life which generated it will constantly be expanding its power membranes, to have the knowledge of how to conduct the constant necessity of the living creature for restoring and reproducing itself.

There's a restricted knowledge for the living creature, restricted to its membrane, making it obvious that the living creature operates inside an enclosure. There is another knowledge, the impulse of life, capable of expanding to the external environment foreseeing and programming the evolutionary sequence of the living creature.

The first living creature does not stop operating inside a closed system (operational enclosure), however, the impulse of life that was generated and made possible, has at its disposal a knowledge of the external environment much greater than can be imagined. The impulse of life is the one that creates the membranes. Who lives is not the living creature, but the life that lives within it!

The evolutionary process of the living creatures is nothing more than a constant expansion of power membranes through the impulse of life. In these membrane expansions, the impulse of life acquires knowledge of the external environment and transmits this knowledge to the living creature so that it becomes more apt to protect itself at each time. The evolutionary process is a constant manifestation of the impulse of life, which will acquire knowledge from the exterior environment and transferring it to the living creature, so that it can capacitates itself to face the challenges of survival.

Colony

At the moment of the initial process, the impulse of life will manifest itself by the launching of a new power membrane which is capable of perceiving the situation that is being designed, with the constant shortage of nourishment and with the protei molecule accumulation in decomposition. At this moment, the impulse of life has to manifest itself again to give the process a sequence.

In places where there's a good supply of nourishment, the impulse of life will join units, as if it had made a survival pact to find equilibrium, so that all of them will not become destroyed by the exhaustion of nourishment. That's how the multi-cellular creatures are formed, which defend the place in which they live and maintain a compatible multiplication process with the environment's resources.

The impulse of life's second manifestation happens through the production of identical substances through all the units. These substances are probably our hormone precursors.

The Plant Kingdom

Still at the initial moment, the impulse of life notices that an important transformation is occurring in the external environment, due to the accumulation of protei molecules in decomposition. Again, if there were not a new manifestation, the evolutionary sequence would be interrupted. Creatures that have the capacity to feed in a different way from the ones that first manifested become necessary. Nothing more logical than to grow living creatures that find nourishment in decomposing proteins.

With this new manifestation, the impulse of life creates a living creature system capable of self-sustaining itself. It is very probable that the plant kingdom appeared then, using solar energy, an abundant source of energy then.

Another possibility would be that the initial units suffered changes so they could start feeding from decomposing proteins. This possibility also, can be viewed as a manifestation of the impulse of life, which in the same way, will create sustainability for itself through the expansion of its power membranes.

Constant Manifestation

The impulse of life manifestation cannot restrict itself to the membrane episode that separated an auto-poietic creature from the environment. If so, there'd be no surviving possibility for the living creatures. The impulse of life is constantly manifesting itself!

The living creatures are necessarily, independent; in another way the impulse of life form could not manifest itself. The operational enclosure that represents the restricted knowledge (confined) of the living creature is crossed by the impulse of life knowledge, which includes all the living creatures. Otherwise, life's manifestation and the existence of living creature's would not be possible.

Every living creature's chain is crossed by a knowledge that transcends the enclosure in which each living creature operates. The evolutionary process is a transmission of the impulse of life knowledge for each living creature. To each one it transmits the necessary knowledge so the next evolutionary step can be given.

Pact

The evolutionary sequence had its beginning with life units. These units grouped around proper places and became multi-cellular forms, initially united by chemical contacts and later, grouping around a new membrane. So the new membrane could be formed, a pact was necessary between all the group's units or colonies. A distribution of tasks was performed within the group.

Again, the impulse of life was necessary to create a pact. The group's rule should be written inside each unit, so that all had the global knowledge of the action of the colony, although each one was in charge of one task. For that, the impulse of life manifested itself again inside each unit's membrane, writing in molecular sequence the group's or colony's rules. Perhaps the basis for our genetic code is in this manifestation of the impulse of life.

This manifestation is so complex that there was the necessity for the impulse's manifestation again inside each unit with the knowledge that was brought from outside. The unit should carry out the task inside the confined system, but each unit should have full knowledge. Besides that, each unit should have the knowledge of all the other tasks of all the units.

Transmission of Knowledge

Each unit possesses a restricted knowledge to its function, but has the potential, or the possibility, of assuming other tasks in the colony, through the new creature's global knowledge. This way, a new living creature is born, which is different from the first living creature because it is composed of a group of living creatures.

The impulse of life needed to be manifested once again within each unit, with the knowledge of the new creature. The acquired knowledge through the membrane's power of expansion, done by life's impulse, will be transferred inside each multi-cellular unit.

Cell

At the instant in which a multi-cellular membrane is formed, what we nowadays call a cell appears. The cell's organelles which as we know, behave as if they had life of its own. Using only the cell's nucleus of a modern multi-cellular creature, it can be reproduced. We have the tendency of analyzing the evolutionary process through what we know as a cell, but the initial units were not like the cells that we know today.

In this manifestation of the impulse of life we find the bases, or the embryo, of the genetic code system and from the formation of the cellular units, which we know today. This moment was the beginning of processes, which continually improved.

Sexual Reproduction

The multiplication of living creatures in the beginning, happened through cellular division, forming two identical units. The transmission of information from one to the other was done directly. With the increase of the living creature's complexity, this type of transmission of information was no longer possible.

To solve the problem, the impulse of life began writing the information on organelles which were found inside the cell (nucleus), which were passed from one unit to the other. The boat (living being) which took in the impulse of life began to have a "log book": the information resulting from the existing experience started to be written in the DNA.

The duplication possibility could no longer happen in the multi-cellular creatures, since all the cell components could not duplicate at the same instance and in the same way. But the living creature did not lose the registration of the period in which it would divide by multiplication. It continued dividing itself in two, but each one with half of the experience information. Two halves would join, forming a new unit: this was the sexual base structure.

This process was efficient, since there could be an exchange of information between the same species. On the other hand, the division with half the information was an excellent possibility for multiplication.

Sexuality

Since cells are formed with half of the information, a strong tendency in these cells of joining the other half appears. The unstable cell that forms the gametes generates a

tendency, inside the enclosure in which the living creature resides, of uniting itself to the other half, configuring the embryo of what will later be called sexual attraction.

An example, so that the idea of the importance of this evolutionary step can be made: in a mammal like a lion, if a medication that impedes the formation of gametes is used, we will be blocking its tendency of uniting itself to the other half, a fact that would modify the lion's entire social group structure.

Social Membrane

With the sexual reproductive appearance separate from the concept of living creature from the concept of operational enclosure and from the organizational concept of auto-poietic operation.

In the unicellular creature there is a limited knowledge about the cellular membrane. The unicellular is an isolated auto-poietic organization from the exterior world by a cellular membrane, and lives operationally enclosed by this membrane.

The multi-cellular creature with sexual reproduction continues being an auto-poietic organization. However, its operational enclosure involves its reproductive partner. There was, therefore, an expansion of the power membrane which doesn't stop being a social membrane in its operationalism.

The operational enclosure of this living creature has no longer a physical limit. It starts having a limit that involves its reproductive partner. The operational enclosure of the sexual creature extends itself beyond the unit, beyond the skin. The sexual animal is an auto-poietic organization, which has an operationalism beyond the skin, beyond its physical limit, beyond its membrane and border. Its operationalism is closed by an intangible limit, which is a power or social membrane. There is the outside, a power or social membrane, and the inside, where two living units are. Its outside knowledge is expanded to perceive what could have interfered in these two units.

The living creature operates in a closed system. In the case of the sexual reproductive creatures, the closed system of its operationalism includes the sexual partner.

Man

This concept will be applied to the animals that have sexual reproduction, including man. The operational enclosure in which man lives does not include, uniquely, the living creature, but, his partner.

Man, as a living creature, operates in a closed system which is not limited to his skin. The impulse of life, which is manifested in man as a living creature expands a membrane, of power or social, and involved his partner.

Man also has many other membranes, either of power or social, which include his operationalism. I'm showing this first membrane connected to the sexual reproduction because it is one of the first membrane expansions, created by life's impulse, which is manifesting itself in the man's living being.

The auto-poietic unit, or the living being, will cover, more and more knowledge of the outside world, because life's impulse that inhabits in it expands its power membranes.

Genes

The union of the multi-cellular units happened through the expansion of the power membrane, which joined a group of cells and began to carry out a new function inside the new membrane. In this situation, each cell began to have a position and a different function of the whole.

The new order had, also, to be written as a pact between the living units. The pact began to be registered in the nucleus of all the cellular units. This pact is what will determine the multi-cellular sequence that might occur when the creature reproduces.

All the cells have within them the registration the group's pact, besides the existing written code of the species. Each cell has, in it, all the existing information, together with the pact which will determine the multiplication sequence of the units to form the auto-poietic organization of multiple rates.

A cell from an auto-poietic unit of multiple rates is also an auto-poietic organization, because it is capable to start forming a complete creature if its nucleus is injected in a gametic cell. The impulse of life is the same that manifests in the multi-cellular creature and in its cellular units.

There's a unique life impulse. However, the operational enclosure of each cell that forms a unity of multiple orders is more limited than the enclosure of the total creature. The unit is united by a power membrane which is manifested in the organization.

Part 2

9

Evolutionary Sequence

After the auto-poietic unit's constitution of multiple rates, which are the multi-cellular creatures, the evolutionary process continued through the membrane emissions to the exterior environment. The extended membranes obtained knowledge that was revised to his multi cellular.

The multi-cellular creature, operating inside a closed system (operational enclosure) and that survived this way had its knowledge extended. The new knowledge already came with the orientation of a transformation to be made in its operationalism.

Having in the external environment a great quantity of certain nourishment, the impulse of life, through the membrane expansion, has this information, transmits it to the living creature and this one adapts its metabolism to the new nourishment. The natural selection really acts in this situation, selecting the capable creatures. But it is not the selection that determines the transformation. It is also not by chance that mutations occur and the most capable mutation is selected. There is an intention of life's impulse, which determines the alteration. Besides that, life's impulse controls the total manifestation so that there is a self regulation.

There's a natural selection between the more capable mutations, which may have some by chance factor, by nature acts in the macro! These are not the motives of the evolutionary process. There's a life's impulse determination that collects information from the environment and transmit them to the living creature. The evolutionary transformations are done inside an expansion knowledge plan, evident in the presiding equilibrium that manifests in the living creatures.

Leveled

(illustration)

The units organize themselves in a plan, so that all have the same exposition to the environment.

The first units that united to come to form the multi cellular probably had a leveled form, so that all could have equality in relation with the external environment. In case they were spheres, the central units would have a bigger protection and less access to the environment. The unities would tend to place themselves in an equality condition, because this would be the planning.

Cord

(illustration)

The units differentiate themselves and form a “cord” which unites them

The first multi-cellular forms of communication were the precursors of hormones. There was the need for greater communication, which was resolved through the formation of a cord which would travel through the entire creature. This cord is the embryo of the nervous system.

The cord units, being differentiated, had the necessity of diving so they could come out in a leveled position. At this instant in which there was a differentiation from the basic units, it stopped having the need of the distribution in level and the multi-cellular folded over itself, forming a tube.

Tube

(illustration 1)

The center cells isolate themselves, forming a neural tube.

(illustration 2)

The multi-cell folds over itself and also forms a tube.

When the initial units began to present morphologic alterations, it happened because life's impulse manifested itself, once again within them, altering its interior, “writing” in their organelles the group's multi-cellular pact. The initial units became cells.

Through the level's curve, the multi-cell assumes a tube format. The level's units transform themselves in cells which will specialize in the contact with the external environment, transforming itself in what would be the skin.

The cells that dived to form the cord have the same cell level origin which will form the skin. This fact can be observed in the ontogeny, because the human embryo demonstrates this evolutionary passage: the nervous system that is immersed in the multi cellular tube has the same origin of the cells that come into contact with the external environment (skin), that is to say, the skin has the same origin of the nervous system.

The knowledge of the exterior environment is immersed in the living creature's knowledge through its nervous system. This fact has profound implications in the knowledge of physiology.

Organs

(illustration)

Internal protrusions will form the internal organs

The internal part of the tube will originate the multi-cellular internal organs. The internal structures are very similar in the great majority of the animals. The human embryo presents, in its ontogenetic sequence, organs such as branchiae and kidneys similar to fish. This fact corroborates the evolutionary process as being a knowledge acquisition process.

Segmentation

(illustration)

When the plaque closes itself the segmentation initiates.

At the beginning of the cellular differentiation, the multi-cell assumes a segmented form. This segmentation must occur concomitantly with the central cord formation which creates the nervous system.

The segments function with a certain independence, which facilitates the multi cellular manifestation. Each segment is relatively independent, but it is united by a cord. The segmentation of man is well represented by the spine.

(illustration)

Cerebellum

Bulb

Appendix

The dislocation improves itself with the appendixes appearance, which leaves from each segment – legs, fins, apprehension organs and tail. The “central” control of movement appears.

Cerebral Pole

The necessity of the formation of a pole in the cord appears so that the segments can manifest themselves as a whole. The first protuberance of the nervous system appears originating the brain.

Members

When an encephalic segmented central coordination pole begins to happen, some present an expansion which will become the members, the fins or the multi-cellular appendixes. The great variety of these appendixes, takes place from the knowledge to solve specific problems in relation to the exterior environment, detected by the impulse of life's membranes.

Nervous System

The initial cord gives origin to the nervous system which will make the communication of all the multi-cellular segments, forming the Central Nervous System. With the protuberances' appearance in the tube's interior, which will originate the internal organs, the cord begins to have the necessity of coordinating these internal organs. Part of

the cord, then, adapts itself to the internal organ's coordination, originating the Autonomous Nervous System.

(illustration)

Central Nervous System

Autonomous Nervous System

Part of the cord will innervate the internal organs, originating the ANS

Encephalic Pole Central Nervous System

Autonomous Nervous System

The cord that unites the multi cellular divides itself.

One part will innervate the internal ANS organs.

The principal will form the CNS.

This same distribution is found in man the internal organs originating the ANS.

(illustration)

Correlations

Anatomy of the autonomous nervous system

Sacra

Cilia Ganglion

Eye muscle cilium

Pupillary Sphincter

Lachrymal glands

Nasal glands

Sub maxillary glands

Parotid Glands

Heart

Stomach

Pylorus

Colon

Large intestine

Ileostomy valve

Anal sphincter

Bladder

Detrusor Muscle

Trigonal

Man has two nervous systems: the Autonomous and the Central. The autonomous was formed before the encephalic pole and became independent. The internal organs function independent from the brain. There's a connection that was formed when a great encephalic pole expansion happened and the twelve cranium pairs appeared. The tenth cranium pair, or Vagus nerve, contacts part of the brain with the internal organs and is responsible for the great number of psychosomatic diseases.

(illustration 1)

Cerebellum

Bulb

Appendices

CNS (Central Nervous System)

ANS (Anatomical Nervous System)

(illustration 2)

Me encephalon

Cerebellum

Bridge

Rachidian Bulb

**The encephalic pole improves:
the cerebellum controls the
movement coordination and
unites the inferior part of the
brain that begins to form.**

Encephalic Pole

The multi-cellular coordination obligated the formation of a center: the encephalic pole. This is the most inferior part of the primitive brain. It corresponds to the controlling of

the auto regulation systems and the movement coordination. In the man it corresponds to the bulb and cerebellum.

Ahsverus' Correlation

The transformations the multi cellular creature goes through are determined by the knowledge acquisition that capacitates the living creature for a new evolutionary task in the constant progression of the impulse of life structured over the ephemeron and unstable molecules.

Ahsverus' legend, the Wandering Jew, is part of the archetypical western culture. Transmuted to the Christian culture, it talks about Jesus, on the way to the Calvary, who had stopped before a powerful Jew and said: "Walk" and Jesus repeated: "Walk". From this day on, the Wandering Jew became immortal and began wandering around the world restlessly.

Our archetypes cannot escape from life's impulse manifestation. Man is not, as imagined, a pensive living creature, independent from his structure and necessities. The legend of the Wandering Jew is a marvelous expression of life's impulse path in its eternal manifestation through the evolutionary process of the living creatures.

Political Correlation

The sequence in which life's impulse orders the knowledge manifestation is always the same. In the multi cellular, many creatures gather to be able to survive. They begin to communicate and assume a leveled form of equality – of tasks and obligations. A differentiation is formed to coordinate the group and the tasks are distributed, with a creature differentiation according to each one's task in the group. The rules are written in each creature's core. The group survives in harmony. They are all different but are able to survive equally.

The Homo sapiens is a multi cellular creature organized in this same order.

When the impulse of life expands a knowledge or social membrane to give a step further in the evolutionary process, man reacts with what he perceives himself as a "Declining Angel". It's as if the evolutionary process had ended in him. Not accepting participation in the group, suppressing that all men are equal, demanding all the advantages in which he was qualified as the whole, and not fulfilling his obligations with the group.

This is the moment that we are living at the beginning of the 21st century. Let's wait to see in which way life's impulse will correct this distortion.

Part 2

10

Reptile

Going back to the evolutionary process sequence of the multi-cell, the brain begins to appear with a bulb and the cerebellum, which coordinates the auto regulation and the movements. With the continuation of the necessities, resulting from the demand of the external environment, the impulse of life's new multi cellular beings with specific capacities in relation to the environment. Some run faster. Others smell more. Others accommodate

in determined nourishment niches, generating a diversity that, on the other hand, does not stop being controlled by life's impulse.

The great necessity of new coordination generates an important increase of the encephalic pole, which will transform itself in what we showed at the beginning. The reptilian brain or primitive brain.

Cranium Pairs

When the encephalic pole expands, the first part of the segmented multi cellular, which is giving support to all the encephalic pole structure, can no longer stand the task.

Twelve new segments form. These new segments form after the creature is structured, and are already born with the specific function of coordinating actions of the own encephalic pole.

These twelve cranium pairs coordinate the encephalic pole manifestation, which coordinates the entire creature. The impulse of life, somehow, or by chance, chose number 12, which is a number which repeats in great quantity of manifestations in our culture.

Meanings

It is written in the reptilian brain a series of automatic behaviors. The vision of certain images should be followed by an escape. An identification of a certain smell ordonates the reproduction sequence and a series of information that should be registered in what is identified as the cerebral cortex. The reptile brain requires a lodging of necessary information for its survival and which should be registered in the cortical region of the brain.

MacLean

Observing the reptile when more advanced, we see that it already has an incipient cerebral cortex.

The division in three evolutionary stages is unquestionably, MacLean's great contribution to the brain study. About the identification of these three units, they seem not to be well defined. The brain is an extremely compacted organ inside the cranium box, which determines a great deformation in its structural orders. The cortex is as if it were a sheet of paper well wrinkled to fit in the cranium box and that's how the whole brain is. The great division between the primitive, intermediate and superior brain, is what should orientate the study of this organ.

In regards for the exact delineation of each unity, we cannot yet have a complete definition. The primitive brain would be the reptile. Over him there was placed a mammals brain and at the last phase we have the superior brain. The distinction must be principally functional and not of location.

Dinosaur Era

The appearance and disappearance of the great reptiles has been one of the great mysteries of the evolutionary process. Many hypotheses have been conjectured and, lately, it is believed that the end of the great reptiles had occurred due to a cataclysm provoked by the shock of a great meteor, causing climatic alterations which made it unbearable its survival. But if the meteor cataclysm had not happened, there would have been another transformation to cause the extension of the great reptile.

Mutation Point

(illustration)

cerebral cortex, where

Primitive Cerebral Cortex

composed of nucleons

Cerebellum

functions, and of the

movement.

Nucleons

Reptile

**The reptile has a little of a
feelings are registered.**

**The reptilian brain is
which command the vital
brain which coordinates the**

With the reptilian brain formation, the first stage of the evolutionary process of the human brain is closed. The main characteristic of this living creature is the fact that life's impulse can transmit, from one to the other living creature, the necessary information for its survival.

Until here, life's impulse behaved the same way: sending knowledge membranes, noticing the necessity in relation to the environment, transmitting the information to the living unity's inside, and orienting its modification in the sense of making it fit for surviving.

The new living creature is already born bringing the necessary information. The outside environment's stimulus is sufficient so that nervous connections are activated, which would be the reflexive actions. The reptile is born with its knowledge written in its nervous connections; it has no necessity of learning. The alligator is already born an alligator.

In the reptile, life's impulse reaches a final evolutionary stage, in which it is capable of transmitting to its decedents all the information necessary. The reptile just grows: it has no necessity of any other learning except the activation of its connections, done automatically through the environment's stimulations.

In the reptile, the possibility of the evolutionary genetic transmitting to its descendents outwears. The living creature, in its reptile stage, has all the qualifications of surviving, within the possibilities followed by its species. The alligator can no longer develop. There are diverse types of alligator, because each adapted to the determined external environment. In the reptile, the possibility of acquiring knowledge outwears. The only competitive advantage that one specimen of reptile can obtain in relation to the environment is to increase its size, maintaining the same structure.

The End of a Cycle

The moment in which the impulse of life created the reptile, the possibility of raising living creatures capable of reproducing themselves, transmitting information through the genetic code, was over. The growing of these creatures, in size, was a detour of the

evolutionary process that would be corrected ahead. If the meteor had not occurred, the big reptiles would have disappeared otherwise. The impulse of life needed to create a new way of promoting its manifestation.

When the big reptiles disappeared, the impulse of life was already experimenting with a new type of knowledge transmission, which originated the mammals. It was not the disappearing of the great reptiles that propitiated the appearance of the mammals: they were already being tested because the reptile had reached the end of its evolutionary potentiality.

The capacity of writing on the DNA molecules was wearing out. The reptile represents the manifestation of the impulse of life in its most complex form, while it can be reproduced from one to the other living creature. The impulse of life needs to create new form of transmitting knowledge from one to the other creature, different from the common multi cellular pact.

Man/Reptile Correlation

The main characteristic of the reptile is the reflexive action written in its multi cellular pact, which we could call primitive instinct. Man brings inside himself the reptile which was his grandfather. In certain conditions, the manifestation of this reptile can be observed. As when, for instance, the man takes attitudes as a creature that reacts, automatically, to an environment stimulus, without considering the other components of his brain.

When man is attacked, he shows a reflexive reaction which is written in his structure, or in his multi cellular pact. Such defense attitudes are commonly observed and characterize an exclusive action of the reptilian brain. Man, as he becomes more egoistic and more egocentric, assumes each time more a reptile behavior. The selfishness and the egocentricity tend to cancel the action of the other two brains.

The world of today is making man become more egoistic and egocentric. Man, as every living creature, operates inside an operational enclosure system. When life's impulse is not expanded by its power membranes, which in this case would be the feeling and the reason, it makes his operational enclosure closed in the reptile man.

The reptile man is the man capable of destroying his fellow man for a dispute of power. Capable of destroying his group, or even his child, in a dispute for food. Even though this might sound terrible, every day we are faced with this kind of reptile man.

It is unconceivable that a living creature that has a triple brain be using the most inferior. This does not happen in nature. It's unconceivable that a living creature does not use his evolutionary competence and qualifications. This fact just occurs because of the distortion provoked by reason. The capacity of thinking can, unquestionably, distort the real necessity that cancels the action of the other components of the brain. That's why man's partial manifest can be crueller and predator than all the other animals.

The form in which man has been raised has not privileged the emotional development. In the emotional lies the structure of reason. The neuron cortex was "mounted" on top of the emotional brain. The pure rationality is capable of creating distortions, as the one which can be observed in the cruelty manifestations of the reptile man, when the natural order in his upbringing was despised. The pure rational man can degenerate the reptile man.

In extreme situations of deprivation, man can behave as the reptile that lives within. In the concentration camps, in certain prisons and in great delusional situations, man reacts to the external world, uniquely, with his instincts, as if he were a reptile. The egoistic or egocentric man, in the deprivation situations, easily, manifests the reptile that is within. While the altruistic man, idealist, maintains this reptile under control.

In the extremely depriving situations, the egoistic man is capable of attacking his child to take the food from him. This fact does not occur with mammals that already have a notion of group.

The idealistic man is capable of giving even his life for a principle that can be religious. An example of this altruism is registered in Austin's concentration camp, when father Maximiliano Kolbe offered to be traded by a family father condemned to death.

Man's reptilian manifestation is set off, as I said, by extreme states of deprivation. However with man, what happens is that when the extreme deprivation occurs, it may not be the reality but a creation of his mind. With the acceleration of the evolutionary process, the genetic transmission was no longer capable of passing to the descendent the necessary connections for survival. The information of the predator was passed from one to the other creature; but with this acceleration the predator started to show up very suddenly, not giving time for the information to be transposed to the genes.

A multi cellular creature recognized the predator by a determined characteristic written in its nervous system. Such as, for instance, a certain roar, smell or behavior which, identified with the predator, made the animal run. A certain type of smell or aspect, however, was identified as food. These identifications were directly written in the reptile's nervous system. It was not a conclusion which would be made, as happens, by the creature that has an intermediate brain.

With the acceleration of the adaptable species to various environmental situations, non identified predators began to appear through the nervous system and there was the necessity of the transmission of knowledge.

Part 2

11

Mammals

The reptile is at the top of the living creature's evolutionary process in which the knowledge transmission is done, uniquely, through the genetic code.

(illustration)

Superior Brain

**Emotional Brain or
Intermediate**

Primitive Brain

When the impulse of life structured the multi cellular, it made the knowledge transmission be written on the cellular units of the organelles. This way, it was transferred to the descendants. This "writing" was made through a proteinic language, using a few nucleic acids, wrapped in a helicoid structure. The genetic transmission of knowledge reaches, in a reptile, the climax of this evolutionary process.

Each reptile specimen has structured its genetic code adaptable to a specific surviving situation. The tyrannosaur is already born a tyrannosaur. A dinosaur is born a dinosaur. The qualifications and the knowledge for the specimen's survival are all registered in their genetic codes.

Membranes

The impulse of life, which lives in the multi cellular, was expanding its membranes constantly to the external world, obtaining knowledge of the exterior world. This was brought to the inside of the multi cellular creature, writing on its genetic code. These are the knowledge membranes which we already showed before. The knowledge membranes, when codified inside the organelles of the multi cellular units, transform themselves into a social membrane, which began to limit the operational enclosure of the multi cellular creature.

The initial unit has its operational enclosure limited to its cellular membrane. The multi cellular has its enclosure limited by the social membrane that involves all the group's units.

The living creature, with sexual characteristics, has a social membrane that involves its partner. The reptile has a social membrane that involves its species and that is registered in its genetic code, determining the specific morphology of the species.

Necessity

The reaction of a determined stimulus to the external world has a definite response in the neuronal circuits of the reptilian brain. The reptile is already born knowing what to eat, where to find the food and how to defend itself. All of this is defined in the reptilian brain structure. The encephalic pole, in the reptile, (reptilian brain) represented an evolutionary moment of greater importance. The impulse of life kept launching knowledge membranes, identifying niches in the exterior environment, transmitting to the reptiles inside the knowledge of these niches and adapting the reptile to this opportunity. This way, the social membranes were created which generated the species.

The reptilian brain permitted a species explosion; each one adapted to the knowledge more or less specific to its habitat. The reptile's structure was totally transmitted by the genetic code. Due to this fact, the physical form of the species became somewhat plastered and, not having great variation possibilities, was left as the best possibility its size growth, which generated the big reptiles.

Knowledge

Due to the appearance of a great diversity of habitat and to a constant and rapid modification of this habitat, the knowledge transmission by the genetic codification became an insufficient resource to give a continuation to the manifestation process of life's impulse through the living beings. The habitat varied and modified with such velocity that the process of transmission of the knowledge purely through the genetic writing became obsolete. This moment coincided with the end of an era of the great reptiles.

The impulse of life needed to create a new evolutionary artifact through which the living creature could acquire knowledge which was not specifically transmitted by the genetic code. There was the necessity of creating an artifact or an organ, inside the encephalic pole, which would permit the living creature to learn.

This new artifact would be a set of nervous cells organized or mounted in such a way that the experience of the living creature would activate determined circuits. The activation of these circuits would represent a new knowledge, not genetically transmitted, but acquired through experience of the own multi cellular unity.

The Evolutionary Artifact

This artifact or new organ permitted a marvelous flexibility so that the living creature would acquire new knowledge from the external environment or from his habitat.

This new organ is the emotional brain, which appeared in the mammals to solve a surviving impasse resulting from the great variety and the constant and rapid habitat change.

The impulse of life created an artifact, the emotional brain, which is nothing more than a “neuron net” arranged in a special way. The artifact is genetically transmitted. The activation of this “net” will be done after the birth and will be activated by the information that the living creature receives from the external environment.

Example

(illustration)

Emotional or Intermediate Brain

Primitive Brain

Not Activated

Trained

The contact of the baby animal’s mouth with its mother’s tits makes the mammal respond with a group of actions which is the sucking act. The calf which is taken during one hour from the mother’s contact and later does not play with the others is an example of this kind of knowledge. When licked by the mother, brain circuits are activated which permit the special playful behavior later on.

New Potential

This evolutionary artifact permitted the living creature to have its adaptable capacity incredibly enhanced. Each living creature began to have an individual knowledge from the external environment and small alterations of the environment began to be incorporated, in an immediate way, to the new knowledge.

When new possible surviving “niches” appeared, it began to be noticed and transmitted in a rapid way to the descendents. The alteration form of a food could be noticed. The places that presented better conditions of survival began to be identified in a much faster way and this was transmitted to the descendents.

This new acquired qualification by the new artifact permitted that the mammals survive the great environment transformations and came to be the top of the evolutionary process after the great reptile era.

Emotions

(illustration)

Basic Emotions

Surprise	Happiness	
Anger	Emotion	Acceptance
Aversion		Fear
		Expectation
	Sadness	

Initially, this type of acquired knowledge, from the external environment by the emotional brain, is very limited.

The “neuron web” from the new artifact, when activated, acquires knowledge. The new environment information can be in agreement or disagreement to the new acquired knowledge. Basically this new artifact limits itself to identifying if the alterations of the environment should be accepted or rejected.

The new artifact does not have an external environment impression as the reptilian brain does. The reptile knows that it should run away when facing of such information. The new artifact functions in a different way. It has a special circuit of neurons activated by the knowledge acquired. It can only tell if the received information is or not in agreement with the sequence of its activation – the heart has reasons that the reason does not know.

The neuron web of the emotional brain does not act in the same way as the reptilian brain. In the reptilian brain, the information that arrives is not processed by the neurons as if they were computer chips. The emotional brain responds through flashes and after the information processing, responds in a patterned way of how to accept or reject.

The new artifact that appears in the first mammals becomes more complex and sophisticated, being identified later on, as the emotional brain. With the evolutionary sequence process, this new artifact gradually became more complex. It began a participation in the selection of partners to reproduce, identifying the most adequate and better qualified for surviving. Besides rejection and acceptance, associated circuits were being created to fear. This generated reactions like fear and anger, expectation, surprise and, much later, sadness and happiness circuits.

Adaptation

The new artifact decided a great surviving problem and permitted a new species explosion, with a much bigger variety of inferior species, which were adapting with bigger velocity.

The mammal could adapt, rapidly, to different habitats, without the necessity of great transformations in the species structure. The bear can be an example of this potentiality: the polar bear, the steppe bear and the mountain bear. The great variety of felines is another example of this mammal potentiality of adapting without a radical structural change.

Straight Line

The evolutionary process cannot be seen as a straight line. The great mammals that we know are not the same from the evolutionary sequence. Man is a result of the evolution of a sequence of living creatures that are to us totally unknown. He does not descend from the ape. It's impossible to find the links of this evolutionary chain. However, it is possible to identify the phases which he went through as reptile and mammal. The reptiles and mammals did not disappear and continue adapting to the new habitat.

It's possible to make comparisons with the living creatures that we know. The alligator of today possibly has some of the acquired knowledge, which is not proper of a reptile, from the evolutionary straight line process.

The species developed in group but, for didactic reasons, I separate here the evolutionary steps until the moment in which life manifested itself in man.

Equilibrium

The species distribution among the mammals followed the same equilibrium law that rules all the manifestation in the impulse of life. At the beginning of the manifestation, an animal and vegetable group was formed which self sustained themselves. In the mammals, again this law was imposed; a predator and herbivore group was generated. This law should have been expressed in the first reptiles.

The predators had their multiplication contained by the offer of prey and the herbivores had their multiplication expanded so that there would be equilibrium in the impulse of life as a whole. The herbivorous mammals, somehow, reproduce in a greater quantity than the carnivorous do.

In this equilibrium we can, once again, notice how the impulse of life manifested itself as a whole and not individually.

Solution

The new artifact that appeared in the mammals, corresponding to the emotional brain, solved the impossibility of the knowledge to continue being transmitted from one to the other living creatures only through the genetic code.

The new artifact is transmitted by the genetic code; however, the acquired knowledge is not. Each mammal began to have a type of self knowledge, resulting from the transmission of information in its new artifact, the web of neurons, which is the emotional brain.

Problems

The competitive advantage that appeared by the acquisition of the emotional brain brought two new and great problems:

1. Education: the necessity of a living creature to have to teach another one. The mother had to "educate" her offspring. This necessity did not exist in the reptiles.
2. Individuality: the knowledge acquired by the living creature was an individual property which guaranteed the exclusiveness of being a unique living creature.

Enclosure

In the reptile ends the evolutionary sequence in which the knowledge is transmitted only by the genetic code. A new phase of knowledge acquisition begins, which happens outside the maternal organism.

The living creature's knowledge is limited to the enclosure in which it operates. To understand the transformation that occurs with the appearance of the mammal, it is interesting to review the evolutionary process of enclosure knowledge in the living creature.

The mechanism of operational enclosure, presides the auto-poietic organization of the multi cellular creatures. Since the beginning, the living creature operates inside an operational enclosure mechanism: it, living creature, and the external environment.

The enclosure expands as the impulse of life creates power and social membranes which will involve other living creatures. The living creature with sexual reproduction has its operational enclosure expanded in a way which includes a sexual partner. As I said before, the searching partner impulse results from the creation of a gamete, which is a cellular unit of the group, with half of the codified information, which generates a desperate search for the complementation.

This type of membrane is present in the reptile. The reptile's operational enclosure involves its partner, and in an indirect way, its species. All this involvement is written in its code. The reptile operates inside a social membrane, which involves the partner, and

another that involves the species in which it belongs. The enclosure, even inside these membranes, is transmitted through the genetic code.

The reptile has neither an option nor choice in its developing process. It's already born with its complete development. All its knowledge was transmitted by the code.

Operationalism

With the appearance of the mammals, and the emotional brain, the living creature begins to have its own knowledge, which will make its social membranes different one from the other individual of the same species.

The tiger, trained in captivity, may have a social membrane different from the membrane of the one raised in the jungle. However, the alligator raised in captivity, has the same social membrane of the one in the wild. The captive tiger can very well live as a pet because its social membrane was altered through training. But, when it becomes very hungry, the brain's emotional social membrane will disappear and it will then operate inside his reptilian membrane, attacking man.

Correlation

Man does not differ a lot from the other multi cellular creatures. It is a reptile, whose emotional and rational structures were mounted upon. The reptile man has his operational enclosure limited by its genetic codes. The mammal man has his operational enclosure limited to a personal knowledge acquired by the emotional brain's activation – activation which is generally deformed.

Training

In the mammal the acquired knowledge appears on the outside of the mother's body. The new social membrane that appears involves the young. This new membrane is not totally written in the genetic code.

The necessity of the mother in training the child generated contrary activity to the tendency until the valid moment in the evolutionary process. The living creature was no longer acting according to a genetic obligation. It began acting according to the social obligations. The social membrane's were no longer imposed by the form, and began to be imposed by the acquired knowledge.

The mammal is much more qualified than the reptile as a living creature. For such qualification, it needed to have its development completed outside the mother. This fact was only possible through the creation of the emotional brain. When activated by the maternal care it generated the necessity to act in a similar way with its young.

Sacrifice

The mammal has to sacrifice for its young and species' survival. This sacrifice was done by the reptile through genetic imposition, and began to be done through an acquired knowledge.

The turtle's effort to lay its eggs on the beach is a genetic imposition. The bear's sacrifice that almost provokes in her inanition to feed its cubs happens because of an acquired knowledge.

The reptile and mammal's sacrifice is manifested in man. The care for the young is first a genetic imposition. Further, an emotional imposition, and only in a later stage is it rational.

Individuality

The other great problem that the appearance of the emotional brain raised was the strengthening of the individuality.

The mammal begins to have a knowledge that is his own, which differentiates it from the others of its species. This knowledge limits itself to the activation of the neuronal web. It is already mounted genetically and is also limited by the type of web response which occurs like a flash. A great variety of information can determine a same response.

This individuality confronts the reptile's genetic individuality. The reptile operates inside its own individualistic genetic enclosure. The mammal will operate inside its own individualistic enclosure, having an individualistic impression of the world that is its own. Different from the reptile that has an impression of the world like all the others from the same species.

Correlation

Man, as a multi cellular living creature, carries the reptile's genetic individuality which impels him in the surviving struggle. This can be observed in extremely restricted cases when the emotional rational brains can become indifferent.

This individual duality impregnates man: with his rational brain, he distorts and creates false situations that can, many times, drive him to self destructive attitudes. He is capable of self destruction, convincing himself that he is doing it to save other creatures.

Holembach

In the Armed Forces Hospital, I had the opportunity of working with Sergeant Holembach. One day, taking a walk at Brasilia's zoo, a teenage boy entered the coypus' pen. When Holembach saw that, he entered the pen to save the boy and was bitten, dying the next day. When asked why he had done such a heroic act, Holembach said he couldn't bear to stand still and watch the boy die. Today the zoo is named after Sergeant Holembach to honor his act.

In this example, the reptilian and genetic individuality were surpassed by the mammal's emotional individuality and the reasoning was put aside.

Holembach's emotional brain activation was done, in such a way, that the acquired knowledge forced him to an irrational act. Man's religiousness is associated to this transition from reptile to mammal!

Conclusion

The emotional brain is an evolutionary artifact which the impulse of life used at the moment in which the possibilities of knowledge transmission by the genetic code had run out.

The "neuron web" of the emotional brain is transmitted by the genetic code. However, its activation happens from an individual knowledge. The emotional brain acts through a flash. The information is weighed and generates an only type of response. This response, at the beginning, is acceptance or rejection. During the evolutionary process sophisticated itself, reaching what today we know as basic emotions.

(illustration)

Mammal

**Primitive cortex
when**

**The Emotional or Intermediate Brain appears
the possibility of purely genetic transmission**

ran

**Reptilian Brain
neuron**

out. Below the Primitive Cortex there is a

Intermediate Brain

Web like, in the middle of the Reptilian Brain

Part 2

12

Man

(illustration)

Superior Brain

Frontal lobe

Primitive Cortex

Reptilian Brain

Intermediate Brain

Man

The Superior or Rational Brain appears in man as a new neuronal web, above the Primitive Cortex, over putting the Primitive Cortex and practically creates the Frontal Lobe.

The group of forests, fruits and bird eggs, generated a new proper environment for mammals. Some species of mammals climbed trees and started to live relatively comfortable in the tree tops of the luxuriant forests, feeding from leaves, fruits and bird eggs.

This group presented a surprising development because they did not have big predators. The reptiles that became birds did not have much capacity to compete with the mammals.

Down on the earth, the mammals developed into two big groups: the carnivorous and the herbivores. This dichotomy, which can be observed in all the evolutionary moments, has the finality of bringing the species equilibrium.

Proto-Man

The impulse of life, as we saw, intervenes balancing the reproduction of these groups. The herbivorous multiply a lot and the carnivorous less. During the whole

evolutionary process, the impulse of life was always promoting this balance so that the living creatures would not destroy each other.

The mammals that climbed to the tree tops were the proto-man. This proto-man was also the proto-monkey or proto-simian.

The proto-man, in the security of the tree top, adapted to this type of surviving, developing a very good eye sight, to identify fruits and nests and abilities from the arm appendices and legs, to shift between the tree branches.

Regulation

Since the proto-man had no efficient predator, which could determine the velocity of its reproduction, the impulse of life, itself, was forced to control its growth by the quantity of food available, different from the other carnivorous and herbivorous mammals, which regulated themselves.

The regulation of the reproduction cannot be, uniquely, attributed to the natural selection. When there is a lack of food, more offspring die and in a short time the impulse of life goes into action diminishing the fertility of the species until equilibrium is reached. Likewise, the impulse of life increases the fertility when there is an abundance of food.

The impulse of life has the perception of the unfavorable environment through the social membranes, and transfers the knowledge to the inside of the living creature, creating conditions of equilibrium. The natural selection alone does not control this equilibrium. The impulse of life intervenes, regulating the reproduction. If it weren't this way, many elephants would be born and many would starve to death. The impulse of life reduces the reproduction so that equilibrium can be maintained.

Part 2

13

Emotional Catapult

(illustration)

Cortex

Intermediate Brain

Primitive Brain

With the mammal's emergence, there was a new explosion of the species and sub species, something rare among the reptiles.

The end of the big reptiles, also, coincided with the great alteration of the vegetable world which, after big destruction, manifested itself with diverse characteristics. New forests in favorable places appeared, like in the middle of the African continent, and new and variable species of fruiting trees. It also coincided with this moment of great dissemination of the arboreal reptiles, which would later be transformed into birds.

Proto-man

On earth the mammals developed into two great groups: carnivorous and herbivorous.

The carnivorous fed from the herbivorous, creating a self adjusting system: the animal and vegetable kingdom. The impulse of life acts through this type of dichotomy, which can be observed in all the evolutionary moments, and has the purpose of promoting the equilibrium of the species.

Special Condition

The proto-man who lived on top of the tree was a very special mammal.

The proto-man mammal lived in the top of the trees in a relatively calm state and had his restrictions controlled by the manifestation of the impulse of life. He could observe from the top the predator attacking its prey, without his participation in the scene. In a general way the proto-man became a big observer. He never had a great necessity to fight for his food, and depended almost exclusively of his identification with the environment. His fight for survival was regulated by life itself.

Correlation

Man has an ingrained concept that he's a protected creature by divinity. This concept is in the scriptures, under the concept that he lived in paradise. The proto-man is truly the image of Adam living in paradise.

Fall

With the disappearance of African's big forests, the proto-man fell from the tree. In evolutionary terms it was, from one moment to the other, that he saw himself in the African savanna. He was totally disqualified. He was not carnivorous, had no aggressive organs and was not an herbivorous that had adapted to such a situation. Had no way of defending himself from aggression and no velocity to run. Could not attack or run! He was a disqualified creature, with different characteristics from the others: a tendency of maintaining himself erect and very developed senses.

Another factor that should be taken into consideration would be the great quantity of this type of living creature. Since they had developed at the edge of the equilibrium of the other mammals, probably they were in a big number, with many sub species.

Competitive Advantages

The proto-man, who fell on Africa's savanna, had one competitive advantage which was not very common among the living creatures. Because he had no predators and lived in a favorable habitat, he was not submitted by the species dichotomy which self regulated them. He was regulated, very directly, by life's own impulse. Fallen on the savanna, he was a living creature with uncommon characteristics, like the erect tendency of walking and a good sensorial ability. It was natural that the impulse of life would use these characteristics.

Blessing

This evolutionary step explains why man has, impregnated in his knowledge of the world, the divine protection. The surviving of his being so disqualified in Africa's savanna only happened due to the direct intervention of the impulse of life.

Man's creation was, somehow, a blessing of God or a divine manifestation of the impulse of life.

Review

The impulse of life, which since the beginning of its manifestation, launched power membranes to the exterior environment, capturing the new possibilities and transferring to the inside of the living creature the new knowledge, had run out of possibilities of purely genetic transmissions in the reptiles. The reptile has written in its cellular units all the genetic code of the living creature. Even more, it has written in its central and autonomous nervous system all the life experience, since it was one unit. The reptile knows all that happened during the evolutionary process!

Having run out of the purely genetic transmission path, the impulse of life created an intermediate or emotional brain artifact. The intermediate brain resource was very limited, structured to respond with a flash, and through this flash acquire an individual knowledge from the external environment.

To save the proto-man who fell on the savanna, the impulse of life would have to create a new competitive artifact which would broaden the knowledge acquired by the emotional brain.

Salvation

Why would the impulse of life try to save this creature?

- 1- The impulse of life presents a tendency to expand the knowledge, creating membranes that capture information from the external environment and transfer the knowledge to the living creature
- 2- The proto-man was a creature with a tendency to walk erect and with well developed senses, controlled by, very closely, the impulse of life for not having predators.
- 3- The appearance of the mammals permitted the appearance of a self living creature's knowledge. There was already a certain individuality, which, in the case of the arboreal mammals (proto-man and proto-monkey) functioned as an incipient conscience.
- 4- In the mammals, the intermediate or emotional brain, when a danger flash is identified, commands an immediate action coordinated by the cerebellum or by the reptilian brain. This type of flash (emotion) was in part resulting from an acquired knowledge by the activation of the neuronal web from the intermediate brain. In the arboreal mammals, there was a different development in this artifact, permitting the appearance of a varied emotional flash. This was because there were no important predators.
- 5- He had a special emotional brain, for being outside of the great mammal fight; was neither carnivorous nor herbivorous. Being arboreal, he could observe the fight of the surface and guess what would happen.
- 6- The brain hemispheric duplicity could have been an important factor for the incipient conciseness brain to appear. The emotional brain acts responding to a flash, informing the action to the brain's hemisphere. The existence of the emotional brain proved conscience duplicity: I on the right side, I on the left side and the image from the outside.

Pause

The interaction from the emotional brain with the immediate reactions in the mammals was fast and efficient. Seeing a predator, he ran immediately.

The mammal had the neuron web of the emotional sensible brain by the acquired knowledge through training. Besides that, the web had its own structure of the species, which identified the predator. Identifying the external situation, the information passed

through the neuron net of the emotional brain and this one commanded a flash of how to attack or run away. The reaction was coordinated by the cerebella and transmitted to the body, resulting in an immediate response: aggression or escape.

In the arboreal mammal, because of the lack of predators, this reaction was not as immediate, permitting a lap between the flash and the action. This lap permitted an intervention, consequence of an acquired knowledge.

Example

In face of a stimulus, the living creature should have a reaction. What was being stimulated could be a predator or a food opportunity. Imagining that the stimulus is information similar to a tiger, the reptile, which already brings written in his brain what a tiger should be, runs just with a little information which is in agreement with its evolutionary and genetic experience. The mammal processes this information in its emotional brain, which analyses and commands a flash, which might be of escape or aggression. Since the emotional brain acquired some knowledge with the training it had when activated, it is capable of recognizing a new opportunity of food which has, for instance, the stripes of a tiger being used to camouflage the prey.

The arboreal proto-man, for having few predators, was able to have a bigger lap between the flash and the action to be taken. This bigger lap differentiated him a lot from the other mammals. It permitted that he weighs his own knowledge which, at this moment, was registered in this primitive cortex.

Anticipation

The arboreal proto-man watched from the top the fight for survival of the remaining mammals and could foresee the result of a resistance for survival. He could see that the tiger, after being satisfied, did not attack the prey; that the herbivorous could run from the tiger if there was space and a series of situations that permitted him to foresee a result.

This possibility of guessing or anticipating what would happen was the embryo for the development of the thought. Such mammal could use its knowledge to determine its next action.

This type of knowledge is the anticipation. The anticipation is the basis for our reasoning and of our thinking process. The arboreal proto-man could use his acquired knowledge, with the observation, and guess or know, acquiring then a new knowledge.

The lap between the flash and the action, and the incipient anticipation, were the resources used to develop a new evolutionary artifact, the human brain.

Big Membrane

At the moment in which the proto-man fell on the savanna, the impulse of life launched a membrane that covered a new and adverse environment to this proto-man.

The evolutionary process had reached a high degree of development and needed a new model of living creature that covered, in its power membrane, all of the impulse of life. This power membrane covered almost entirely all the manifest of life; all of the multi cellular was involved in the question of the proto-man's survival.

He was neither herbivorous nor carnivorous. He needed weapons to attack and defend. He needed canines and molars. He needed a larger digestive tub, as the herbivorous, and short as the carnivorous ¹. He needed to run or escape. And he needed shelter.

The solution could not be given by genetics, like the reptiles; neither by the development of the intermediate brain which responded in a flash. It was necessary that the proto-man have at his disposal an artifact which could give him more knowledge.

1 The carnivores have a shorter intestine because of the fast putrefying food cycle and to the easier absorption of the proteinic material. The herbivorous have a longer intestine because of the difficulty of observing the vegetable food cycle.

Evolutionary Knowledge

All evolutionary knowledge was written in the reptilian proto-man. Each experience from the impulse of life was written on the reptile's genes. When the reptile adapted to a determined niche of survival, this experimentation was written in the organelles of all of the reptile's cells. The reptilian proto-man is like a big book where all the experience of the impulse of life is written.

With the resource of the individual knowledge furnished by the emotional brain and the knowledge perceived by the anticipation, the impulse of life created a new neuronal web capable of using all the evolutionary knowledge.

Read the Book

For this knowledge to be used it would have to be transformed in information – the information is registered by the nervous cells, the neurons.

The impulse of life should create a new web of neurons with the possibility of reading man's reptilian book. The new artifact would have to have conditions to register this evolutionary experience at the moment it became necessary. It would not be possible to decode the entire evolutionary experience book, which was written in the genes in a neuron web.

(illustration)

Growth of the Neo Cortex, creating a storing mechanism.

Anticipated mechanism

The new web would have to have potentiality to understand the entire book, but would only have access to the pertinent information in time.

New Artifact

The impulse of life created a new artifact with a great quantity of neurons which, interlinked by their axons, form an infinitive quantity of combinations, making it possible to codify any evolutionary experience that is registered in the reptilian man and, specially, in the reptilian brain. This way the neo cortex was created: neurons gained.

Neo Cortex

Basically, the neo cortex, which identifies the man and differentiates him from the other mammals, is composed of the adding of two great neurons.

1. The great development of the front lobes, which practically appears in man. All the structure of the front lobes was created at this moment. The frontal lobes are the part of the human brain that really makes the difference from the proto-man's arboreal brain.

If we analyze the simians of today, we'll see that there is a great difference, exactly, in the skull's structure where the frontal lobes are. The frontal lobe is responsible for the anticipated mechanism, which appeared incipient in the proto-man and defined itself in man.

2. The adding of the neurons in the other cortical lobes of the mammal. The other brain cortex lobes, which already existed in an incipient way in the reptile and, a little more developed, in the mammal, were, at this moment, very enlarged. There was, this way, a great gain of neurons in the other brain lobes, which came to configure the human brain.

Genetics

The necessary genetic change for the creation of the new artifact was not a very big one. There was the formation of a neuronal web above the emotional brain. This is, practically, the genetic difference between man and the big apes. The difference between the genetic codes of the chimpanzee and of man is inferior to 1%.

The genetic alteration is not big, but the secret of this transformation is in the way in which the impulse of life articulated, or set, the neuronal web of man's neo cortex.

Part 2

14more

Mechanism

The living creatures, ephemera or unstable, sheltering the impulse of life, through an evolutionary process, transferred each time more knowledge to the living creature.

The impulse of life needed some adaptation to be able to use the sheltered evolutionary power stored in the reptilian brain, making it possible for the proto-man to make a more qualified decision. This way it generated the necessity for life to have the knowledge of this experience.

The conditions found in the arboreal proto-man proportioned a unique condition in the developing process, that is to say, the transferring of all the evolutionary knowledge to a new structure, the new organ being called neo cortex. This organ would have the possibility, although not the capacity, of the awareness of all the information accumulated in the primitive brain during millennia, originated in the evolutionary process.

The knowledge transposition from the reptilian brain to the neo cortex was not the transposition of the information but of the possibility of making such information aware, furnishing options so that the proto-man, if threatened, be able to decide between escape and aggression.

Pointing out that this transposition happened in a very fast way, if compared to the evolutionary sequence. Man showed up almost suddenly, some million years ago.

The sudden transposition can be compared to a catapult effect, projecting the knowledge in a new organ. This catapult effect resulted from the interposition of the intermediate or emotional brain, between the two brains: the reptilian (inferior) and the neo cortex (superior).

In a try of adaptation and survival life dared to make the accumulated experience in the evolutionary chain conscientious, permitting that the proto-man use the information for its own survival. This way, the neo cortex came to express this information.

Man's neo cortex is an adaptation for life's survival, which consequently has the necessity of perpetuating itself.

Mammal web

So that we can understand the way in which the impulse of life organized the new neuron web, it's important to review the neuron web which forms the mammal's emotional web.

When the impulse of life created the mammal's neuron web, it used a very special and different mechanism which was used in man.

The emotional brain has its own structure for each species, transmitted by a genetic mechanism. Each mammal has his own emotional brain from its species. The bear has a type of emotional brain different from the wolf or from the emotional brain of the tiger. The neurons are disposed according to the species of mammal. The activation of the circuits through training will represent the acquired knowledge. The stimulus which has nothing to do with the species will be interpreted in a tendentious way to the species.

The external stimulus gets to the brain through feelings and goes through the emotional brain. The passage through the emotional brain happens by an articulated web according to the species. This passage will determine an emotional flash which, basically, will be positive or negative.

The mammal has a less developed cortex than man. According to the flash test the information will be put in the cortex and placed in a determined area.

Laboratory Mouse

The laboratory mouse is capable of learning how to remove food from a trapdoor. He learns that determined action releases food. This information reached his emotional or intermediate brain and was accepted with a positive flash, being then stored in the cortex. From that moment on, it begins to feed from the trapdoor. This same mouse, if abandoned in his habitat, will not be able to survive because it had the activation of its emotional brain distorted in relation to its species.

The emotional brain was made to be activated inside the habitat context to which the mammal was raised.

Evolutionary Knowledge

The impulse of life, to save the proto-man who fell in the savanna, had to give the opportunity of using the evolutionary experience acquired during millenniums to survive. With this knowledge it would allow him to use the knowledge of the impulse of life when confronting an adversity.

The option was made through the creation of a new neuron web, constituted by two arrangements in the mammal's brain: the increase of the existing lobes and practically creating the pre-frontal lobe.

Catapult

The evolutionary knowledge of the impulse of life written in the reptilian man was transposed to the new artifact (neo cortex) capable of saving the arboreal proto-man. This

transposition happened, intermediated by the mammal's emotional brain which functions as a catapult, intensifying and regulating the transference process.

Neo Cortex

It was not possible to create a new artifact as if it were a computer, with an infinitive number of chips capable of saving all the evolutionary process information. And besides, the impulse of life had no interest in this information, except when it was related to the problem to be confronted. There was not the necessity, nor the possibility, of putting all the evolutionary process information in the chips of a new artifact.

To save the proto-man who fell from the tree, an access of information held in the core of his being needed to be furnished, registered inside each unity in his genetic code. The action of the impulse of life would have to be fast because his survival in the savanna was impossible.

There couldn't be, suddenly, a web of neurons that contained all this experience. It was necessary a mechanism that could go, gradually, catapulting the experience for the new artifact. The impulse of life created a web of neurons which should be disposed or placed in a way to be transferring the information to the new chips as they became necessary. The new web would have to be taught!

The proto-man already had a web below this one, capable of identifying exactly the necessity of certain knowledge in a determined moment. The emotional brain is capable of identifying the relevance of the question which is being analyzed and inform in the form of a flash. The new artifact began to have, as an entrance mechanism of information, the flash originated from the emotional brain.

The resources used by the impulse of life to save the proto-man were, in the first place, the personal knowledge acquired that showed up in the mammal and, secondly, the anticipation that was made possible in the arboreal proto-man.

The Brain Lobe Enlargement

The new neuron web should not respond through a flash; it should have its own register information.

This new artifact or neuron web would have the function of keeping the information and the web should function as if it were composed of binary computer chips.

In the mammal's cortex there were already neurons disposed as chips that contained the personal information of the mammal. It was a big advance of the mammals in relation to the reptiles. This artifice did not constitute something new. It would only be necessary the increase in the number of neurons, as the impulse of life did with the monkey's tail or the giraffe's neck. In a simplified way, we can say that there was an increase of many chips in the brain lobes of the arboreal proto-man.

Frontal Lobe

The second transformation was the most important: the establishment of a new functioning front lobe mechanism and the creation of a pre-frontal lobe.

This alteration, that really characterizes the human brain, required a genetic and correspondent modification of 1% of our existing difference with the monkeys. The impulse of life practically created the pre-frontal lobe. This lobe is directly connected with the emotional brain and functions in a different way from the rest of the brain cortex.

The frontal transformation did not happen in the same way as the previous step, which was a simple increase to an alteration already existent. Here, a new functioning

mechanism of the brain was created which relates with the anticipation and with the interval between the emotional brain flash and the action.

Morphology

The impulse of life functions inside certain characteristics that can be seen when we observe nature.

The tail growth suggests the necessity of a bigger equilibrium. The giraffe's neck gives access to the leaves of the tree. The configuration of the skull box of man with the increase of the frontal region has a relation with the anticipation.

The necessary alteration for the transformation of the mammal's brain of the monkey in the human brain becomes evident when the conformation of the skull boxes is compared in these living creatures. There's a progressive increase in the capacity of the box and in the last stages, between simian and man, there's a projection of the anterior part of the brain, corresponding exactly with the frontal lobe.

There are some hominid races that disappeared and in which the Archeology obtained the skulls. In them there is the absence of the frontal lobe, which shows us how it must have been important, the appearance of the lobe alteration.

Free Space

For not being possible to transfer all the evolutionary knowledge to the new artifact, the impulse of life had to set a system that was gradually transferring the knowledge, as survival made itself necessary. A space needed to be created that would not be used immediately, but would be occupied when necessary.

It's as if a hard disk from a computer were being created. A disk with billions of gigabytes that would be being used gradually but capable of accommodating all the evolutionary knowledge registered in the reptile man.

This hard disk is spread through the brain and corresponds to the increase of the brain lobes that appear in the human brain.

Double Artifact

The impulse of life capacitated man to be able to make use of all the evolutionary knowledge. The capacity happened through the creation of the rational and neo cortex brain. This new artifact has a rigid disk that is being fed as the knowledge necessary makes itself essential – and a mechanism which is like a processor locates the frontal lobe.

Interval

The frontal lobe was created similar to the emotional brain, as a web of neurons specific of the human race. It has its connections arranged to take advantage of the interval that was happening in the mammals between the emotional and action flash, giving the possibility that in this interval the information received can be compared with those that are in the hard disk.

The Anticipation Process

The other resource installed in this frontal processor was the improvement of the anticipation. Through the formation of the neuron assemblies, similar to the one that occurs in the emotional brain, it is capable of producing assemblies analyzing the

information that is received from the exterior world. Since it is more advanced than the emotional brain it can manipulate this information based in the data that is already stored in the brain.

The neuron assemblies create an impression of the exterior life on top of which the possibilities of anticipation are weighed. This anticipation process assumed a continuous character that generated a great knowledge development. Day and night, awake or asleep, the process does not stop. There's a constant anticipation in which man can not rid himself but, on the other hand, this assures an incredible development for the human brain.

The impulse of life created the neo cortex, or superior brain, to save the arboreal proto-man that fell in the African savanna. He was given an artifact with the potentiality to understand the whole evolutionary process, so that the experience was used as a surviving resource. The artifact required a constant development, which was furnished by the anticipation. In the frontal lobe, the anticipation functions as a continuous movement, playing with the data stored in the new brain.

I am

It is shocking the placing of the superior brain as a surviving artifact used by the impulse of life. Even more chocking is its purpose of being the knowledge of the evolutionary experience. We are in such a way used to the omnipotence of our thought and of our individuality that this information will be hard to accept.

Origin of Knowledge

There's a constant progression of the accumulated knowledge in the living creature as the evolutionary progress advances. The evolution is a knowledge acquisition process – which in the last chapter I denoted as Origin of Knowledge.

The reptile, mammal and man are the big steps of this knowledge increase. In each of these steps knowledge is focused in opportunities, or surviving niche, generating a great diversity of species.

Man is in the last step. When the impulse of life expanded to the whole planet, occupying all the possible spaces, there was the necessity to expand a last and great membrane that covered all the acquired knowledge during the process.

Two Lines

(illustration)

Anticipation – Thought – Self-Regulating

Interval – Knowledge – Expansion

The impulse of life, during the whole evolutionary process, has been acting the same way; expanding membranes to the exterior environment, perceiving the possible alterations and transmitting the information to the living creature through genetic or behavior alterations which materialize as social membranes. It owns great knowledge and, progressively, transmits it to the living creatures.

The great knowledge includes a vision of the whole manifestation of life. Life self regulates itself since the first manifestation. The animal and vegetable kingdoms

complement themselves. The carnivorous and herbivorous balance themselves. The reptiles and insects also do. There is always a dichotomy regulating itself. Before I showed that this happens from an impulse of life's direct intervention; not being purely a consequence of natural selection.

The manifestation of the impulse of life has been made in two lines: the knowledge transference and the self regulating.

The Two Brain Lines

Man is at the tip of the evolutionary process. The most qualified living creature to carry forward the impulse of life through the ephemeron and instable structures of the living creatures.

In this last evolutionary act the impulse of life capacitated man to have the knowledge or conscience of his whole experience until that moment. And this was done through the creation of the superior brain.

It gave the last step inside those same two lines of manifestation of the impulse of life. On one side it increased the number of neurons so that it could have the potentiality to accumulate the evolutionary knowledge. On the other side it increased the frontal lobe which, through anticipation, could conduct the self regulating mechanism of life manifestation.

The impulse of life could not capacitate the living creature with the whole evolutionary knowledge because it would not have an auto regulated mechanism. The capacity of expanding and storing knowledge is practically unlimited allowing the expansion of its own living creature man. Man's thought, however, is impregnated by the knowledge of the self regulating necessity.

The impulse of life gave man the possibility of acquiring all the evolutionary knowledge. It also gave man the responsibility of self regulating.

Access though the Emotional

The main characteristic of the new artifact, the superior brain, is that the access to it is done through the emotional brain. The emotional brain functioned as a catapult which projected the reptilian brain knowledge to the superior.

This has been largely shown in the last decades. Through the emotional, man can take a bigger advantage of his superior brain. There's a whole literary of self help based on this premise. With the emotional intelligence presentation Daniel Goleman started a series of studies demonstrating that the potentiality of the intelligence is exponential when associated to the emotional.

Through the emotional brain, then, is where the access to the evolutionary knowledge can be made.

Michelangelo

(illustration)

The Impulse of Life furnished the Knowledge

Knowledge
(Gnosis)

The Impulse of life created a Self regulating mechanism

In the first part of this book I showed Michelangelo's creation of man in the Sistine chapel as an example of the access to knowledge beyond the rational; acquired through the emotional brain.

Here, again, Michelangelo gives a demonstration of the evolutionary access through the violent emotional crises during which the Sistine chapel was painted.

The fresco which symbolizes the original sin, the creation of the human brain is shown by the impulse of life, presents the two lines over which the brain was mounted: the knowledge and the responsibility.

The serpent gives man knowledge. Rolled around the tree, it is the representation of caduceus of Hermes and gives man the apple of knowledge. The symbolism of caduceus is the manifestation of force which, coming out of the earth, crosses man's vertebrate spine, very similar to the reptilian knowledge being catapulted to the superior.

The sword which the angel points to the man's head when leaving paradise has given rise to many interpretations without a significant knowledge of its understanding. Analyzing the way in which the human brain was mounted the corresponding knowledge is the developing of the thinking power and in this power is the responsibility sword of life's self regulating transferred to the human brain.

The painting of the original sin (see table page 272) identifies the two lines shown before that detailed the creation of the evolutionary artifact, the superior brain. Once again, Michelangelo, in an emotional crisis is capable of accessing knowledge beyond his rationality. Even if he had dissected innumerable anatomical parts and put anatomical images in the ceiling's painting of the Sistine chapel, the knowledge of the god of creation in the brain and the mechanism in which the brain was created transcend his rationality.

Duality

The formation of the human brain brought man the duality which dominates his existence. There's the potentiality of the evolutionary knowledge but, also, the responsibility to administer and perpetuate life using this knowledge.

Individuality

The appearance of the new and wonderful artifact created by the impulse of life through the expansion of the great membrane that covers all the evolutionary knowledge brought us a great problem: the individuality.

In the reptile all the knowledge is written in the genetic code. All the reptiles of the same species have the same knowledge and there's no individual knowledge that separates it from its species. In the mammal there's an incipient independent knowledge consequence of the activation of the neuron web of the emotional brain. This individual knowledge, does not differ very much either from the whole specie because the web is mounted according to the necessity of the specie. In man there's an ample and varied individual knowledge. Each man has an individual knowledge different from the other.

There is, in man, an exacerbation of the individuality consequence of the own way of being of the evolutionary artifact with which the impulse of life gave it. This individual exacerbation is so big that man is dominated by it. He forgets that he is made of protein molecules ephemera and unstable which can dismount at any moment and that will surely disappear. He is shocked when he is told that his superior brain is an evolutionary artifact created to perpetuate the impulse of life. He believes that this wonder was created especially for his individuality.

The human brain is not an artifact for the survival of man as an individual but for the perpetuation of life. The human brain was created to think about the whole and not about the individuality!

The pathological exacerbation of the individuality is in Descartes' expression: "I think, therefore I exist".

I do not think. My thought is an evolutionary artifact and I do not exist. Life is what exists in me!

PART 3

GNOSIS

Part 3

1

Rethink

The best way to keep an elephant tied to a post in the back yard is to maintain it tied there since it was recently born. This is an oriental story that simulates the actual situation that man is living.

Nazism

During the Nazism power capture in Germany a fact similar to this one of the oriental tale happened in relation to the Jews. The Jews were German. They were raised and educated in a Germany which was their country. They were born and raised like the yard elephant. The Jews were powerful like the elephant. It seems incredible that the Nazism had grown in Germany where the Jews had an immense power that never mobilized, resulting in the holocaust. Also, in the countries taken by the Nazism, similar fact occurred. The Hungary and Czechoslovakian Jews were in a passive state like the elephant tied in the yard watching their neighbors being imprisoned.

Today

At this beginning of the 21st century, man is going through great transformations and continues tied like the yard elephant. Man has to try the post that ties him. Challenge that which is established. Innovate! Create! If he does not use all his potentiality he will die in the yard that he is destroying himself.

The planet is being polluted reaching a point where there will be no return when nature will no longer be able to restore its systems. There are clear confirmations of these facts and man has done nothing.

Individuality

Each man acts as if only he existed. The societies have not been able to prevent man from being an individualistic being.

Each man is like an elephant cell. Each cell believes that only it exists. But if all the cells don't mobilize all will die with the elephant.

Religion and patriotism, during some time, were capable of, with some groups, acting outside the individuality. Both, however, are disappearing. Patriotism, mixed through the globalized world, and religion, are refugees of science.

To Have and To Be

The economic help put itself over the humanitarian help! The goods exceeded man! The being was swallowed by the having!

The great economies maintain programs of humanitarian help to the underdeveloped but continue to maintain the poor in a dependence state not giving them real opportunities of self development. The world today is capable of producing sufficient food to feed all and almost half starve. Violence is exposed on TV as an amusing source and everybody is shocked when a child shoots people in a school.

Each one lives involved in his personal worries and problems as if what were happening to his neighbor were not going to influence his life.

The terrorism is originated of a revolt resulting from a social exclusion and not from a religious war. The terrorist episode of September 11 2001 exposed to the world this wound to our society. The surprise of the American society with such an episode is an example of the isolation in which the humanity is living in.

Last year (2005) the United States was surprised with the flooding in New Orleans exposing the misery and discrimination existent inside its boundaries which the very American population practically did not know.

Essential Change (Metanoia)

Man is tied to the planet's yard. Like the elephant in the story. He has to react to change his mind. Make an essential change to discover his destiny. Recover the greatness that exists inside him. Through this essential change put the everything back in order.

This picture is like what happened to the Jews in Nazi Germany. The individuality needs to be transmuted by the feeling of collectiveness. This is very difficult because we are used to making, constructing, edifying, coaxing and taking. Such attitudes can be computed as facts. Our science is structured in facts. These are the facts that should be transmuted in values.

I Exist and Think

To make an essential change it is necessary to abandon what we call science which is focused in the neo cortex – I think, therefore exist - to enter a new and more truthful science focused in man and in our brain as a whole: science focused in man as a living being – I exist, therefore think!

The first step for this essential change is the integration of the emotional brain. Perceive that the neo cortex functions attached to the emotional brain. Man needs to begin to feel more, to get more emotional, and accept that having a little less can make it be a little more.

The present moment is a mutation point in which there should become a new world, not more structured by facts but based in values.

Rethink

Man needs, urgently, to be rethought!

He developed a marvelous technology and this did not make him happier. He is destroying the planet by polluting; knows this but can not control the devastation. Developed political models of democracy but can not create equality. Improved the cultivation of the land and today obtains more food than necessary but one third eats all the food and two thirds starve.

The family model that always was society's cell is deteriorated. Religion, which always functioned as a controller to the satisfaction of the individual needs, disappeared. The population grows in an uncontrolled way. The ones that can a lot don't reproduce. The ones that don't have the minimum conditions multiply. The youth, in face of such uncertainties, satisfy themselves with drugs.

Man was able to increase life survival in a fantastic way and does not know what to do with his time. Was able to become old but does not accept old age.

Man needs to be rethought!

Mutation Point

This moment is similar to the mutation point when one electron concentrates a quantum of energy and leaps to an uncertain orbit.

Creativity, mutation point, Quantum mechanic and uncertain principal define the moment that man is living through at this beginning of the 21ST century.

Part 3

2

Man as a Whole

Man is a result of an evolutionary process that started in the first life manifestation. Went through the multi cellular, reptiles, mammals and finally reached the human being. Cannot be seen separate from this context!

He is an unstable and ephemeral living being like the protean molecules that constitute him. He lives in equilibrium with the other living creatures, animals and vegetables, which proportion his sustainability. He is subject to the temperature and environment alterations such as earth quakes and volcanic eruptions. He depends on water for his survival. He depends on other men more than any other living creature!

Human Body

Man's body was mounted by the impulse of life the same way as the other living creatures. It was bestowed with qualities to survive, like the tiger, bear or wolf, making the most of surviving potential niches that appeared during his evolutionary process.

The arboreal proto-man received from the impulse of life an artifact, the superior brain, to be able to survive when he fell in the African savanna. The artifact made it possible for him to use the experience of the impulse of life acquired during the evolutionary process. This knowledge was written in the reptile man and was transposed to a new artifact through the measuring of the mammal's brain (emotional).

The knowledge can be acquired because the new artifact has neurons which comport it but which have to be carried on, gradually, as surviving makes itself necessary. These surviving necessities are measured by the emotional brain.

Responsibility

The knowledge of the evolutionary process includes the self regulation which the impulse of life exerts since the first manifestation. Knowledge implies responsibility!

When the arboreal proto-man fell in the African savanna he would have no condition of surviving if the impulse of life had not saved him with the artifact which permitted him to think. Through this artifact he could use the experience of the whole evolution to survive in a hostile environment, dominated by mammals and which self regulated them between the carnivorous and the herbivorous.

So that man could free himself, of this self regulation, the impulse of life put in his new artifact the necessity of a self regulation of the manifestation of life.

Self Regulation in the Brain

The surviving potentiality offered to this living creature, through the new artifact, was somewhat extraordinary: it could provide food and surviving to the whole group with an abundance never seen before. This potentiality also implied a capacity of destruction which somehow needed to be regulated. A self regulating mechanism from the manifestation of the impulse of life was this way inserted in the new artifact.

Man's thought happened from the anticipation mechanism which proportioned an extraordinary expansion of the frontal lobe and the creation of the pre frontal lobe. The access to this part of the brain happens, basically, through the intermediate brain, mammal, emotional or Limbic System.

The intermediate brain which had its origin in the mammals is a self neuron web for each species. It is activated through experience or education of the species itself. In the intermediate brain there are registered the self regulating mechanism of the species.

The intermediate brain (mammal), on the other hand, was created to enlarge the possibility of adaptation of the inferior brain or reptilian which responds automatically to the manifestation mechanisms of the impulse of life; including the self regulation.

Man's thought is not as free as can be imagined. And his knowledge is tendentious in relation to his necessities of surviving. The neo cortex is compromised with the brain as a whole and, this way, participates of the self regulating mechanisms that are registered in the mammal and reptile.

Independence

Man does not think what his neo cortex wants but what his intermediate brain judges that is in agreement with his necessity for survival. This judgment is, in part, permanent by the web structure of the neurons of the intermediate brain and, in part, a result of the associations made by the experience of activation of these circuits. The judgment made by the permanent part of the intermediate brain, the Limbic System, is always in agreement and is coherent with the manifestation of life. But the succeeding judgment from the activation of the circuits can be corrupted.

Intuition

Intuition functions as an impulse that comes from the intermediate brain and, many times, puts itself over our rationality. This intuition impulse should always be right – many self helping books teach us to trust intuition. Practice shows us, however, that sometimes it is wrong.

When intuition happens from an emotional flash, result of the passing through the permanent structural web in the limbic system, it is true; when it happens from the passage of the activated circuits, through education or through experience, it can or not

be correct; it depends of the education having or not been made according to the species' necessity.

Mythology

The neo cortex has, uniquely, the potentiality of filing the knowledge of the whole evolutionary experience. But it does not have this knowledge. This knowledge is acquired gradually by the frontal mechanism.

Each neo cortex has its own knowledge that may seem individual. But it is mounted on the same structure created at the moment in which the neo cortex was formed. The new cortex formation happened with the catapulting, formed by the emotional brain of the evolutionary knowledge written in the reptilian brain.

The neo cortex has pigeonholes where the knowledge is stored. These pigeonholes are the same for all men. The more each ones knowledge seems individual it is kept in a same structural order.

The mythology of the most different cultures keeps a same structure which exemplifies clearly how diverse knowledge is filed in the same pigeonholes.

Part 3

3

Several Men

What man has that is more developed is his reasoning, or his superior brain, which gives him the power of thinking. This power, however, cannot be seen in an independent way because it is a result of the evolutionary process where all of man is involved and manifested.

Man has, therefore, to be seen as if he were four men on top of the other. When he is manifesting it's not one or the other but the four as a group.

1. The Vegetating Man

He is responsible for the vital functions of the human body and corresponds to the segmented man before the encephalic pole was formed.

The human brain is placed inside a rigid cranium box which communicates with the rest of the body through a short bony opening called "foramen mango".

When there's a very big edema in the brain, the foramen functions as a guillotine, sectioning the brain. The brain is constituted of a very soft tissue and the brain edemas are extremely common to almost all the brain pathologies. That's the reason for there being so many sick people named brainless. In these cases all the organs function and the body can be maintained alive for a long time if food and assistance are supplied because the endocrine system is capable of functioning alone.

As we saw, it is the first coordination system of the multi cellular. This is the vegetating man living in each one of us.

2. The Reptilian Man

The reptilian man is mounted on top of the vegetating man. He is represented by the primitive reptilian brain; also named "archi-brain" (ancient brain).

It is constituted by the encephalic pole which coordinates the manifestation of man in the environment in which he lives. It has a very special small brain which coordinates all of man's movements: the cerebellum.

Twelve other segments were developed to be able to command the external part of the encephalic pole which is the head. They are the twelve "skull pairs"; nerves with highly specific functions among which are the optic nerves, hearing, olfactory, the nerves responsible for the movement of the eyes and muscles capable of articulating the speech.

The reptilian man also has nuclei neurons which coordinate all the reactions of survival written in the genetic code. These reactions do not limit themselves to the automatic reactions which we show when we are attacked. They go way beyond coordinating the primitive surviving impulses and the ones of the perpetuation of the species.

Man, in a state of complete deprivation, can come to manifest the reptilian man that lives in him. In a concentration camp, where a man can attack the weakest and take, by force, food; like a reptile would have done. In a straining life threatening moment man can behave and defend as a reptile does.

In states of great emotional instability, as when the activation of the emotional brain web suffered extreme corruption that can practically annul the neo cortex action, the reptilian man can be manifested. In the serious sexual disturbances we can, sometimes, find this man in a very evident state.

The reptile man is extremely important in each one of us; it is responsible for the impulse of survival and for the perpetuation of the impulse of life. The basic impulse that made me study medicine, marry and have children was furnished by the reptilian man that lives within me. My two other brains serve to orient the manifestation of this reptilian man inside a context that goes beyond the genetic knowledge that my being has.

The reptilian man has in it, written, all the evolutionary knowledge. Its competitive advantage will depend only of its size and not of its quality. The bigger the reptilian man the stronger and easier will he conquer the survival and the perpetuation of the impulse of life. This is only true, however, until the point in which there is an increase in his necessities. Because he is bigger physically he surpasses the advantage of being big. This fact must have had an influence in the disappearance of the large reptiles.

To go beyond he needs to be coordinated and contained by some new evolutionary artifact created by the impulse of life; capable of reintegrating him in a new group and a more ample manifestation of life; the mammal world.

3. The Mammal Man

Until the reptilian man knowledge was passed by the genetic code. The alligator is already born an alligator. To broaden knowledge it was necessary to make it possible for the living being to acquire knowledge after birth, already outside the egg or the maternal organism.

This knowledge, coming from the outside, would have to be conducted or directed. It could not be obtained by chance. Each species has its own necessities and specific interests. The living pre mammal being, living in the open, would have to be suitable to run and to perceive the surviving niches that would give competitive advantage to him; like discovering new food sources among the herbs of the field. Yet the pre mammal that fed of meat would have to be suitable in dominating its prey and discover new ways of not being seen. The knowledge of one does not serve the other.

To solve this necessity knowledge question, specific to the species, the impulse of life created the encephalic pole; an artifact which corresponds to the intermediate and emotional brain. The artifact is transmitted by the genetic code as a simple encephalic appendix, practically neutron (virgin). It needs to be activated so it can function.

is composed of
(illustration)
for each species.

birth, according to
Emotional or Intermediate Brain
some circuits which

them, is in agree-
Primitive Brain
artifact of the mammals

being received. It
Not activated
necessity of the

Educated
is being tried is a

The artifact of the mammal's encephalic pole
neurons disposed in series, forming specific groups
These series of neurons are being activated after
each species ritual, permitting the formation of
will check if the information, which moves through
ment with the necessities of the species. This
is not capable of identifying the information that is
simply informs if it is or not in agreement with the
species.

The emotional brain does not know if what
new smell, a new image or a new taste. It only
informs if the new experiment is or not in agreement with the species necessities. The
answer of the mammal's emotional artifact is given under the form of flashes. In the first
mammals, the artifact should answer with an acceptance flash and another of rejection. In
the superior mammals, and in man, there are other types of flashes; but the emotional
brain continues functioning in the same way.

The information that comes from the external environment, in the reptile, has a
response already programmed by genetics, not comporting new experiences or new
knowledge. In the mammal the information that arrives from the external environment
passes through a sieve; the emotional brain, which will tell in which way this information
will be incorporated to the knowledge as a new experience.

Circuit Activation

The calf, which is licked by its mother soon after its birth, activates a series of
circuits in the emotional brain which will guarantee a behavior pattern unique to the
species.

The smell difference between two types of herbs will be relevant for the deer that
grazes on the savanna but not for the lion. In the deer this information is accepted as an
experience and transforms it into a new knowledge; in the lion it is blocked by the sieve of
the intermediate brain.

The emotional brain functions selecting and identifying the relevance of all the
information being received by the feelings at each moment creating flashes which, in the
superior mammals and man, assume more complex forms, each time, until they can finally
be identified as what we call *emotions*.

Emotions can be compared to neuron assemblies which are formed in the brain and
determine a subjective state of perception of the external environment. The same
assembly can be activated before a varied conjunction. An image or a sound can
determine the formation of a same assembly.

Third Eye

The emotional brain functions as if it were a feeling. It is constantly receiving information from all the other senses and cataloguing its relevance according to the species necessity to which it belongs. It functions as what can be called a "third eye". An eye, that is constantly, watching the external world under a very special prism, which is its own, and is related to the necessities of its species.

Despite the complexity of the human brain, it does not stop presenting these mammal characteristics. Man is constantly noticing the world interpreted by this third eye. If I find myself in a threatening situation I have the information selected, which tend to protect me from the danger, and I won't notice the information which will help me choose a partner. In the beginning of the book, I told the story of the Cava vein where the fear that I had of a surgery accident would make me select the information.

Education (training)

The mammal man is born with his emotional brain relatively virgin. The circuits will be activated through the own species rituals.

The same way as the calf, taken from the mother at the first hour will not play when bigger; man should have a series of unknown behaviors which are related to the first contacts with the mother. A whole adult man behavior is related to this period of our formation, when the circuits of the emotional brain are activated. It is impressive the little importance that has been given to this emotional formation. The big majority of man's problems and suffering is related with deformities in the activation of these emotional circuits.

Our neo cortex diverted us a lot from the specifications of our species!

Myths

As all the mammals man has an own emotional brain for its species that can be activated in many forms. But always has the same types of circuits. The emotional brain of the Brazilian Indian, the Greek and the Scandinavian are the same, which makes their myths always similar. The Greek mythology, the Scandinavian and the Brazilian have a lot in common.

The collective imaginary (archetype) of Jung is, in last analyses, the identification of the emotional circuits which are activated in our emotional brain. In Jung's analyses, these emotions are already contaminated by the superior brain and, because of this, are similar to the feelings, but in its origin they are originating from the emotional brain; happen with the activation of the own circuits of the human species; are equal in all men.

Lion

To show how the brain works in an entwined way, I'll use the example of the lion in the savanna. The lion, when hungry, is impelled to the prey by its reptilian brain. When its reptilian brain feels the necessity of reproducing it is impelled in search of the female. When threatened prepares it to fight. When the reptilian brain has no necessity the lion can rest in the savanna.

If the lion is impelled to its prey its emotional brain will be selecting the possibilities of the hunting. It will organize the hunt according to the activated structure though training, using the females to involve the prey and send it to this lion.

If impelled for reproduction, its emotional brain will select a female more suitable for the survival which differs from the reptile.

If it is impelled by a threatening situation, its emotional brain will select the added information through its experience, superior to the pattern reactions of the reptile.

When the reptilian brain is not needed and is resting, the emotional brain will exercise the own circuits of the species, which constitute in positioning inside the group, as definitions of male, female and young positions.

4. The Sapiens Man

The *sapiens* man was the resource that the impulse of life found to save the arboreal proto-man who fell in the African savanna. In a similar way to what happened when the mammals appeared man's appearance also happened because of the creation of a new artifact in the encephalic pole.

Individuality

The mammal's artifact represented an advance in relation to the reptile's encephalic pole because it gave a possibility of acquiring knowledge out of the egg or mother. This advance created the possibility for the living creature to acquire an individual knowledge which, until this moment, had not occurred in the evolutionary process. This individual knowledge required an increase of the neurons, capable of accumulating exterior information. This information was personal, and was not a reality of the outside world. This was the beginning of the individuality tendency which, since the mammal, began to happen inside the evolutionary process. Although the reptile is an individual, all its manifestations are tied in a genetic code and nothing in it is personal. In man, this individuality will assume great relevance.

Limit

The individual knowledge is a threat for the evolutionary process of the impulse of life. Until this moment, the impulse acted in a direct way in the formation of the living creature and, from the mammal, a new individuality manifest began, with the personal knowledge. This new expert needed to be contained. That's why the emotional brain is an artifact that has specific circuits of its species, which gives it a very limited individuality. The increase of this capacity of individual knowledge implies imposing limiting mechanisms so that the expert behaves inside the same evolutionary line as the manifestation process of the impulse of life. Creating the new artifact in man's brain, the impulse of life had the necessity of maintaining this control.

Catapult

The resource, used to save the arboreal proto-man, was given to him; the possibility of having as personal knowledge all the acquired experience through the evolutionary process. It would be like transferring the written information in the reptilian man to the new artifact. Create a new brain computer with the capacity of storing all the information to be able to use it in the surviving and perpetuation of the impulse of life. It was not sufficient to create a bunch of neurons with infinite capacity of interrelation to save the information. This bunch of new neurons would have to have a pre fixed structure and arrange a mechanism that would feed it with information. Besides this, the new artifact would have to have a control so that the new living creature, with its personal knowledge, would not interfere in the evolutionary process.

The solution found was to use the emotional brain as an inter-mediator for the formation of the new artifact. This way, the continuity of the process of individualization

was given, which had started in the mammal, since this process was already somewhat restrained and tied in the emotional brain structure.

The written knowledge in the reptile man was transferred to the new artifact. This transference happened through the emotional brain which functioned as a catapult sending to the top the knowledge which was below.

Anticipation

The new artifact can, in a certain way, be considered a creation of a bunch of neurons with possibilities of connections that could store infinite information, but was assembled under the tutoring of the emotional brain. Its structure is an enlargement of the growth of neurons which was already appearing in the mammals: the brain lobes, already existent, were amplified. The feeding mechanism of the new structure was furnished by the frontal lobe hypertrophy and practically through the creation of the pre frontal. This process of the information feeding is also impelled by the emotional brain and took place through the anticipation perception which was unique in the arboreal proto-man.

Limitation

The superior brain was created inside the emotional brain molds, maintaining an individual knowledge control. The same way in which the emotional brain permitted a limited individuality to the specie's circuit structures, the superior brain, being catapulted by the emotional, also keeps similar limitation to the emotional brain. Man is born with a superior brain assembled in the same way as the mammal is born with the emotional structured for the species.

This assembly determines a structural limitation. There are determined places for the information to be catalogued and webs that communicate these places as preferable routes. Man is forced to think through a process that is assembled in his brain's neuron web and this assembly happened because of the catapult by the emotional brain. Man is not as free to think as he imagines; his thought is very oriented by his "third eye".

Dynamic

The information transference process to the new neurons will be happening as they are necessary for the survival. That's why the human brain has great potentiality which is not used.

The new filled information is also being related to the emotional brain flashes so they can be used as new knowledge and be optimized. From this process a new exchange with the emotional brain happens which will generate more complex neuron assemblies than the emotional brain flashes, which we call feelings.

The anticipation process of the frontal lobe makes man always anticipate the events. It happens day and night, awake or asleep. It functions like a windmill, spinning continuously with the wind of time.

This windmill will continually feed the information to the neuron connections of the superior brain, generating a very peculiar knowledge, which makes each man different from the other. It's impossible for two individuals to have the same knowledge, but all of them have their knowledge assembled inside a same structure. Information is weighed through the same webs.

Pseudo Knowledge

The feeding of the information through the anticipation process, which happens in the frontal, becomes much overstated in today's man. The collected and catalogued information in the superior brain will constitute a personal and proper knowledge that little has to do with the external reality. This makes man's necessity of survival unreal, generating the transference of distorted information.

The distortion resulted in an inaccuracy of the purpose of knowledge. Man created an own knowledge that has little to do with his real necessity of surviving. This pseudo knowledge generates amazing lucubration, moving him away from his true man necessities. From what we know as "science", there's a great quantity of knowledge which is a result of this distortion. Even more, when this pseudo knowledge is associated to the emotional and identified as feeling, the aberrations can become astonishing.

Summary

In man there are four assemblies. The visceral or vegetative, the reptile, mammal and sapiens man. All function in an entwined way. The most inferior is more important for the survival and assume the general command when threatened.

The man with a renal colic or with the pre-cordial myocardial pain infarction is, basically, the visceral man; the others become annulled.

In an aggression threat, reptile man assumes the control to attack or run.

In decisions that involve the social group, especially the family, the mammal man assumes the control.

The sapiens man, through his superior brain, can affect all the other men, but cannot dominate them! Each one of them has his function and these functions are stratified and linked by an evolutionary process. Evolutionary process, in which man is part as a living creature and, as all living creatures, is ephemeron and instable as the proteinic molecules in which he is constituted. The sapiens man received an artifact which gives him the capacity to access all the evolutionary knowledge, but this does not free him of being a living creature in the manifestation chain of the impulse of life which first started in the cellular unity.

Man, group of these four, can at any moment, be one more or another, but the four will always exist. Because of the individual knowledge and extraordinary person that the neo cortex gave man, he judges himself as the Man – I think, therefore I exist.

Part 3

4

The Neo Cortex Dictatorship

Man is the union manifestation that functions as the four men described; in which one is not more important than the other. The individual manifestation of any one of these men will always be an aberration.

We have lived a dominant manifestation time of the superior or rational brain and we almost don't notice the aberration. Time is running out because the rationality wears out and man has noticed that he is not rational himself.

Man is neither a separate being from the other men; he's part of a union of a species in which he inter relates and is also part of the whole manifestation evolutionary process of the impulse of life which maintain an equilibrium relation with the exterior environment.

The Mammoth Fable

The neo cortex is an artifact given to man to be able to survive in a hostile environment in which he was not physically capacitated. He was given the capacity of using the evolutionary knowledge, weighing this knowledge to find ways of survival.

The neo cortex permitted that the primitive man weighed (think) the surviving resources. A group of man could scare a mammoth and make it precipitate over a bank. The big animal, this way dead, was used as food by the whole group. Its fur was used for protection from the cold. And all would live in peace and harmony. There were those who ate first and others later. Others used better fur and others not. This difference was the same as the social structure that there was when man lived in the tree and observed the mammals. This way the new artifact fulfilled its purpose which the impulse of life had given man.

The new artifact continued functioning and man began to invent ways of conserving the meat; bettering the mammoth fur and using the bones, which before were left for the animals of prey.

Today a group of men continues leading the mammoth to the bank but only a few are fed – they keep the rest of the meat in refrigerators. A few cover themselves, taw and prepare the mammoth's fur to make sophisticated clothes. The same few produce weapons utilizing the mammoth's bones, provoking instability in the prey; using these same weapons to guarantee his priority over the meat and fur of the dead mammoth.

The dominant manifestation of the sapiens man for the misrepresentation of the evolutionary artifact's function is stereotyped in this mammoth fable.

There's Something Wrong

The development of science brought a series of comfort to man; multiplied his possibilities of producing food; prolonged life's expectancies and, with all this, did not make him happier!

There is something wrong with the scientific and technological development. All this development should be used to make man happier.

Uncle Julio

My mother told us that when she was young and lived in the interior, my uncle Julio came from the capital informing of the great novelty. There was new equipment that allowed people to listen to the opera at home, with people dressed in their pajamas, without the necessity of dressing up elegantly and going to the theater.

In the 90's, instant transmissions could be watched instantly of the Olympics, on the other side of the world, through a color TV.

If we could quantify the satisfaction on a scale, I believe that my uncle Julio's would be much bigger than mine.

I still remember that at my grandmother's house there was no refrigerator. The water, fruits or anything that needed fresh conservation was put in a steel basket immersed in a well. I can still remember the taste of the fresh fruits taken from the well which we ate in the afternoons. The beautiful fruits that I buy today at the supermarket, keep in the refrigerator, do not compare to the fruits of my grandmother's house. Many people still believe that the food made on a wood burning stove is tastier than the one on a gas stove.

T Model

Today we are standardized by the consumption and we almost don't notice.

When Henry Ford established the first production line to produce his first T model cars, the salespeople asked what they'd do if the client wanted to choose the color of the car. Ford's answer summarizes what we are going through. "The client may choose the color as long as it's black!"

In face of the line production industry, of the uncontrolled consumption and aggressive marketing, which dominate our society, Henry Ford's answer defines the standardization in which we are being submitted.

Man's interest is subdued by the production interests, consumption and marketing.

Reason

The dictatorship of the brain cortex initiated after the Renaissance, as a Reason Era, in contraposition of what would be the Dark Era of the Middle Ages. Neither one is so rational, nor is the other so dark! Man became so inebriated with his rationality potentiality. And his emotional component was abandoned and despised because apparently the emotional decisions were deprived of rationality.

Religion

The religion that followed man since his birth was abandoned completely by the rationalists because it was disassociated from reason. Although the thinkers tried in all the ways to demoralize the religious creeds, possibly, due to the abuses committed in the anterior cycles, in the middle Ages, man maintained his faith. Religiousness is ingrained in man as a whole and cannot be fought against, even less, contested by reason.

Politics

With the end of the Absolutism, which had nothing of rational, the Illuminists started to preach the social equality between men, which culminated in the French Revolution terror. The republics and the democracies followed trying to force equality. But they kept privileging the differences between men. Rationally men should be socially equal. But they are not. As in the mammal society, they are socially equal. Man is not purely rational; he's a mammal that has a thinking artifact. Napoleon could not be socially equal to a sans-culottes.

Spiral

The manifestation cycle dominated by the superior brain is reaching its end at this beginning of the 21ST century. We are reaching the end of a cycle of thought, or of a way of thinking which we know as scientific thought.

This way of thinking – egocentric – began in the 17th century, with Descartes, was soon structured by Francis Bacon in his book *Novus Organum*. The structuring of the scientific thought generated a development of science which, in the 20th century, sent man to the moon.

The scientific development happened like a spiral where each turn was amplifying the circle and in the passage of the millennium is reaching such a big circle that the own scientific thought is unable to follow.

At the beginning of the century, with the discovery of the relativity, a series of concepts fell to earth thought of as scientific. The development of the Quantum mechanism and the sub atomic particle discovery, which joined the anti matter concept, began to require a new structure for the scientific thought. At the end of the century the information revolutions of communication and knowledge came to join the creativity revolution which is changing the world in a way that we do not totally notice yet because we are like the elephant stuck to a pole in the yard.

Part 3

5

New Vision

During the evolutionary process the individual knowledge happened in the mammal and expanded in an extraordinary way in man. The individualism, however, is contrary to the evolutionary process which tries, in many ways, to keep the individuality. In the mammal the individual knowledge limited the emotional brain structure of the species; in man, the neo cortex was molded by the emotional and made the accesses to the new knowledge by the frontal lobe, which functions in a similar way to the emotional.

Individualism

The exacerbation of the individuality is, therefore, an aberration of the evolutionary process which manifested in this reasoning era!

Man is not a unique being. He is part of an evolutionary group through which life has manifested itself. He is part of an eternal birth and death of the living beings that perpetuate the impulse of life which one day manifested itself in the initial cell. It is very difficult for the actual man to accept that he is part of a perpetuation process of the impulse of life that one day manifested itself through proteinic molecules of Carbon; ephemeris and instable as him.

Man does not live; life is what lives in him!

Knowledge

The evolutionary process is a constant growing transference process for the living creature. The knowledge transference goes on forming itself according to the perpetuation necessities of the impulse of life. Knowledge is furnished to the living creature so that he can better adapt to the surviving conditions and, this way, perpetuate the impulse of life.

The shaping of all knowledge is in the impulse of life which distributes a partial and tendentious knowledge in relation to the survival of each type of living creature. The wolf disposes of a different knowledge from the bear. Man, despite the neo cortex, has a knowledge limited by the arboreal proto-man's emotional brain structure.

As an understanding exercise we could imagine how man would be if the impulse of life had put the neo cortex artifact in a proto-carnivorous. This way we could understand in which way man's knowledge is restrained and limited by the emotional brain of the proto-man.

Man does not have the freedom that he thinks he has in the use of his thought! The true knowledge is in the impulse of life and it is not given for our understanding.

Emotion

All the evolutionary knowledge is written in man and, somehow, was catapulted to the superior brain through the emotional.

When the mammal confronts an evolutionary difficulty, it has an access to its individual knowledge by a flash, native to the emotional brain. In man too, the access to the knowledge is linked to the emotional brain because the emotional brain is responsible for keeping the experiences and they are what allow the access to knowledge. A same experience can furnish knowledge in many areas. Man who has one positive experience of affection is capable of foreseeing the conduct of a dog which is treated with affection. The mother who loves her child is capable of orienting better than a psychologist in the solution of a problem. Man during emotional crises can show true knowledge which is not rational.

The emotional experience opens an access to knowledge beyond rationality.

Science

Science is a lucubration of the superior or emotional brain to justify its individuality. There's a pretension of science in finding a rational understanding to all of man's questions. While human rationality of the rational brain does not join the other brains, our science will always be tendentious and truncated!

The true science has to integrate man as a whole; as a living being.

Continuity

Man is not the final point in the evolutionary process. He is simply a moment in this process. When the impulse of life gave him the neo cortex artifact, the evolutionary process was not finished – the neo cortex itself has developed since the moment in which it was created.

The process of constant anticipation functions like a windmill (time), feeding the almost infinite capacity of information transferred to the neurons.

The impulse of life has not stopped acting during this transference process in a try of saving its creation. The try of saving the neo cortex is oriented in the sense of forcing man to find his balance with the evolutionary process.

The impulse of life will not let man spoil the evolutionary process. It will maintain the balance from the manifestation of life where man is a very weak link – at any moment it can be destroyed by another manifestation of life like by a simple virus.

The evolutionary process is superior to man!

Part 3

6

Politics

Since the appearance of sexual reproduction, living creatures began to have the necessity of coexisting with their equals. This necessity began to be emphasized as survival became more difficult. Some species developed this coexistence so much that they began to form social organisms with different creatures such as ants and bees.

Besides the necessity of coexisting with the same species the impulse of life always maintained a control of the manifestation of life among the species, forcing a self-regulatory coexistence. This type of regulation also forced a coexistence among all living creatures regulated by the impulse of life.

Man having forcefully acquired the potential of usage of evolutionary knowledge, the need for developing the coexistence among its peers and living creatures made itself necessary.

The coexistence among its peers was established through language and made it possible for an extraordinary exchange of information. Language led man to group actions easing his survival in a very important manner.

The arboreal proto-man was already a social mammal which, having the neo-cortex artifact, had his social structure catapulted to a more elevated dimension. Politics is therefore, an evolutionary necessity of man's survival. Man is a social and political animal.

The individualistic knowledge process has corrupted the political concept in such a way that when I say man is a political animal it seems as if I were referring to the politicians' astuteness and the astuteness in which man should go through to survive in today's society.

Equality

The process of individualistic knowledge generated a deformed concept of equality among men. Men are not equal and should necessarily be distributed in a different way in a society. The alpha mammal of a group is not equal to the other members of the group. However he takes responsibility for the other members of the group.

Man, with his individualistic knowledge, has invented the concept that human beings should be equal so that the responsibility, which individual knowledge imposes on him before the group, does not have to be assumed. The impulse of life creates unequal beings inside a group so that, as a whole, they will have a positive survival result. There are men with unequal aptitudes who, within a group, complete themselves. Besides that, the own complex self structure generates an inequality, which, even though not physical at birth, becomes physical during its existence.

Men cannot be treated equally because they are not equal. They should have the same rights and access to a full manifestation and happiness. They should be treated with equality so that they have the same opportunities, but in an environment where physical strength is a competitive advantage, the strongest have more opportunities. The same happens in the intellectual and affectionate world.

The Origin of Politics

Free men from the Polis gathered at the square (Agora) to decide the laws and actions of the group. Each free man decided for his family unit which included, besides the family, the aggregates and the slaves. There was a sense of responsibility of the free man towards his dependents. There was a question of honor in this making a decision from the free man. The same way as the Alfa male from a group of mammals is responsible for its unity. Within this system equality was created; free man truly sought what was better for those who lived under his responsibility.

Mythology

The sense of responsibility that the free man had in his decision at Polis occurred in a culture based on myth. Mythology was an integral part of the way in which the free man thought. The Odyssey was the book which oriented man's conduct; his life was, somehow, integrated with the gods of the Olympus who participated in men's lives.

There was, therefore, an integration of the emotional man (mammal) with the sapiens man. The conduct of the free man in the decision making process involved his group and its unity.

The Political Fall

With the dialectic development in Agora, men who sold their word to force their opinion in Polis appeared. They were the Cynics. At that time the Sceptics, Epicureans and the Stoics appeared reinforcing the individualistic philosophical thought. This fact generated the appearance of the Despots who, with material power, began to command the Polis. The responsibility of the free man with his unity was disrupted and, right after, so was the group responsibility. The despots no longer had any responsibility with the group, which was being dominated by personal interest (individual).

Philosophy

In the beginning, the Greek philosophy thought was associated to mythology and had a strong emotional component creating equilibrium in man's manifestation. Philosophy was associated with politics. Man was seen integrated to a society and this integration was provided by the emotional component of mythology. Philosophy was Politics.

The merchants' word destroyed the existent structure and created the despots. Socrates appeared exactly at this time and developed a personal and individualistic dialectic. Philosophy is divided nowadays into before and after Socrates.

Socrates wrote nothing. His ideas were written by Plato who was the true person responsible for the Philosophy of those so called later Socratics. The Despots appearance shocked Plato who definitely abandoned Politics to dedicate to Philosophy.

Plato could have developed Politics because he was a man with an important emotional manifestation component; he was an upright man. He was followed by Aristotle who, not being so upright, strengthened Philosophy as a form of individualistic mercantile, personal and even egotistic thought.

Philosophy is an exercise of thought with the purpose of developing a personal and individual knowledge strengthening the distortion resulting from the partial and defective manifestation of man. The true philosophy should be the Political!

Dictatorship

The rational Greek thought cycle, initiated with Socrates, disappeared during the Middle Ages, which was a dominantly emotional thought cycle. It reappeared in the Renaissance, retaken by Descartes and Bacon who implanted the dictatorship of the neo cortex brain.

This dictatorship is reaching its end at the beginning of this millennium and should be transformed into a new cycle; not emotional but integrated as "reason and emotion".

This new cycle should be similar to the previous Socratic philosophy period when there was an integration between reason and emotion; when men were not equal but all had a commitment with their family and group unity; when there was liberty and responsibility; when men dreamt of gods integrated to their lives.

Part 3

7

The Ludic

The superior brain was an artifact, which the impulse of life implanted in a mammal to give it the possibility (potentiality) of using the evolutionary knowledge. The evolutionary

knowledge was not put in the brain directly but the frontal windmill was assembled to transfer it, gradually, to the cortex.

Through the anticipation mechanism, the frontal cortex anticipated day and night, potential situations in which, most of the times, had nothing to do with the external reality. For all the anticipation made new information was transferred to the cortex. So that the new information, that was being installed, would be filed as knowledge they had to reverberate in the emotional brain, being then catalogued according to the species necessity.

Through this mechanism a new personal world was formed, which is the world of dreams or the interior world. Each man has his own world. However, all have the same structure, proper to each species.

Copy

The frontal mechanism was copied from the emotional brain in a much bigger dimension. The small lion, for example, plays and fights with its mates exercising an ability which is not transmitted by genetics but in exercises to acquire a self and individual knowledge.

In the superior brain the anticipation creates a series of possible situations which will or not be later used.

Information and Knowledge

Information can be kept in the brain like it is with information in the computer; through binary connections which correspond to neuron connections by the neuron plates.

This information, to be accessed, should be related with emotional brain flashes which give them meaning. The information should reverberate in the emotional brain and this way a group of information transforms itself in knowledge.

Knowledge is a group of united information through an emotional flash. Knowledge is linked to the emotional and even the information, to be accessed, should participate of some emotional flash. This fact can be confirmed through the discovery that man's memory is located in the emotional brain especially in the hypo camp.

Structure

Considering the term "ludic" as a playful imagination as in the sense of game we can say that each man has his ludic environment but all with the same structure. This way, even when the information coming from the outside is different, they assemble themselves in similar knowledge flashes. The study of the compared religions shows that almost all have the same assembly where different information is used.

Every man, even the ones with diverse cultures, has the same knowledge structure which was, in a certain way, identified by Jung as collective unconscious. When asleep they dream in their playful environment using the most diverse information which is also assembled inside a same construction. An emotional knowledge can be expressed in a dream in many ways. An insecurity conflict can be dreamt in many ways.

The Thousandth Monkey

Monkeys, in separate islands, present communication among their brains. They learn, for instance, to clean the sand from the food by washing it with water. When the quantity of these monkeys reaches the critical mass knowledge in monkeys from isolated islands appears.

Unity

Men are all interlinked by their emotional brain which is the same for the whole human species and the same of the arboreal proto man. The existing unity among the mammals' brains can be perceived in the social behaviors.

When a group of cranes relocates it is done in a precise order; when a baboon group is attacked by a lion the oldest surrenders saving the group though sacrifice. This type of social behavior is due to an existing unity among the emotional brains. This unity can be due to the structural unity of the emotional brain which had transferred the social behavior to the mammal or from intercommunication among brains.

Constant Change

The superior brain is in constant transformation. When a determined knowledge reaches a critical mass it is then incorporated by the brain.

There are also a variety of superior brain peculiarities from one to the other individual. It's as if the impulse of life were acting to maintain control of the whole. Einstein would have had a temporal lobe slightly bigger; which gave him a special thought capacity different from others. Some men with this condition might have been born that were not availed by society. The impulse of life makes it possible that in a population of a thousand men some ten men have a brain structure which gives them a leadership condition. But among these ten only one will be the leader and the other nine will live a frustrating life.

Part 3

8

Gnosis

Men own two types of knowledge.

The first knowledge in which we are used to is expressed by our rational or superior brain. It cannot be said if it is conscience or unconscious because the conscience happens due to some momentary factors which I will not discuss here. This knowledge is decurrent of personal experience because of the frontal anticipation. It is a knowledge which can be expressed by the language.

The other knowledge is the result of the emotional catapult which transposed the potentiality of evolutionary knowledge to the cortex. It is basically the evolutionary knowledge; it is in the superior brain but under a ludic or emotional form. To differentiate it from the frontal knowledge I will call it Gnosis.

Two Understandings

The gnosis is the understanding of the evolutionary process at man's total disposal so that he can manifest himself and survive leading further the impulse of life that lives in him. This understanding, although it is in a potential form in the superior brain, is strongly related to the emotional brain which allows its access. It is expressed in a ludic form being difficult to express though language.

The gnosis is not an individual or personal understanding. It is a group understanding of the species, of the kingdom and of the manifestation of life. It has no preoccupation with the individual's preservation specifically but with the whole. The gnosis

makes a mother sacrifice herself for her youngsters which would be contrary to her rationality.

Coronary

The gnosis knows everything that happens in the organism. It knows exactly if an artery of the myocardium is obstructed before causing pain and if this obstruction is or not significant for the organism.

This information is generated in the visceral and goes through the reptile and mammal man and is expressed in the cortex in a ludic form. It will generate a series of actions which, many times, we do not perceive to protect the visceral man. The visceral man calms down, the reptile man does not move, the mammal man noticed the danger and emitted a flash and the sapiens man had an impression that the walk that he was making during the afternoons was not agreeable to God because it did not go in front of the church.

The information path until the emotional is fixed. The great variety will occur in the superior brain where infinity of thoughts can determine the execution of the order that should be emitted to abort the process.

The very advanced yoga affirms that they are able to stop thought and abolish their persona or the necessities of their personality. This way they'd be able to affirm: I will end the walk because my right coronary is obstructed.

Crutches

A normal man can have access to the gnosis during strong emotional crises in which the superior brain becomes almost inactive.

When working in the interior I operated on many indigents. I couldn't give detailed information of the procedures for lack of time and also because of the lack of understanding of the patient. Once, operating a rural worker, when the spinal anesthetic was done he noticed that he had become paralyzed from the waist down and freaked out. He yelled that he was paralyzed and would have to walk with crutches. It was interesting that he said that he would walk with crutches. I ordered the anesthetic to sedate him so I could operate the prostrate.

The prostrate surgery bled a lot in the pos operation and required a lot of care with the probes. In the second pos operation day the patient had a vascular brain ischemic accident and half of his body was paralyzed. During the pos operation the movement of the arm recuperated and he was discharged in fifteen days with a leg partially paralyzed and using crutches.

Psychosomatic

Did man know that his brain artery was on the verge of an obstruction? Or did he cause, unconsciously, this obstruction? Both hypotheses are acceptable.

In the first case the process which I described as the coronary could have happened. In the second the inverse process could have happened. A strong fear of the sapiens man could have initiated a crisis in the mammal man which descended with the information for the reptile man and materialized it in the visceral man. This path would be the cause of the so called "psychosomatic" sicknesses.

The Vagus Nerve

The inverse path, however, is not very easy to be traveled. The superior brain can, in a retrograde way, generate an emotional flash because both are connected. The passage to the reptile man can generate an action of escape or aggression; but the passage to the visceral man is totally blocked. The visceral man developed before the encephalic pole.

There is an exception in relation to this reptile/visceral communication which slips by the fact that one of the reptile cranium nerves units the visceral system which is the tenth pair or Vague nerve. This is the only passage known by the psychosomatic diseases.

A great brain tempest, like a big fear, may reverberate in the emotional and determine a reptile action. This reptile action may provoke a vagus nerve shock and cause a urine loss. A chronic fear or stress may generate constant chouse which require a continence effort which ends up manifesting itself as a physical disease. This is the colitis mechanism, gastritis, prostratitis and many other psychosomatic diseases.

Protection

Medicine does not know the form in which other diseases like cancer, for instance, can be determined by a retrograde information path transmitted by the gnosis. It seems like this path is possible and it can be noticed that the cancer happens, more frequently, with unhappy patients.

There is a protection system that prevents the sapiens man to cause harm to the whole man. My impression is that this retrograde path which causes disharmony in the man as a whole is possible when the disharmony or the sickness is of the man's interest as a whole.

The Manifesto

The relation of the gnosis with the visceral man is extremely complex and leads us to a blind end. The gnosis manifests in the visceral man in a still unknown way that does not happen through the genes.

Gnosis is a very special knowledge which manipulates information not registered exclusively in the nervous system.

We can understand that the nervous system is a structure assembled by the genetic code which contains infinity information that is transformed in knowledge when materialized in this nervous system.

We are totally unaware of how the impulse of life manipulates the knowledge which is not materialized in the nervous system. The impulse of life is capable of making an arm or leg grow on an amputated man using the cell stem. It is capable of making a paralytic swim or a blind man to see. Our knowledge and our gnosis restrict themselves to what is manifested in our nervous system.

Against such ignorance I come to think if it is possible for the manifestation (of the nervous system) to know what is being manifested (the impulse of life). It is only possible for us to analyze the gnosis, from the reptile man above, because his knowledge is manifested in the nervous system.

Total Man

The manifestation of the total man can be compared to a bull cart. The cart is the visceral. The bull is the reptile. The one who conducts the cart is the mammal and the one who goes on top of the cart, being conducted, is the Homo sapiens.

Reptile Man's Gnosis

Reptile man's gnosis is an evolutionary knowledge which the genome manifests in the living creature. The reptile man's knowledge tends to be taken for granted by the Homo sapiens. However the reptile has its way of manifestation which cannot be changed. The reptile man has the necessity of feeding, reproducing, is made to recognize some dangers and to initiate an escape action or aggression.

When the reptile man initiates an aggression reaction the sapiens man cannot abort the action directly because his control over the reptile man is indirect! Against a danger the reptile man reacts and only later will the other men discuss the action. There is, therefore, a series of immediate actions that depend on the gnosis of the reptile man. It would be what can be identified as instinctive reactions.

Many of the reptile man's reactions are authorized by the man above. The authorization implies in the action imposed by them. The feeding must be made this or that way. The reproduction must follow this or that pattern. But the impulse, the mobile of action, is from the reptilian man.

The sapiens man who "I think, therefore I exist" is no more than a reptile man who has his necessities modulated to live in society.

When man seeks a partner to reproduce is because the reptile man commanded the necessity of reproduction. He will choose one with big legs because this one can run to defend the young from predators and this was an adaptation made by the mammal man. He may give a wedding party, invite friends, live with a partner and may never have children if he decides this rationally. But if he has no children he will be unhappy because the reptile man was not obeyed!

Besides the direct action of the reptile man in the group there is, still, the gnosis of the reptile that was catapulted to the cortex and is installed there inside the ludic. He knows, indirectly, of the entire struggle in which he went through to reach the reptile man. An example of this is that the sapiens man knows that it is constituted of twelve cranium pairs and attributes a magical significance (ludic) to the number 12.

All the reptilian men are exactly alike and the reptile man's gnosis contaminates all our actions!

Mammal Man's Gnosis

The mammal man's gnosis is different from the reptile gnosis. It transports the reptilian gnosis to the cortex. Its importance happens in the way in which this was done by the neuron assemblies which are the emotional brain's flashes. These flashes are defined according to the species in which the mammal belongs. It defines what is in agreement with the species.

The emotional brain has its own gnosis which functions as a third eye. It is constantly analyzing the outside world and defining neuron assemblies which will rule the superior brain's function. The third eye is always distorting the information so they can make the most of the species.

The mammal's proper gnosis is constituted from a series of evolutionary knowledge that occurs with the emotional brain activation and represents a self evolutionary knowledge of the arboreal proto man species that, in the end, is the same knowledge of the human species: over this mammal was the neo cortex assembled.

In the case in which the reptile ordered the reproduction the mammal chose the woman with the longest legs because she could run more. This was a decision decurrent of the mammal gnosis acquired when already a mammal.

The emotional brain flashes (third eye) are extremely powerful. They define the information that should and should not be taken into account. The optical illusion phenomenon which occurs in the deserts is an example of the third eye.

Man sees what his emotional brain wishes to see.

The Sapiens Man's Gnosis

The superior brain has, in its ludic environment, the registration of all the evolutionary knowledge! This gnosis is accessed as survival makes itself necessary. The access is done in a ludic form which cannot be defined in words and represents knowledge.

For the whole man to be complete he has to manifest himself through his gnosis or through his evolutionary knowledge and not through his accumulated personal knowledge in the superior brain. The man of today, result of the scientific development, is a deformed man who can never be happy!

The dictatorship of the cortex is conducting man to a disharmony position with the impulse of life. If he does not fix himself for the suffering that the misrepresentation will cause the impulse of life will correct it in some way.

Gaia

Being true the hypotheses that the planet has life the World would be a living creature whose name was Gaia. Man would be her child; generated by her and would feed from her breast. Man grew, grew teeth and is biting his mother's breast. It is quite possible that, soon, the Gaia mother throws her son out!

Part 3

9

Religion

Since the period of the caverns man has had a tendency to religion. This tendency is the result of his evolutionary structure. When he received the neo cortex, or superior brain, he began to have important individual importance which was easily accessed through intelligent actions. On the other hand he owned the gnosis, or evolutionary knowledge, which was lost in the ludic of his neo cortex.

Dichotomy

This duplicity knowledge, the individual or intellectual and the evolutionary knowledge, or gnosis made man, since the cavern period, present dichotomization in his way of thinking and being.

Man always had a tendency to believe that he is his superior brain valuing the individuality in detriment with the whole man who is the four men together (visceral, reptile, mammal and sapiens).

The first actions of man in relation to religion were the care with the dead and the belief that a superior being or god existed. These two tendencies happen from the shock with the gnosis. The care with the dead intended to prolong an individuality that the reality showed at every moment as transitory; the belief in one god justified the transitivity of life.

To find an equilibrium and a certain degree of happiness man, since the beginning, had to live with this duality between religion, which identified him with the man as a whole, and the rationality decurrent of his superior brain, which through the individual knowledge tends to manifest him as a misrepresentation of the man as a whole that he is in fact.

Man is an unstable and ephemeral being like the protean molecules over which the impulse of life was installed initiating the evolutionary process. During this process the impulse began to transfer knowledge to the living being through a process which I will denominate Gnoseogenia.

The last act of this knowledge transference process was the creation of the superior brain which gave man the potentiality of using all the evolutionary knowledge (gnosis). The potentiality, however, is not the gnosis knowledge. To access the evolutionary knowledge (gnosis) he has to behave as a unity or a man as a whole and not as a freak which happens to be man uniquely sapiens. The access to the gnosis can only be made by the emotional (mammal man) which, on the other hand, should be in synchronization with the reptile man which is sustained by the visceral man.

Religion Evolution

Religion was developed and became a creed based on the rational. There were a series of attitudes in disagreement with the rationality which man was forced to make like the young's sacrifice or the sacrifice in the struggle for his territory proper to the gnosis. Man knew that such attitudes were correct because they gave him equilibrium (certain facility). Since these actions were in disagreement with his rationality religion was developed in a way that it would accommodate his onrush decurrent from the evolutionary process of his rationality.

That god became anthropomorphic. Many gods began to exist who cared for specific problems. The care with the dead generated an imaginary world appropriate to each culture where the imposed attitudes of the evolutionary process were rewarded with a better life, full of satisfaction and desires. Each culture created their gods and their post-mortem life.

Contamination

The ludic world became contaminated by this religion that was being created accommodating man's gnosis. The Gnostic knowledge (or evolutionary) began identifying itself with what we began to call religion.

The access to the evolutionary knowledge is done by the emotional brain that does not have at its disposal the information but neuron flashes which the rational man has difficulty of reasoning with.

With the contamination of the Gnostic knowledge made by religion man began to translate this knowledge through the information and the images of his religion.

Miguel Angelo

This process can be identified in the two frescos which Miguel Angelo painted in the Sistine chapel commented before.

In the creation of man an assembly manifested itself with neurons (emotional) which express the creation of man through the religious concept Jew Christian: iod-he-vau-he. The first fresco shows, inside the religious knowledge, the impulse of life transforming the arboreal proto man in the sapiens man through a blow of life.

In the other fresco, leaving paradise, the evolutionary process on the snake that climbs the tree in the form of caducei is expressed; the gnosis (apple) being offered to man and the struggle that man has to face to find equilibrium (happiness) in the duality of his mind.

The gnosis (evolutionary knowledge) is present, however, was adapted or contaminated by the active religion.

Ecstasy

The Gnostic knowledge access is done by the emotional brain which is, in reality, the evolutionary sequence of the superior brain.

There is, however, a series of obstacles which reasoning puts obstructing this access. First there are the five senses that are constantly giving information and occupying attention. Secondly, the personal knowledge occupies the center of the common man's attention.

During an emotional crisis, like the one that occurred with Miguel Angelo being forced by Pope Julius II to paint and finish the ceiling of the Sistine chapel, the gnosis access happens. This state is commonly called "ecstasy". During the ecstasy man loses contact with his senses. It's as if he was isolated from the world and his personal knowledge was left out of his attention. This way a gnosis is accessed with more freedom but it brings with it the culture and religion contamination.

Static

The ecstasy state that permitted a privileged access to the gnosis is the basis of all the religions. The static brings, from the ludic, fables that are received and understood by all men of our culture. Everyone understands the fable because they all contain the same emotional brain and one same culture or a same ludic deformation of the gnosis.

Since primitive man there are shamans who, in ecstasy, transmitted the gnosis to all tribe. Among the Greeks the oracles were manifestations of static pythonesse. In Christianity the static manifestation appears as being a divine perception. Pentecost, as revealed in the Apostles' Acts, is a group manifestation of static.

Message

The static brings a message which, although it is contaminated, is a gnosis expression (evolutionary knowledge) and because of this is accepted and understood by all even if it is out of the parameters rationality. The Gnostic message comprehension occurs because everybody has the same emotional brain and the same culture. There is still an echo or reverberation phenomena which make the message, echoing in other brains, complement and arrange it mounting a new fable which might not be exactly the original ecstatic message.

Global Mind

The emotional brains of all men are equal and the ludic rational brain is also equal to the same culture because it suffered a similar deformation.

When the static brings a message with Gnostic knowledge everybody has, immediately, the notion of the importance of the message because it is in synchrony as its whole. When transmitted to the others the messages will be adapted to the understanding and the reality of the culture by the resonance with the other brains being then known as fables.

Through this mechanism we can notice, in Mythology, specially the Greek, a deep Gnostic knowledge (from the evolutionary process) because our culture is from the Greek culture. There is a global mind from our civilization that perceives the importance of each mythological passage because each one of the passages is part of the human gnosis.

Personality and Individuality

The human brain is the component of the three superposed brains that function as a group as if they were interchanged computers. Because of the neo cortex dictatorship we are raised to be rational or only as Homo sapiens, forgetting that we are mammal, reptile men and that we still carry the visceral man who, in the end, is the true creature that manifests the impulse of life in us.

We become perplex when we notice that our personal knowledge and that our personality (sapiens man), which we judge as our individuality, are nothing more than an artifact that life's manifestation used to give pursuance to the evolutionary process. What we believe to be our individuality is our personality, which is nothing more than a true individual artifact which is in the global man.

Species Union

Our personalities are really very separate. But our true individuality is very similar, if not the same, as every man. We are totally equal visceral men. We are reptile men and we have an intuitive behavior written in our genotype exactly the same way. We are mammal men with an emotional brain equal to all the species and we judge ourselves as equal for all our species and we think of ourselves as special because, at the tip of this whole structure, we have a neo cortex that functions almost like a hat!

The gnosis is the true understanding of all of us. Because one of us has a feather hat (the one who is the master); and the other a cap (the one who is a slave). We believe we have different knowledge.

We are all a species that behaves as such. The same way as the gnu walks in the savanna in a fixed formation. The same way as the older baboon sacrifices himself for the group. The same way the bear becomes squalid to feed her cub. We transfer these conducts to our culture, making adaptations, disguising our animal reality.

Besides this specie's union we are committed with the evolutionary process, perpetuating the impulse of life that wants to constantly be transferred to other groups of instable and ephemeron proteinic molecules. This way we are committed with the whole life and the whole planet.

Comfort

Man cannot live with a freak which is a uniquely sapiens man. The personality which man developed, allegedly rational, cannot lead him to a position of balance (happiness). Man can only manifest himself completely when he accepts his emotional condition, which connects all men that live in him.

Rationality impedes man to have a global manifestation and, because of that, he is unhappy. To find some equilibrium man has the tendency of throwing himself to religion, where the Gnostic knowledge (evolution) can be brought to conscience, giving him a certain comfort.

When suffering some evolutionary embarrassment, as the loss of a family member or the difficulty of providing or reproducing, man gives himself into religion. Men are sensible. The one that has an active emotional component tends to be religious.

Religion is, basically, the comfort that man finds to be able to manifest himself in a more complete way in a world dominated by the dictatorship of the neo cortex.

Part 3

Christianity

At the beginning of Christianity there was a group of religious people, the Gnostics, who noticed that the true knowledge was a superior one (evolutionary) and not an individual personal knowledge. This sect began to encourage the access to the knowledge through the emotional and was attacked by the church, which could not accept the access freedom to a superior knowledge because this jeopardized its divine intermediation. Almost all the documents of the sect were destroyed. The Gnostics became, in practice, known through the writing of a bishop from Lyon (Irineu) who misrepresented totally the sect's truth.

The Gnostics

Many writings and concepts were left in which the Church seized and which were obtained as messages native of the Gnostic knowledge (evolutionary), accessed through the emotional ecstasies.

Church

The four gospels of the New Testament, chosen by the Church, make reference to the four men that form and live the integral man: Matthew, Mark, John and Luke.

The access to the Gnostic knowledge was always combated by the Church in a drastic way and, even, cruel. There was a new try near the year 1,000, in south France, which was known as the Albigenses or Cátalos (pure) sect that was cruelly smashed by force in an unprecedented slaughter. The crusade was commanded by Simon de Mont fort, supported by the pope Inocencio III in whose order is known to decimate the Cátalos: "Kill all. God will recognize the ones that are his". It was during this time that the padre Domingos de Gusmão who, to evangelize the heretics, created the Dominican religious order; later responsible for the Inquisition.

Saint John

The gospel, according to Saint John, is a marvelous message from the Gnostic message, translated to the Judaic culture. It is not known for sure if it was written by Saint John or by an initiated group. This gospel is basically an emotional access to a Gnostic knowledge (evolutionary), translated as a message to the Judaic culture.

Saint John wrote two books: The Gospel and the Apocalypse.

The Gospel begins with the pure divinity, identified as logos, which will be manifested and ends with the divinity imprisoned on the cross. The apocalypse initiates with the divine resurrection and ends with the return of the logos. I will talk about the Apocalypse in the next chapter (Salvation).

The Gnostic writings of Saint John dramatize the evolutionary process. The impulse of life, at the beginning, is something in nominate which will be manifested through a series of living creatures, reaching a bigger complexity and ending with man's appearance, through his neo cortex, who was imprisoned, crucified in the material world by the five wounds which are his five senses. In a dramatized way man's evolutionary process is here described since the impulse of life manifested itself in the first cell, until the man of our era, imprisoned by the neo cortex's dictatorship.

Saint John believed that a god manifested the impulse of life and that this occupied, gradually, living creatures increasingly more complex, to which the Gnostic knowledge (evolution) was transmitted. At a determined point the evolutionary process was stuck to the materialistic by a perception of one's world, individual, that was reached by the five

senses (five wounds), giving the impression that its personality was the individuality. The message symbolizes clearly the deadlock in which man finds himself in this moment of the evolutionary process. Man cannot be saved if we analyze the parameters of our personalities, which maintains him trapped to the world by the senses. The neo cortex, alone, does not have the capacity of saving the whole man from his earthly personality.

Devil-Satan

What is happening is as if the impulse of life were betrayed by the materialization of the human personality, as if divinity had created a power (neo cortex) which was trying to be on a par with it, divinity.

Here the model of opposition is symbolized. Satan means "opposition". The one that judges himself as similar to divinity; like a decadent angel who has the pretension of being equal to God. It is written in the gospel that when Jesus, when leaving the desert after going through all the temptations of personality, left as a complete man and becomes Jesus Christ. Jesus symbolizes the material body of this man and Christ the impulse of life that lives in him. When leaving the desert Jesus told his opposition, Satan: "Step back, Satan". He says: follow me, behind me. Not: disappear or leave. It is necessary that the opposition, the neo cortex or Satan be marching together with the complete and full man. The sapiens man must serve the complete man and not the complete man to be serving the sapiens man. This is what is happening in these transformation times in which we live in.

As can be seen through this passage there is in the gospel the true understanding of the gnosis which is here expressed by a symbolism appropriate of the emotional.

Crucifixion

Saint John's gospel is the true story of the impulse of life, which manifested itself in the living creatures through an evolutionary process, transferred the potentiality of its knowledge (gnosis). The living creature, man, with this potentiality wanted to be the very own divinity.

During history narrated in the gospel we have the man (Jesus) son of a woman, which represents the human body or the living creature in which the impulse of life lives, we have Christ who is the impulse of life himself, the light that lives in all men who comes to this world; we have Jesus Christ, the son of man, who is the man completely manifested, the total man, that harmonizes all the unities.

When leaving the desert, where all his rationality (neo cortex) is submitted to man as a whole, or to the impulse of life, Jesus leaves christened (**cristificado**) as Jesus Christ. From this episode Jesus begins speaking in the name of the impulse of life or in the name of Christ.

The gospel's message is, some way, terrible because it shows us man as a living creature, dying crucified through the senses in which he perceives the world.

In a practical sense the gospel wants to show us that man, behaving as he is, like a sapiens man, like a misrepresentation of man as a whole, is not feasible and will be destroyed by him. What we see in today's world, at the beginning of the XXI century, is Saint John's prophesying in the gospel. Man, through the hyper-wounded development of his personality, of his selfishness, of his understanding that the world rotates around his senses, is destroying himself, life and the planet. The crucifixion of the living creature man is symbolized through his five wounds which are the five senses in which he communicates with the world.

Gnostic Translation

The impulse of life is not correctly a god. It is a verb which in the beginning was in god and manifested itself using the protein molecules, specially the ones of carbon.

Everything was done by the impulse of life and without the impulse of life nothing would have been done. In the impulse of life is where life was! The knowledge of life, the gnosis, was man's light.

The gnosis shines in the living creature which is darkness, uniquely, protein molecules. These molecules did not recognize it.

"In the beginning was the Word, and the Word was with God, and the Word was God. He was in the beginning with God; all things were made through him, and without him was not anything made that was made. In him was life, and the life was the light of men. The light shines in the darkness, and the darkness has not overcome it." John 1: 1-5

John the Baptist was the man who abstained of all the materialistic necessities so that the sapiens man would serve the total man. He lived in the desert, feed from grasshoppers and dressed with an animal skin. He was a total man who wanted to overpower the sapiens man through force. He baptized using water, soaking the desires.

John gave his testimony that the impulse of life had manifested in the world through the light (gnosis) that shines the ones who come to the world. The men who received the light (gnosis) did not recognize the impulse of life. "The impulse of life comes after me because it already existed before me and will manifest itself as the Christ."

"There was a man sent from God, whose name was John. He came for testimony, to bear witness to the light, that all might believe through him. He was not the light, but came to bear witness to the light.

The true light that enlightens every man was coming into the world. He was in the world, and the world was made through him, yet the world knew him not. He came to his own home, and his own people received him not. But to all who received him, who believed in his name, he gave power to become children of God; who were born, not of blood nor of the will of the flesh nor of the will of man, but of God.

And the Word became flesh and dwelt among us, full of grace and truth; we have beheld his glory, glory as of the only Son from the Father. (John bore witness to him, and cried, "This was he of whom I said, 'He who comes after me ranks before me, for he was before me.'")

And from his fulness have we all received grace upon grace. For the law was given through Moses; grace and truth came through Jesus Christ.

No one has ever seen God; the only Son, who is in the bosom of the Father, he has made him known.

And this is the testimony of John, when the Jews sent priests and Levites from Jerusalem to ask him, "Who are you?"

He confessed, he did not deny, but confessed, "I am not the Christ." John 1: 6-20

This initial Saint John's gospel resounds inside all the men because the knowledge of the evolutionary gnosis or the true history of man. John Evangelist wrote the Gospel during an emotional or ecstasy crisis, having this way access to the Gnostic knowledge or the evolutionary process knowledge of the man.

Apollonius

The story of Jesus Christ is so characteristic of the gnosis which is not possible for us to know if it really occurred or if it were created, afterwards, by the resonance of the evolutionary

knowledge. It could have been built on top of true facts, which with certain adaptations generated a perfect story. If there were no Jesus Christ the same story could have been assembled, for instance, over the life of Apollonius of Tyana, a thaumaturgy in the same level as Jesus.

Saint Paul

The Church hoists being the one responsible for propagating the doctrine preached by Jesus. However, the doctrine described in the gospels is very distant from what the Church preaches. And even more distant is its behavior found. The Church is a Pauline institution!

In a research done to discover who influenced the humanity more, Mahomet was chosen first, Jesus second and because the third would be Saint Paul, the two would be the first.

Saint Paul was an extraordinary man. When he persecuted the members of the Assembly of the Way, he suffered an emotional crises. There are those who believe that is was because of an episode of epilepsy from the temporal lobe. The fact is that this crisis opened an emotional channel for the gnosis knowledge which permitted that a formulation of a doctrine where the gnosis was adapted to the men of that era.

After the crises on the way to Damascus, Saul went to Tarsus, which was an important commercial center where all the religions were represented. After seven years of fermentation, he left Tarsus with a Church doctrine ready to be diffused. He made a Judaism adaptation (which he was a specialist), a Mithraism, Zoroastrianism, Egyptian religion. A religious synthesis of the period based on the history of Jesus Christ and expressing a gnosis (evolutionary knowledge), in the end.

Saint Paul noticed that a change was happening in the men. They were having a necessity of perceiving something beyond the roman and mosaic justice; something more subtle; like the compassion or love. In the Salvation chapter I will treat these changes happened cyclically among men.

Saint Paul adapted Jesus Christ's doctrine, basically emotional, to the neo cortex. The progressive rationalization began then to create the Catholic Church which was progressively rationalized, transforming it into the Roman Church, where the chief looks more like a Caesar than with Saint Peter, to whom the first chief was attributed.

Christ and the Impulse of Life

The gospels are clear messages of the gnosis. If the word Christ were substituted for the impulse of life we will see that many of the obscure affirmations will become clear.

When Christ said: "I am the truth, the way and the life", he was not using metaphors; not through parables; he was talking about a crystal clear truth. The impulse of life is the truth, the way can only be done through the gnosis of this truth, and the impulse of life is life itself.

"That which is asked for in my name will be attended". Anything that is the interest of the impulse of life will be resolved by it. Many miracles happen because of the interest of the impulse of life. The impulse of life is capable of making an arm grow through the activation of some stem cells, if this is the interest of the impulse of life. But it will never do it if it is only because of man's interest.

"Those who have a lot, a lot will be given, and the one who little has, little will be taken from." This affirmation becomes clear at the moment in which we know that the impulse of life acts in the macro, in the sense of a manifestation of the whole and not of the personality.

Sensitivity

Man today lives a transition moment in which science reached the peak of its manifestation and did not bring the harmony or happiness that it promised.

Man cannot be happy while he does not manifest himself completely through the four men that constitute him. The sapiens man, with his personal and individual knowledge, cannot be happy until he does like Christ and commands that science go behind him and serve him. Not like now because, since Descartes, man has been serving science.

To compensate this unhappiness the most sensitive men dedicate to religion. I say the most sensitive because the quantity of information, of things and the necessity to do, which the modern man has, makes many of them completely absorbed in him, as a freak manifestation from the sapiens man.

Evangelicals

The need for religion happens from the structure of the human brain. The neo cortex is not capable of manifesting man completely because it is the mere evolutionary artifact of man as a whole.

Living in a world in which it was established that survival depends of the rationalization resulting from the neo cortex, man has manifested himself in a very deformed way, personal and individualistic. Man is not his neo cortex. He is a whole. He has the necessity of giving vent to the emotional perception and because of this he needs to dedicate to religion.

Many people that live with us inside a world marked by rationality, who work with their intellect and are brilliant in their profession, when they “devote to Jesus”, they assume an irrational and not logic behavior, but which permits that they balance their manifestation.

In the Christianity Christ is the impulse of life. When they devote themselves to Jesus they are integrated as a whole, manifesting the impulse of life. The interesting thing is that this devotion, besides the comfort and balance that it brings, makes it possible for the religious man to manifest himself successfully in the rational world and, from this; many pastors take advantage, claiming they are intermediates of such a success.

Rationalization

Jesus Christ's doctrine is an expression of the gnosis (evolutionary knowledge).

Saint John, in state of ecstasy, has an access to this gnosis and writes his two books: Saint John's Gospel and the Apocalypse. In the Gospel it begins with the manifestation of the impulse of life and ends with the incapacity of the neo cortex to save the man which is crucified through their five senses.

The message is dramatic: man cannot be saved by the neo cortex. Science and the technologies will not be able to save man because it accentuated his personality making him move away from the man as a whole from what it truly is.

The behavior of the neo cortex compares itself to the Opposition (devil). The perfect man, Jesus Christ, when leaving the desert, orders the neo cortex to serve him: “Walk behind!”

The neo cortex beats the man Jesus, who dies crucified by the senses.

Saint John writes the gnosis in a clear way and writes his second book: the Apocalypse (Revelation) which will show how the Impulse of Life (Christ) will save man.

Saint Paul begins the rationalization of the doctrine which will transform in a new Roman Empire transvestite in the Christian Church. This Church, which destroyed the Gnostics and decimated the Cathar, inevitably, was condemned to die on the cross, as Saint John showed. The salvation will come from the outside!

The great static is unanimous in predicting the end of the Church in this transition moment in which the devil is all around! The neo cortex brought an immense development from science and technology, which didn't make man happier, because the cost of this progress was the stalemate of the personality and selfishness.

Part 3

11

Cycles

The impulse of life did not remain still at the moment in which it manifested itself through the neo cortex. It continued to transmit to man the gnosis, or the evolutionary knowledge, by the activation of determined regions of the neo cortex. These activations gave man a more specific comprehension of determined evolutionary problems.

Continuity

The neo cortex has evolved since its appearance until the present moment. Its evolution did not imply an adding in the structure but in the activation of certain regions that were opened specific understanding horizons.

The impulse of life acted in two ways in the transmission of man's gnosis:

1. Individual adaptations, capacitating individuals from the group so that the group could manifest itself in a complete way.
2. Adaptations in all individuals occur cyclically when the impulse of life notices that man will be capacitated for a new understanding on the gnosis side.

Nesting

The impulse of life does not stop. It is always transmitting the gnosis to the living creature, even when this one is established in a degree of the evolutionary process.

If we observe a dog nesting we will notice that there are differences in the activities among each one of the puppies. The impulse of life manifests itself making some with more leading characteristics than others. The manifestation of the impulse of life which is capable of making creatures with diverse abilities inside a same species and it is related with the group of manifestation of the species which needs creatures with diverse aptitudes.

The manifestation of a species needs a certain percentage of diverse creatures so it can manifest completely. The balance between males and females in all the species respects a proportion that depends of the social structure. The ant hill and the bee hive are a demonstration of how the impulse of life determines and regulates the species' aptitudes.

The Code of Being

In the own human species the impulse of life generates different beings from a same group so that the group has a better capacity of organizing itself. In a universe of one thousand men the impulse of life gives birth to some ten with a more activated region of the neo cortex, giving to each more leading possibility.

Among these ten men prepared for leadership maybe only one will become a leader and the other nine will suffer an embarrassment which they cannot identify. The same happens with leadership. There should be other qualities necessary to the manifestation of

the group in which life prepares some individuals activating determined regions from the neo cortex.

The impulse of life behaves as if it put in each unity of a group a code, character or a vocation for a determined characteristic necessary for the manifestation of the group. Those who are able to manifest their objectives or vocations find better harmony in their lives.

Today we know that Einstein had the temporal lobe a little bigger than the average of the other men, which gave him a superior special reasoning possibility. There might have been in his generation some other men with the same characteristics who did not find their destinations and lived misfit in their social environments.

Men were born with a personal code which the impulse of life established considering the manifestation from a group or from the species as a whole. The observation of this manifestation characteristic's impulse of life was identified by the English psychiatrist James Hillman in his book *The Soul's Code*. It is a search of the character and personal vocation in a different way from this but inside the same context.

It seems like each man has a code and, consequently, a destination with a project to fulfill in his society and is happy when he is able to find his path.

Abilities

The impulse of life prepares individuals with different characteristics according to the necessity of the whole. This code is not from a genetic origin. The children of a leader do not have, necessarily, qualities of leading. The distribution from these characteristics is made according to the necessity of manifestation of the species in determined groups. Leaders are born among rich, poor and the most varied segments of our society.

The Christians, being a gnosis means, knows of these characteristics and thought it conventional in naming them "abilities". Saint Paul, in his epistle to Corinthians 1, chapter 12, identified such characteristics and attributed them to a blessing from the Holy Spirit. The abilities are distributed equally.

The impulse of life retains a manifestation control of the species or of the group. This control seems to be exerted by the expiation of knowledge membranes that involve the group and the species.

Mediumship

The impulse of life prepares some creatures with a bigger preparation of the knowledge membrane which involve the specie and the group. These creatures make their access to the ludic where the evolutionary knowledge (gnosis) finds itself. They are emotionally very unstable and, fairly, this instability permits the perception of the emotional flashes. Through these flashes they access the evolutionary knowledge being able, many times, to notice the knowledge membrane which the impulse of life expanded at that moment.

Joan of Arc

During One Hundred Year war France was being annihilated in their own territory by the English and their allies. The French people represent a group of important human beings in the manifestation of the whole human species, probably decurrent of a proper way of being and thinking of the Celtic people. The impulse of life expanded a membrane which detected the possibility of annihilation of this group that should be preserved. Such a knowledge membrane involved all the French and manifested itself in the ludic or oneiric form of a legend in which it affirmed that a virgin would save France.

From this period there are reports of a great quantity of virgin girls which went mad. Among them was the sister of King Charles the VII. These virgins had the perception of something very important to which they were being called. The perception could not be translated to the neo cortex because it was an evolutionary Gnostic knowledge in the emotional brain. It occurred through emotional *flashes* which disoriented them and many went mad.

To make the transmission from the gnosis to the neo cortex possible it would be necessary a journey which would have its neo cortex adjusted with the emotional brain; a neo cortex that would know how to behave as a man's full server; submitted to the total man without the vanity, selfishness or pretention of being the center of man. This purity condition was manifested in Joan.

Right at the beginning of puberty, during the hormone revolution of the age, Joan of Arc had the consciousness of the emotional *flash*. The simplicity of her formation permitted that the gnosis were received in her neo cortex and decoded to the superior brain knowledge. She began to hear God's voice who told her to be good and to obey the Christian tasks.

The impossibility of translating an emotional knowledge from the neo cortex makes it possible for the sensitive to notice the knowledge through the activation of some brain region responsible for a rational impression from the external world so that what is being felt, by tem, has some rationality. There are mediums that see things and others who listen so that this way the neo cortex rationalizes the perception.

With the neo cortex unblocked, because it is participating of the gnosis, being good and following the Christian duties, the membrane knowledge, to save France, could manifest itself fully, permitting that Joan of Arc give proof of being the deliverer of the salvation message. The king had her admission; talked to Saint Miguel; heard the voices of Saint Catharine and Saint Margaret. It functioned as a catalyzer so that there could be the union of the French people, around a Gnostic knowledge, and France was saved by Orléans. Joan had no battle tactics and knew nothing about war. Her victory was only due to the fact of being able to transmit the gnosis to the French at that moment.

Human Cycles

The impulse of life acts continually over humanity. In an individual way we saw this in the abilities furnished to compose the group. It acts over all humanity activating neo cortex areas as man acquires conditions to understand the gnosis.

This global action is done through the human cycles, when man has a determined area of the neo cortex activated, going through a different perception of the world, of him and the society in which he lives. It is not, exactly, an important modification from the neo cortex but an activation of areas that cyclically modify the human perception. The cave man, although having a neo cortex like the present man, had a totally different perception of the world.

Since we have known the history of humanity we can identify social behavior cycles. The impulse of life, periodically, activates the neo cortex permitting man to advance in the evolutionary sense to come to form an integrated human species among them and integrated to the planet.

First Cycle: Strength

The first history registration remounts to the peoples from the Mesopotamia and Egypt. These peoples had a society notion where strength dominated. The stronger ones dominated the weaker ones; there were masters and slaves. Both masters and slaves believed that was the world's natural world.

The impulse of life, until this moment, had the interest in selecting the strongest men for the species' survival. The concept of justice did not exist among them; it was fair for the strongest to dominate the weakest. Such a form of noticing the world and society remained for millennia.

When this cycle selection of the physically strongest was completed the impulse of life began to have interest in organizing the social group to be able to permit a more effective manifestation of man in the whole planet. It expanded the social membrane that began to cover the groups of men.

Quite suddenly creatures that had some neo cortex connections activated began to appear with the perception that there should be right equality among the groups of men. The concept of justice was activated in men's neo cortex.

Intellectual Mutants

This impulse of life's action will start making men begin to appear who, having connections from the neo cortex activated in a different way from the others, behave as mutants which are not genetic or physical mutations. The mutation is basically intellectual and is imposed by the impulse of life by the activation of the brains areas. They are not real mutations. They are intellectual mutations.

Hebrew

The passage of the strength cycle for the justice corresponds to an important Gnostic knowledge. The Egyptian man, for instance, we can see that had a structural society understanding by the strength. There were the strongest ones and the slaves. The slaves also found it natural that the masters would sustain them.

The Egyptian could not understand the Hebrew who, inside his land, came to manifest the justice concept. Justice for him, Egyptian, was that the strongest submitted the weakest. With the coming of a new concept of justice, inside this society, the activation of the Hebrew neo cortex happened, which was the slave in Egypt.

It is natural that with the coming of the mutants had happened in the slaves and not in the master because the necessity generates the evolution. Justice is an emotional abstract concept hidden in the ludic of the neo cortex which, at a determined moment, can be perceived by man. When the perception of justice was permitted, the impulse of life gave an important step in the human society's integration.

The end of the strength cycle, or Egyptian, is marked by the exodus of the Hebrews.

Exodus

The book of Exodus is impregnated of Gnostic knowledge and this must have been the reason for its inclusion in the Bible. The departure of the Hebrew people from Egypt is a ritual simulation of freedom which the impulse of life, constantly, executed to save man.

Moses who was Hebrew, was raised as an Egyptian. He married Jethro's daughter, the master of the Mesopotamia culture, gathered the whole world's knowledge. With the possession of this gnosis he noticed that the impulse of life would act determining a mutation among the Hebrew slaves. It is impressive how Moses leaves with empty hands to free the people, without armies, based, only, in the gnosis in which the impulse of life was expanding a new social membrane. The plagues which convinced the Pharaoh might have been, as some affirm fruit of the biology and geology knowledge of Moses but, undoubtedly, are connected to the gnosis which permitted him the integrated perception of life as a whole.

Right at the beginning of the desert's peregrination the Hebrew's revolt is described. They wanted to return and go back to being slaves which show that not all of them were mutants. The exodus lasted sufficient time so that the generation that left Egypt would disappear.

The Hebrew's revolt, made clear with the confection of the golden calf, is also impregnated with gnosis. The bull, symbol of the Egyptian culture, represented the strength which was being traded by the law tables. It was the passing of a manifestation time of the impulse of life.

At Egypt's apogee period the Earth passes in the place of the zodiac corresponding to the Taurus constellation. The Egyptian bull symbol Apis, the confection of the golden calf symbolizing the return to the strength and coincidence change dominion from a zodiac era are facts impregnated with gnosis. It makes us believe that the impulse of life has a knowledge which involves interests beyond our planet.

Second Cycle: Justice

The mutants' appearance with cerebral areas activated which covered a more ample social membrane marked the beginning of a second human being cycle.

The beginning of the cycle was represented by the leaving of the Hebrew people from Egypt, through the peregrination in the desert, until all the prior generation disappeared. When they reached the Promised Land all Hebrew were already mutants and had the justice concept impregnated in their convictions which made them believe they were the chosen people by the impulse of life to give continuity to humanity.

The justice mutation happened in all humanity and men were traded by new men, more capable of uniting themselves in societies and of manifesting their own culture.

The mutation happened among the fishermen in the Aegean Sea and gave origin to the Greek and Etruria culture, which were the cradle of Rome.

The new mutants had a bigger capacity to manifest themselves as a group and to survive they dominated the world.

The second cycle culminated with the Romans domination of the world. This fact occurred because they were the most capacitated. Their cerebral function for justice implied advantages in the social organization, such as the discipline.

When creating this justice mutation the impulse of life gave a big step in the transference of the gnosis to the human species.

This justice cycle coincided with the passage of the planet through the zodiac constellation of Aries.

Saint John Baptist is the symbol of the justice era inside the Christian gnosis. His justice notion decapitated his head through Herod's hands because he opposed his relation with his sister-in-law. John Baptist is represented accompanied by a lamb, which is the Aries symbol.

Third Cycle: the Compassion

The manifestation of the third cycle's impulse of life's forthcoming, by the neo cortex's areas of activation, occurred with the Christian appearance.

The impulse of life felt the necessity of broadening the relation among the human beings, increasing each being's participation in the whole manifestation. This necessity happened because of the fact that humanity, composed of justice mutations, began to create great empires. These empires disfigured the gnosis in the manifestation of the impulse of life in equilibrium with the exterior environment.

Empires

The great empires began to be ruled by old strength laws. The strongest dominated the weakest who disappeared and an important loss of knowledge already manifested a patron of knowledge.

The solution found by the impulse of life was to expand a new membrane to give an understanding, besides justice, that every man formed humanity; that each one's life was connected to the life of the others. This was done through the activation of the connections which permitted the compassion perception. The humanity union factor would no longer be the great empire value, result of the Justice era, because men would begin to unite themselves through emotion.

Compassion Mutant

The impulse of life created a compassion mutant that had other connections in the activated and catapulted neo cortex to its neo cortex.

The new mutant presents a compassion feeling which is not proper to the gnosis but the result of a resonance of a Gnostic knowledge in the neo cortex which determine the union of all men around a species. The same Gnostic knowledge through which a baboon sacrifices for the group.

For more than two thousand years compassionate mutant men began to appear. There are not truly mutants because they have no genetic alteration but they have encephalitic connections activated in a diverse way, different from the anterior man. These new men had connections which created a union necessity to the group through emotional attachments which did not occur to the anterior man.

The phase "Love your neighbor as thyself" had no signification to the justice man because making justice had no necessity of love. Affection or emotional attachments were restricted to family ambience.

Martyrs

The compassionate mutants formed groups who thought the same way; who understood themselves and at the end adapted themselves to a new doctrine to justify their intimate necessity of compassion.

The initial necessity was noticed in a very confused way. This created many uncontrolled attitudes making them believe that they were tuned in with an interior necessity.

The coming of these martyrs can be identified among the martyrs from Church. They threw themselves at the beasts like the baboon did with the lion.

Jesus Christ

At that time there was a change that occurred inside men's head and not, as is thought, that it had been determined by Jesus Christ's appearance. Christianity was assembled over the transformation that happened among men.

It is quite possible that the Passion of Christ had been a staging of the passage of an era; from the justice to the compassion; from the Aries era to the Pisces. This hypotheses is well justified by Spencer Lewis in his book about Jesus.

The theater act would have been assembled by Joseph from Arimathe'a, who belonged to the Essenians. Joseph, Jesus' father, was Essene. Pontius Pilate was the son of the Roman governor in Egypt when Jesus was there a refugee. Joseph from Arimathe'a asked to remove the body from the cross and bury it in a tomb in his property.

The resurrection was the bases of the new perception that life was involved in a bigger context than the individuality. The Jews from Sanhedrin's preoccupation with the resurrection is symptomatic in this articulation.

Composition

It is quite possible that the history of the Passion of Christ and great part of the gospels were composed over Jesus Christ's life because of an intimate necessity of the new mutant's psychic which showed up among men. The new mutants had the necessity of finding rationality for the emotional transformation which was presented. In Saint Paul's case, not having the light on the way to Damascus and spreading his Christianity, it is possible that a completely different Christianity were composed and adapted to another thaumaturgy, such as Apollonius of Tyana.

All the Christianity's history was sediment of almost 100 years after Jesus' passage through Galilee. It seems like the synoptic gospels (Mathews, Mark and Luke) are based on a text which lost itself and which could have been the notes of Andrew. John's gospel is clearly a result of a Gnostic knowledge accessed in a state of emotional ecstasy.

Christianity is the result of a transformation rationality through which man went through during the Era change. The rationalization is impregnated with the true gnosis which is adapted to the comparison conditions of the man from his period.

The Centurion

"After he had ended all his sayings in the hearing of the people he entered Caper'na-um. Now a centurion had a slave who was dear to him, who was sick and at the point of death. When he heard of Jesus, he sent to him elders of the Jews, asking him to come and heal his slave. And when they came to Jesus, they besought him earnestly, saying: "He is worthy to have you do this for him, for he loves our nation, and he built us our synagogue." And Jesus went with them. When he was not far from the house, the centurion sent friends to him, saying to him, "Lord, do not trouble yourself, for I am not worthy to have you come under my roof; therefore I did not presume to come to you. But say the word, and let my servant be healed. For I am a man set under authority, with soldiers under me; and I say to one, 'Go', and he goes; and to another, 'Come', and he comes; and to my slave, 'Do this', and he does it". When Jesus heard this he marveled at him, and turned and said to the multitude that followed him, "I tell you, not even in Israel have I found such faith." And when those who had been sent returned to the house they found the slave well."

Luke 7:1-10

This passage described by Saint Luke, who was not a Jew, is impregnated with gnosis. We have, in the centurion, the man well manifested for his time. He had a clear perception of justice and especially for this he could have the perfect understanding of the compassion. He directs himself to the impulse of life which he perceives in Jesus and asks for compassion. He knows perfectly well that the impulse of life can cure his slave, as he does his part, participating of the gnosis. His request becomes concrete (whatever is requested in the name of the impulse of life will be done). Christ (the impulse of life) confirms that not even in Israel has he found such faith.

There is a new perception which appears among men and not exclusively among Jews, as was later demonstrated by Saint Paul preaching the gentiles. There is no Judaic sequence doctrine which became concrete in Christianity. There was a perception transformation which adapted the Judaic doctrine to the new perception that appeared in man.

The centurion, for having a justice sense, could notice correctly the compassion mutation, which did not occur among most of the people from Israel.

Compassion and Justice

Man can only have compassion if he knows justice. To have compassion it is necessary that before he be just. If not the compassion will be distributed in a corrupt and selfish way, no longer representing the gnosis.

It is only possible to exercise compassion those who exercised justice! The compassion not preceded by justice is always tendentious. The true compassion should be exercised equally among affectionate and unaffectionate. Among sons and those who are not. It should be exercised, finally, equally among men. The son rises for all!

Not having the perfect understanding of the transformation in which they went through, the martyrs threw themselves to the beasts.

Church

The great majority of men had no justice experience to have the perception of the compassion in the form shown in the centurion's story. An adaptation of justice became necessary so that the great majority of men could find harmony with their new compassion perception.

The justice experience was expressed by Rome. That's why the compassion adaptation perception occurred (which would be the true Christians) with the roman justice. From this fusion the Catholic Church was born.

The Church was begun involved with the purest compassion concepts, manifested by groups known as Assemblies of the Path, the first Christian nucleons. Right after Saint Paul's transformations were made, adapting the compassion to the valid religious concepts. Later the Church took Rome through an emotional involvement which culminated with the temporal power seize at the time of Constantine.

Through the Church the gnosis was manifested during these 2,000 years, carrying with it the distortions resulting from the lack of experience from a true justice among men. With the Church compassion associated itself with power. The pope is more of a Caesar than Peter!

It was exactly this distortion which proportioned the reappearance of the gnosis arising from Greek culture, during the Renaissance period. The dichotomy science/religion happened from the delay of humanity in living the true justice. Among the Greeks the dichotomy happened when Platoon abandoned politics and stimulated philosophy, which came to privilege the individuality and the selfishness. Among the Jews and the gentiles' religiosity without justice generated selfishness among men, materializing in the Church, which dominated our culture until Renaissance.

Part 3

12

Egocentricity

Man is going through a very difficult existing period. The way he behaves, exasperating the individuality through an uncontrolled consumerism, disrespecting his fellow men and destroying the planet. He became an impracticable living creature. The impulse of life can destroy him at any moment the same way as it did with the reptiles. The planet presents cycles to which the living creatures must, necessarily, adapt to.

Man can be substituted, in the evolutionary process, for another living creature which will adapt to the new conditions of the planet. He judges himself as the center of creation and with his neo cortex he believes that he is dominating the manifestation of life. But he forgets that a simple virus can, at any moment destroy him or at least cause a cataclysmic transformation in the world.

The impulse of life can save man, destroy him or reduce his influence over the planet in a drastic way. It will act because man, behaving as he is, will become impracticable for the evolutionary process of the manifestation of life.

The mystics that access, in a distorted way, the Gnostic knowledge are unanimous in identifying the actual moment as a great change or an end of a period. Some point to destruction; others to a selection by catastrophes and others the salvation.

The great problem man is going through can be simplified by his egocentricity. Each time more the man of today is becoming selfish and less participant of his family, group, species and Life. The scientific progress in spiral at the end of the millennium emphasized his individuality.

Man is living and operating each time more inside a closed system. This is how Maturana and Varela defined in the biological knowledge bases of the living creatures.

Operational Enclosure

Every living creature, since the first manifestation of life until man, is constituted by a group of protei molecules, which shelters the impulse of life.

The organization of these molecules goes through an orientation given by the impulse of life, according to an evolutionary knowledge which is the gnosis. Each living creature is assembled according to a gnosis plan to react in a determined way from an environment stimulus.

The construction plan of the living creatures is written in the nucleic acid through a language which is still not understood by us, despite the immense Gnostic advances. The impulse of life obstructs the knowledge of how these molecules are manipulated so they can keep and orient the structure and the assembly of the living creatures.

The true gnosis is a whole in the first cell as it is in man, who is the peak of the evolutionary chain. The gnosis is being transmitted to the living creatures in homeopathic doses, controlled by the genetics, which permits an arbitrary limit to the living creature's structure. The true knowledge (gnosis) can be restrained in a tree. But it can only manifest the knowledge to which it was assembled. It will blossom flowers in spring and hibernate in winter. The gnosis transmitted to it a limited knowledge which, as strange as it may be superior to man's knowledge many times, who is at the top of the evolutionary chain.

When I told the pau-d'água it was evident that the tree had a superior knowledge to mine to resolve its problem. Gnosis' transfer is not linear! A bear can know, more precisely than man, when the Earth is at the extreme point of translation movement through its metabolism for its hibernation.

The gnosis transfers knowledge to the living creature as it judges necessary that the living creature disposes of this knowledge.

Nervous System

In the segmented living creatures the stimulus reaction from the environment was decided through a knowledge transmitted genetically to each segment. Each segment reacted in an independent way.

With the appearance of the nervous system it was possible for the living creature to show a reaction involving the whole being. The reaction of every living creature to an external stimulus generated a new kind of knowledge transmit ion (gnosis) for the living

creature. The impulse of life began to assemble, genetically, a structure which would have a specific knowledge. The nervous system of the worm is assembled to order the contractions of the body's segments allowing its dislocation.

The reptile's primitive brain, specially the cerebellum, is assembled so that the animal can have coordinated actions according to the necessities of survival of the species. The knowledge is not in the cerebellum but in the group of the reptile's nervous system.

The nervous system is assembled through genetic information so that it can function as a whole, coordinating the actions. Separately its components have no function.

When this type of knowledge passage expired, with the great reptiles, the impulse of life created the intermediate brain where the knowledge could be transmitted from the father to the son through the upbringing. The upbringing was also limited to the own circuits of the species intermediate brain. A determined stimulus furnished by the mother, which activates a certain behavior in the gorilla does not have the same effect on the lion.

Eyes to see

As can be seen, so far, the transmission of the Gnostic knowledge is tendentious for each kind of living creature. Each living creature has its own knowledge of the external world. It sees the world through its own eyes. Each living creature is capable of knowing what was prepared for it to see.

The eagle has an eye capable of detecting movement kilometers away because it feeds off preys which move on the ground. But it is not capable of telling apart an elephant from a bull.

The antelope feels the sent of lion kilometers away but does not perceive other stronger aromas that impregnate the savanna.

The mammal has a nervous system prepared to know a selected part of the gnosis (evolutionary knowledge). This phenomenon which every living creature presents is an operational clause. Each living creature operates inside a closed system to which it was assembled. Here is the biological base of knowledge.

The living creature is assembled to only know what is important for its species, living imprisoned in an operational prison.

Limitation

The operational prison concept applied to man is terrible and unsettling. It destroys the image that we make of ourselves and of our knowledge. Man is a living creature and cannot deny this prison. He was assembled to see and know one selected part of the knowledge held by the impulse of life. He was assembled to know a part of the gnosis! The whole knowledge of the gnosis was not given to man!

Emotional Catapult

When the impulse of life saved the arboreal proto man who fell in the savanna it was given to him the potentiality of knowing the whole evolutionary process. This was done with the creation of a neuron artifact with the possibility of storing the information collected during the evolutionary process and, also, of weighing this information to find solutions of survival.

This process was done by an emotional brain orientation of the mammal (emotional), which directed and organized the superior brain (neo cortex). The evolutionary knowledge (gnosis) written in the primitive brain was, this way, catapulted to the superior brain.

Total Gnosis

Man's superior brain has a potentiality of expressing, in the form of knowledge, all the evolutionary process (gnosis). This fact places man ahead of a paradox. Man's body was assembled to operate in an operational prison system as every living creature; it was assembled to see and know a specific part of the gnosis that might help it in its survival. How is it possible for man to know all the gnosis if he was assembled to know only a part of it?

A paradox

Man was assembled to see a part but with a possibility of knowing the whole!

Knowledge Membrane

This paradox is resolved when we introduce the membrane concept.

The impulse of life develops the evolutionary process through knowledge membranes which are launched to the exterior environment where information is brought which is transmitted to the living creature.

The first membrane seems to be launched when the first living creature appeared. If the first living creature were to indefinitely multiply itself, it would have terminated with the food of the environment. Some knowledge membranes were launched and determined alterations in these initial beings, where they were oriented so that there should be a manifestation balance in the impulse of life. This fact can be remembered when I referred to what would have happened in a try to colonize Mars.

In the living creature, which reproduces through the gamete formation, the evolutionary knowledge (gnosis) is not restricted to its physical body and transcends its operational prison. In its prison it has the necessity of completing the gamete. But this complementation will be done with the gamete from another living creature which has another prison.

Another important moment of knowledge membrane expansion occurs during a formation which already behaves as a social membrane where the unities are involved in a same type of manifestation.

It can be considered that society initiates with a sexual reproduction form in which it is necessary the association of two beings, unstable and ephemeron, to create a new living being who will continue to pass ahead the impulse of life.

As can be seen, the operational prison that defines the living creature's biological knowledge base has an expansion outside the living creature since life's first manifestation.

The knowledge membrane, in the reptile, involves all species. Every turtle lays its eggs in a determined beach. The salmon returns to the same river where it was born. The birds travel in flocks and orient themselves with perfection in their migrations. This information might be written in the genetic code of the reptile but it involves a knowledge which is of all reptiles of the species and not exclusively from a living creature's unity. The knowledge membrane detains a knowledge which is not imprisoned in the living creature's operational system.

Social Membrane

In the mammal the knowledge membrane expands and transforms into what we know as social membrane. All mammals live in some type of society. They are involved in a growing membrane which transcends its body and its emotional prison.

Even the lonely carnivorous, like the tiger, seek for matting and behave in a definite way to raise its cub. Certainly this was defined during the emotional brain's activation. But its partner, cub and the other tigers are involved with it through a social membrane. The social membrane has its own knowledge that might not be in each unity.

Social Operational

The social membrane of the mammal can, occasionally, be so strong to the point of destroying it to benefit the species. The female bear reach exhaustion and fatigue to a point of being destroyed during the period of nursing of its cubs. The living creature's operational prison, the bear, is subjugated by an exterior action; the bear's operation in this case ceases from being imprisoned in the living creature, bear, to become imprisoned in the bear species.

The older baboon also sacrifices his operational enclosure in order to save the band. We see, this way, that the concept of operational enclosure for the living being should be broadened to include other living beings from the same species like what occurs with the mammal.

There are even cases in which the social membrane can include another species like the case of symbiosis and of parasites. The social membrane involves a knowledge which is out of the operational enclosure of the living being.

Language

At the moment in which the neo cortex was formatted its neurons had a more or less definite distribution in determined areas so that the activation of these neurons could furnish an impression or knowledge. This might have happened because, when the evolutionary knowledge was catapulted, the social membrane functioned as guidance in the new assembly.

The orientation furnished by the social membrane came to force the communication which specialized in the vocalization.

The Broca lobe (brain lobe responsible for speech) is the place where this oriented format happened through the social membrane of the specie. A determined group's neuron activation creates a flash which corresponds to knowledge.

The knowledge of what is being a mother corresponds to a determined neuron assembly in the Broca lobe which will be vocalized in the same way. In almost every language the knowledge of the mother is identified with the vocal sound MA, which is a breath (of life) articulated by the lips (that suck).

The extraordinary development of the human society happened because of the fact that the new artifact (neo cortex), when formatted, took with it a social membrane that permitted the vocalization and the exchange of knowledge.

Man

Man is basically the arboreal proto man mammal who received the neuron artifact with a potentiality to use the evolutionary knowledge (gnosis).

The arboreal proto man was assembled to operate in a cloistered way, having its external environment perception deformed by its closure. It was assembled to see and know what is of interest for the species survival. The operational closure of the proto man is expanded by a social membrane that involves its species.

When the evolutionary knowledge was catapulted to the neo cortex, the impulse of life already worked with the living creatures that had their operational closure expanded.

This expansion was being made by knowledge membranes and social membranes which involved the exterior environment.

Social Catapult

Man was formatted based on the arboreal proto man who no longer operated in a closed knowledge system because it had bigger knowledge amplitude through a broad social membrane group, the species and at last the impulse of life's relation with the exterior creature.

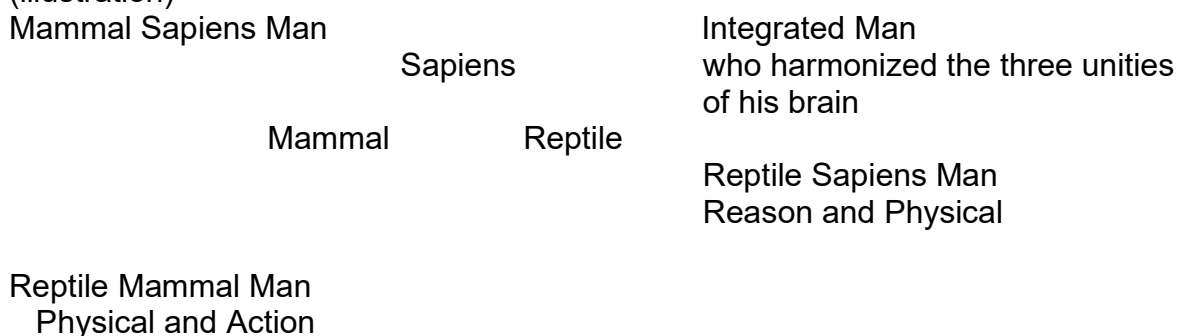
When the impulse of life catapulted the primitive brain's gnosis to the neo cortex, forcefully, it enlarged the social membrane to amplitude of the whole evolutionary knowledge.

Man's social membrane covers all of life's manifestation. The paradox of being a living creature, imprisoned in its operational ways and having a knowledge potentiality of the gnosis can be understood through these membranes.

Social Man

Man is a living creature who stores the impulse of life in a certain evolutionary process stage. Man is also involved in the social membranes which the impulse of life expanded during this process.

(illustration)



Man is not a being capable of living and surviving alone; he is part of a group (family) from a species (human) and of a world where the impulse of life manifests itself. Since it was formatted in parts, we can identify each one of these men in the human brain's components.

The visceral man is linked, uniquely, to his vital functions. This would be the man in the turned off state that happens in the sleep. Such condition is necessary for a reward of the anticipation mechanism and of the knowledge transference to the neo cortex. The sleep and dream are situations in which the neo cortex rearranges his connections. During the sleep it is common for man to have the true access to the Gnostic knowledge because he is canceling his dependency situation from the rest of the brain.

The reptile man is the one only worried about the satisfaction of his needs.

The mammal man is the one who sacrifices himself for his group because he participates of a membrane involving the species.

The sapiens man is the one who has the capacity of thinking and of using his knowledge to make decisions. When he uses the Gnostic knowledge he finds harmony. When he uses his individual knowledge he becomes egocentric and selfish.

Man can manifest himself through his two brains, as can be seen as follows.

The reptilian sapiens man is the one who uses the brain to take advantage of others. This type, to be more specific, is the thief: the astute and devious man with no scruples.

The mammal sapiens man is the one who has an exaggerated emotion to the point of jeopardizing even his condition of life when manifesting his emotion. In this group we find the artists who consume themselves for their art. There are also the mothers and fathers who annul themselves for their children.

The reptilian mammal man is the one who uses his emotiveness to gain or perpetuate his interests of survival. The type of man who involves, emotionally, others to take personal advantage. He is the parasite type of man who survives through schemes involving others emotionally. In this category we normally find relatives and friends. We also find in this category those who expose social fragility to take personal advantage.

The Whole Man or Sapiens – Sapiens

The whole man or sapiens – sapiens is the one who harmonizes the three men: the reptile that imposes itself through strength, the mammal that lives for its species and the sapiens that fulfills itself through reason. The perfect man would be the one whose strength of life harmonized with the emotion, manifesting itself through reason.

There is a great difficulty for man to manifest himself as a perfect living being. In his neo cortex he fights a titanic fight between the individual egocentric knowledge and the global knowledge which is the gnosis. This difficulty does not exist in the reptile which is a pure manifestation of the gnosis. Inside the membrane, in which it is manifested, the reptile has no arbiter. The mammal will have at its disposal a very restricted arbiter, limiting the species' and group's necessities.

When man received the neo cortex artifact, he began to dispose of a more ample arbiter. He could develop his individual knowledge. However, he has a responsibility or a necessity of maintaining the equilibrium of life's manifestation which is formatted in its structure.

It might be selfish or egocentric, but it has the necessity of harmonizing the whole evolutionary process which led it to this power. This is shown by Miguel Angelo in the Sistine Chapel with Adam's expulsion from paradise and the placing of the sword over man's head. The tree where the serpent rolls around (caduceus) represents the Gnostic knowledge pointing to man's head, symbolizing his necessity in adapting to the evolutionary format.

Man's thought is submitted to the sword's evolutionary process. The individual egocentric knowledge is limited by the Gnostic knowledge. Few images like Miguel Angelo's painting express the human thought dichotomy so well between science and religion.

Man's Operational Enclosure

Man, as every living being, operates inside a closed system (operational enclosure), not limited by a physical form but by the membrane which the impulse of life form but by the membrane which the impulse of life expanded at that evolutionary moment in which the living being is manifesting itself.

The reptile lives to feed and reproduce itself; its necessities are written on its body. It only seeks satisfaction from its necessities. The reptile is imprisoned in its operational enclosure.

The mammals, with their emotional brain, possess a self knowledge of the group which obliges it to a clear social behavior: it has a life, family and group necessity. Its operational enclosure involve social membranes inside which are its partner, child and group. The mammal's partner is not only defined by the genetics' writings. He seeks the

one who can generate the best offspring for the condition of the environment in which it lives. It sacrifices itself its offspring and in certain situations for its group.

The operational enclosure of the sapiens – sapiens man involves the whole evolutionary process. It doesn't involve only humanity as every life manifestation. This would be the perfect enclosure for man.

Progress

The great majority of men develop an egocentric knowledge to the detriment of the Gnostic knowledge which tends to imprison him in the reptile man's operational enclosure and from the mammal man's.

The neo cortex's dictatorship with a science and technologic development emphasized modern man's operational enclosure.

Progress has created more necessities than solutions for man. Facing extreme necessities, man is capable of abandoning his offspring, partner or group to survive and, this way, he becomes a prisoner of the reptile man's operational enclosure.

A sad paradox that occurs with the egocentric or personal knowledge development is that the egocentric knowledge is a generator of unlimited necessities. A very similar image of the Gnostic knowledge identifies it as the falling angel who wanted to be similar to God.

The "endless" necessities of man generated by the egocentric knowledge, determines a rationality distortion with the manifestation and the prevailing of the mammal and reptilian man as a necessity of survival.

The egocentric man starts behaving as the reptile! He becomes prisoner of the operational enclosure of the reptile man that lives inside him.

"Progress" has made man even more imprisoned in the reptile and mammal which live inside him.

Criminality and Violence

In our society violence is the most common manifestation of the reptile man. Murders and sexual aggressions are characteristics of imprisoned individuals in the reptilian operational enclosure.

Since the woman is in charge of the pregnancy, she has, during her upbringing, the emotional brain more activated than man to be capable of sacrificing for her young. Because of this bigger activation from the emotional brain it is uncommon for women to be imprisoned in the reptilian operational enclosure. Criminality among women is more linked to theft, cunning and defense of the young which are behaviors appropriate to the mammal's operational enclosure.

My reference here is to pathological situations of both sexes when man becomes a prisoner of the operational enclosure.

New Millennium

At the beginning of this millennium man goes through a difficult moment in his evolutionary process. Science and technology have put him in a situation in which his individual or personal knowledge reached such a dimension that it gives us the impression of power through this individual knowledge; dominate the manifestation of the impulse of life.

Man manipulates gametes and believes he creates life! He identifies the genetic code and judges that he can alter it. The agro industry creates food to supply humanity's necessity and little benefit from it. Man discourses about peace while making war!

The situation happens from the individual knowledge or egocentrism which is a result of the rationalization practiced by the neo cortex without the harmony that happens from the evolutionary knowledge.

Man is not only the sapiens man. Man is represented by the four men over which he was formed, which are represented in the neo cortex by the Gnostic knowledge. The obligation of participating of the Gnostic knowledge has manifested itself through the religion knowledge, which, in a disrupted way, expresses this knowledge.

Man's Enclosure

The false necessity that happens from the egocentric knowledge has made man prisoner of a restricted enclosure. Modern man is becoming prisoner of his own and private world. To get out of this restricted enclosure he would have to access the social membrane which he is part of that involves humanity and life. Religion helps very little because it also became enclosed in its dogmas.

Encouraging the noble dreams and elevated ideals is the way to access the great social membrane which man is part of and leave the egocentric restricted enclosure.

Dream and Art

Man, even when I manifest partially, does not stop being, completely, constituted by all his evolutionary components. The reptile man does not stop being a whole man who is manifesting in a deformed way due to a defective individual knowledge. Every man has a physiologic necessity of manifesting his evolutionary knowledge; his gnosis, because this is part of his physical formation.

The necessity of manifesting the gnosis is done, in the first place, through sleep when the gnosis manifests itself in the dreams, seeking for equilibrium. When man is awake he also has a necessity of dreaming, imagining solutions and situations for his conflicts. This necessity is the result of the anticipation which is also regulated by the gnosis. Man feels good in seeking for the realization of his dreams. The great success of the self helping literature, which encourages the realization of the dreams, is the demonstration of this necessity.

Another great resource for the gnosis manifestation is art! Art is the birth of the coded gnosis knowledge through the specific culture of each person.

Modern art is the demonstration of the turmoil in which dictatorship of the cortex took to the ancient art that manifested the Gnostic knowledge. Art, in the remotes of time, was used as an access to the known Gnostic, as can be seen in the ancient stone structures.

Religion and Gnosis

Religions were assembled over Gnostic concepts. With the passage of time religions began nationalizing their manifestations in search for material power. They became, at this beginning of millennium in deep distorted doctrines which manipulate partial Gnostic knowledge in order to obtain domain and power.

Examples of this distortion can be seen in the orthodox Islamism which preaches a holy war. The Gnostic principal of the species was distorted by the interest of a group. The holy war has deep inside an interest of manifestation of power groups. In the Catholic Church the celibate of priests goes over the procreation gnosis and the family.

The inquisition authorities are the most famous example of Gnostic knowledge distortion. They were commanded by priests who denominated themselves as disciples of Saint Domingo (Dominicans) who was a man of great wisdom and kindness.

At the beginning of this millennium, the advance of the Pentecostal religions have been observed, which try to exacerbate the emotional brain to such a point that a state of ecstasy to which a Gnostic knowledge is accessed may happen. The own Catholic Church gave up beginning the Charismatic movement.

In reality, extraordinary Gnostic knowledge revelations may happen with the use of this method. However the revelations have no value because they are assembled over religion dogmas and contaminated by the experience and existence of the static which transmit them.

The way that they are structured the churches and the religions have imprisoned man more through tendentious manipulation of the Gnostic knowledge. Religions would help a lot abstaining themselves from their dogmas and beliefs. They would be more effective simply by encouraging the group actions, the noble dreams and the elevated ideals.

Part 3

13

Impasse

Man is a complex proteinic structure, result of an evolutionary process of proteinic structures which, transitionally, has sheltered the impulse of life.

The proteinic structures, being ephemeron, force the impulse of life to change the structure constantly and continually; being unstable, they force the impulse of life to a constant adaptation to the new conditions imposed by the environment and because they occupy the whole disposable space the proteinic structures need to be controlled by the impulse of life in its manifestation.

The proteinic structural sequence has developed following a plan which is the evolutionary process. This process cannot be accepted as "chance". It has a knowledge sequence that is transmitted from one structure to the other, reaching man in the most complex proteinic structure already inhabited by the impulse of life. For man the impulse of life transmitted not only the acquired knowledge during the process but also the potentiality to perceive (notice) the own impulse of life.

This last evolutionary step was given at the moment in which the impulse of life, to save the arboreal proto man, gave him the potentiality (possibility) of knowing the evolutionary process.

This step was given through the creation of a new evolutionary artifact; the neo cortex. The neo cortex is an artifact constituted of neurons disposed in a way of being able to use the evolutionary knowledge to resolve the challenges of survival which the exterior environment may present.

The evolutionary information had a representation in the visceral nervous system and in the primitive brain. So that this knowledge could be transposed to the new artifact (neo cortex) it went through the intermediate or emotional brain. The disposition of neurons in the neo cortex is related with the intermediate brain which functions as a catapult.

Man's neo cortex functions as a binary computer which has its chips organized in the way in which it respects the emotional brain. This was done to limit the functioning of this computer to the needs of the species that are established in the emotional brain.

The thinking form of man is oriented in a way to resolve the new challenges which the exterior environment might present according to a line or guideline. This guideline is the species to which man belongs and the impulse of life that lives in the proteinic mass which is the human body.

Man's thought is not as free as imagined! "I think, therefore exist" should be substituted for "I live, therefore think"!

Science and Gnosis

When the impulse of life created the neo cortex there was the strengthening of an individual knowledge which began to oppose the evolutionary knowledge. There will be fortification because the individual knowledge will already be in the mammal.

The individual knowledge developed an own structural for each group of men and the individual knowledge of each group gave origin to different cultures.

However, every culture is based on a same evolutionary knowledge (gnosis).

Eros and Psyche

The Psyche princess (I am, I think) was one of the most beautiful women. Her beauty was commented at the Olympia, bringing out Aphrodite's jealousy (goddess of beauty). Aphrodite then sends her son, Eros, to bewitch her.

Eros was a being of extraordinary beauty who threw arrows releasing the passion of those who were wounded. When he had the ark stretched to dart Psyche she awoke. Seeing her he became disordered with her beauty and wounded himself with his own ark. This way Psyche bewitched Eros. Psyche married him and lived with him in a distant castle which displeased his mother, Aphrodite.

Psyche lived alone in her distant castle that reached the clouds. Aphrodite imposed that Eros could only show up at the castle at night and his face could not be seen by her. But Psyche's sisters tantalized her into seeing his face one night. At night when he appeared, Psyche lit a candle and saw his face. Eros was truly marvelous. He looked nothing like the image that her sisters tried to make her believe in. Like Eros' incident when he was bewitched by the arrow, Psyche was also out of control by his beauty and ended up burning his chest, which made him desperately run away.

Their union was undone. Aphrodite demanded that Psyche do a series of tasks to reunite her union with Eros. The last of these sacrifices was to go to Hades (hell) and bring Aphrodite a box containing the beauty. This way the goddess' beauty would never again be contested.

To accomplish this mission Psyche had to confront a dog that guarded the gate of hell. Cerberus, the three headed dog!

After a fierce struggle Psyche leaves Hades with the box. On her way to deliver it to Aphrodite she thought, "Why give Aphrodite the box if I can open it and possess eternal beauty?" So she opened the box.

In rage, Aphrodite decides to destroy Psyche, who is human. However, impressed with the love of Eros for Psyche, the great Zeus comes down from the top of Olympus to save her transforming her in a goddess.

Gnosis

The neo cortex gave man the capacity of thinking. Thought developed through sexuality, the basis of the species' reproduction. Despising the origin of the species (gnosis), Psyche wants to understand the evolutionary gnosis, inside its patterns (science) and carries a candle searching for an unnecessary truth that would not make her happier.

Science's search happens when stimulated by the intrigue of her own sisters. Psyche's vanity and egocentrism turn her away from her gnosis. Psyche suffers fighting against the goddess but does not submit to Aphrodite. She goes to hell fighting against a three headed dog (trine brain).

At a determined moment, after a knowledge acquisition spiral process, Psyche reaches the extraordinary power from the box. Her knowledge (science) puts in her head the power that belongs to the gods. She sees herself on a par with the impulse of life and opens the box. She just wasn't destroyed because a divine strength intervenes altering her human state.

Psyche is not capable of reaching the next stage on her own of the evolutionary process of her gnosis. She has the pretension of being able of, because of a personal knowledge (science), altering the manifestation of the impulse of life (divine). She doesn't disappear because divinity itself saves her.

Salvation

The impulse of life is about to alter man's nature to save Psyche. The knowledge (gnosis) of salvation is in the Greek culture. It is flawlessly expressed in this mythological episode.

The gnosis of this salvation is also in almost every culture expressing itself in diverse forms. God comes to save man! Christianity is based on the Gnostic knowledge.

The impulse of life will modify man in the same way in which it already did when it created the justice and compassionate mutants.

Religion

Man knows, instinctively (gnosis), that he is subjugated by the impulse of life (divinity). He knows that he is, simply, a bunch of protean molecules which move according to the Olympus gods' game.

Like Psyche he has the vanity of judging himself independent of his biological necessities. He affirms, "I think, therefore I exist! If I think and can possess the eternal beauty box, why can't I be the divinity?"

Man nowadays is living this evolutionary moment. He knows instinctively that he merely a reference of the impulse of life. But this understanding is incompatible with his rationality.

Man is the manifestation of all four men contained in him. He can only find harmony (be happy) accepting the manifestation as a total man. But his superior brain developed a perception of itself incompatible with the whole.

Man fights between a total manifestation, which he does for religion, and the partial manifestation, which he tries through the means of science.

Neither science nor religion is the true gnosis (evolutionary knowledge).

Religion is the gnosis contaminated by rationality (superior brain), to be able to express itself in a rational world. Science, on the other hand, is the rationality contaminated by the gnosis, to try to express the total man in the rational world.

Faith

The superior brain becomes divided between its own knowledge (science) and the total knowledge (gnosis). Like Psyche it thinks that it can be like the gods but knows it's human.

To equate this duality, science x religion, the superior brain found two ways: the religious and the skeptic.

The religious path accepts the impulse that comes from the totality and rationalizes (as it can) the Gnostic knowledge. It assembles a history that justifies the justice of the mutant, as is expressed in the Pentateuch, and believes in a fair god. For this it accepts the impulse of its totality and calls it faith. Later it assembles the history of a new god of

compassion to rationalize the compassion mutant that showed up in him, as was expressed in the New Testament.

The most difficult thing for the superior brain is to bend itself and accept the impulse (faith) that comes from man's totality (gnosis). In order to accept the gnosis the superior brain has to accept its insignificance inside a bigger process which is the evolutionary process. This way each culture articulates its own religion! Every religion preaches the abandonment of the individuality to benefit a group (species).

The path of the Skeptic is the other form of confronting the duality: deny the impulse that comes from man's total intimacy. To be able to choose this way of resolve the duality man is obliged to encourage his own individuality. This resource is obtained through the exacerbation of the contact with the exterior world through the excitement of the three senses. Maybe here we find an explanation for the diffusion of drugs in this millennium change. With the impulse of its totality annulled the skeptic is capable of seeing and criticizing the religious dogmas irrationalities.

Incompatibility

The great majority oscillates between these two forms to confront the dual impasse to which they were exposed through the evolutionary process.

The acceleration of the individual knowledge (science) that occurred in the last centuries generated the dictatorship of the cortex and favored the skeptics. Science found itself in its rationality limit. Matter is contested by the anti matter. Time space was unstructured by relativity. Biology was shocked with the reality that nothing more can it do than discover processes that the impulse of life uses for millenniums. The computer and the communication show that the whole world is tangled. The butterfly that flaps its wings in China can be the beginning of a typhoon in Florida. The Quantico theory and the uncertainty principal of Heisenberg were a shovel of limestone in the pretension of rationality of man's superior brain.

The acceleration of technology and science left the man of this beginning of millennium in a situation of so radical and rapid transformations that the only certainty of this period is the change.

On the other hand religion had to move away in such a way from rationality to confront this advance that it fell into a state of fanatics. It's the case of the so called fundamentalist religions arrogated in ancient concepts. Some religions set up pseudo rational systems over the true gnosis to harmonize with the present world. The religious man is also lost!

The path for salvation will be found by none of these men – the impulse of life will create again another mutant.

14

Salvation

Man finished this millennium at the bulge of a science and technological spiral transformation. With the computers he could then have at his disposal of possibilities almost infinite of crossing data. He had access to a quantity of information which is no longer compatible to his brain. Computer memory stores almost infinite information which communication revolution put out for men's disposal.

The agro industry and cattle-raising made it possible the production of sufficient quantity of food to feed the whole humanity.

Medicine allows us to live proportionally longer and better than the other living beings. It gave man resources against pain and suffering which precedes the inevitable death.

Engineering made it possible the construction of houses with water and sewers capable of sheltering every man. Technology created clothes and heaters and refrigerators that ease the climate.

Economy produced wealth capable of proportioning men all these basic resources of surviving and comfort.

Commerce created means and resources to be able to distribute the conquests to humanity.

The impulse of life seems to have done right at the moment in which the arboreal proto man was saved when he fell on the African savanna, giving him the neo cortex brain artifact.

Arcadia

This was the world programmed by the impulse of life. Men would live all in peace having their basic necessities satisfied, dedicating themselves to beauty and arts. This world existed in the imagination and dreams of men and was called Arcadia.

The concept of Arcadia is very ancient. It is rooted to the Greek thought before the birth of Socratic's philosophy. A time in which mythology, the dream and the gods dominated the world prior to the installation of the egocentric philosophy thought which we know as post Socratic.

In the XVI century, still in the beginning of the science spiral, this was a dream expressed in many works of art. Nicola Poussin, for example, left paintings that showed what the imaginative world of Arcadia would be like.

The Error

Something went wrong when man began to develop an individual and selfish knowledge. It was different from the Gnostic knowledge to which he had been planned programmed.

Men began to consider other men as their opponents and not as allies.

The man who cultivated the land became an enemy of the one who cared for the animals. Agriculture and cattle-raising crashed at a first moment. The Gnostic knowledge left this register when Cain murdered his brother Abel.

Men separated, habiting diverse regions which generated different cultures and languages.

The language was an important factor in the human's diversification. The difference of language creates a communication barrier which cannot be transposed.

The story of the heaven-reaching tower of Babel is the Gnostic register of this dispersion which separated men. Men wanted to construct a tower so big that would reach heaven: the register of the neo cortex pretension of man being equal to god. God determined that, at mid construction, they would no longer understand each other through the language and the Tower could not be completed. From then on men separated and wandered about through the world.

The impulse of life did not give up and began to act in a way to correct this error.

Cycles

To correct the detour that happened in the neo cortex's development of the personal knowledge the impulse of life began to activate cyclically parts of this neo cortex.

The membrane of knowledge of the impulse of life detects a possibility and transfers a new Gnostic knowledge to a new kind of man who then becomes a mutant. This occurs through the appearance of a different perception altering his behavior in relation to the exterior environment.

Cyclically the impulse of life activates brain areas which make an individual more capacitated for a determined understanding. The other individuals do not understand him and normally despise him. But on the long run his ancestry shows itself through his superiority.

The action of the impulse of life over man reached us through the Gnostic knowledge that remained registered through the ones, in ecstasy, who wrote the Genesis. This was the first book done known as the Bible.

The cycles are real although written in a timeless way and oneiric characters, in a ludic atmosphere proper of the emotional brain catapulted to the neo cortex.

Noah

At the moment in which the impulse of life, through the knowledge membrane, thought that the human species was ready to proceed its evolution as a definite species, it locked the genetic code.

Before Noah there was the crossing between men and other species determining the appearance of distorted beings like the Cyclopes and others that might be the result of inactivated evolutionary experience.

The impulse of life activates Noah's neo cortex so that in him there is defined the knowledge of a human species: while he builds the ark Noah is despised by his contemporaries who did not have the activation of the neo cortex.

The symbology is wonderful. The water, in the oneiric world, represents the desire, occupies the spaces in a constant way, molding itself to each place. It requires a great effort to be contained. Because Noah had an illumination (activation), locks himself in the arc with his family and a couple of each species. Those who were not illuminated lost themselves in their desires (water).

Starting from Noah man began to have a locked genetic code and every man is his descendent.

Abraham

Noah's descendents occupied the world. The Tower of Babel's episode, which defined men's dispersal through the language, is described in Genesis during the period of the 16^o Patriarch.¹

The membrane of the impulse of life for the environment detected the necessity that man should have the knowledge of his existence. The impulse of life's next step was to activate the neo cortex so that man could have a unity of life concept. When this neo cortex area is activated man began to have an instinctive notion that he participated of a Whole. Such activation determined the perception of an only all mighty god.

¹ The gnosis is in the middle of all the emotional and mystical knowledge and there is a correspondence between the great arcane from the Tarot and the Genesis Patriarchs. The 16th Patriarch was named Serugue, which means Tower; the 16th card of the Tarot is the Tower. The arcane 21 from the Tarot corresponds to Isaac, which means The Laughter and this card is The Crazy. The arcane 22 corresponds to Jacob from where the 12 tribes from Israel came from and the card is The World.

In the middle of Ur land a mutant with the name of Abraham appears. Desperate with the perception of something different, which is happening in his mind, he leaves in search of a new land where he can find equilibrium with the new perception which he will have to decode, somehow, so he doesn't go insane.

The impulse of life manifests itself giving him proof of its strength making his wife, Sara, already at an advanced age, give birth to Isaac. The impulse of life would have to have a full proof that it would be the mutant Abraham that was creating and from him raise descendents like the stars in the sky or the sands of the desert.

The episode where God orders Abraham to kill his son Isaac is the proof that the mutant really had acquired the perception that it had the impulse of life acting and manifesting itself above its physical necessities (reptilian) and mammal (family).

In the episode of Isaac's sacrifice the Gnostic knowledge is so strong that both Arabs and Jews (the main believers of an only god) call themselves the children of Abraham and declare episodes as if Ishmael became Arabic and Isaac Jew.

The impulse of life created mutants with some difference in their neo cortex to permit them an impulse of life perception as a conductor of the evolutionary process.

Moses

Men were being ruled by the strongest man's law. The strongest dominated the weakest who therefore became his slave. It was not the interest of the impulse of life to select the strongest. The most qualified to carry ahead the impulse of life was not always the strongest. It was necessary to build into the neo cortex a seed of equality which materialized itself into the concept of justice.

Again a mutant was created by the impulse of life.

Moses was an extraordinary wise man in his days. Perceiving the mutation, he decodes his new perception talking with the impulse of life which is in the burning bush.

He removes the Hebrews from Egypt after a demonstration of strength from the impulse of life. He travels for forty years with the Hebrews through the desert until every generation that left Egypt is submitted to the new mutants of Justice. The justice cycle extends itself until the Romans and it corresponds to the Aries Era.

The anterior Era Taurus has a strength connotation. The Apis bull was a cultural symbol of Egypt at the time when the Hebrews left. In the desert, when the Hebrews rebelled against Moses, they worship a gold bull.

The gnosis keeps a correspondence in all the knowledge; these transformations occur with the passage of the Era.

The Taurus Era corresponds to the strength of Taurus. The Aries Era corresponds to the Lamb justice. After comes the Pisces Era which corresponds to Jesus' arrival.

Mutant of Compassion

With the passage of the Aries Era to Pisces, the impulse of life made an intervention, creating the Mutant of Compassion. Justice had favored the formation of great empires which continued to oppress man in detriment to the manifestation of the impulse of life.

The knowledge membrane of the impulse of life detected the necessity of a bigger exchange among men, with a bigger interaction from their actions. It

activated the neo cortex connections which gave the man of this time the concept that there is something beyond justice. There is a feeling that the impulse of life wants something more than justice among men. This feeling forcefully needed to be decoded somehow to make it possible for man to understand and practice it.

For a Roman it would be very hard to understand such concept because his neo cortex concepts had not been activated. It would be uncommon for a Roman to understand compassion. The same way as it was impossible for the Egyptian to understand that his slaves had some rights besides serving their masters. The crucified Christ was an abomination for the Romans.

So that this transformation that happened in the head of the new mutants could be decoded, understood and applied an adaptation it was necessary that an adaptation should happen. And this happened with the birth of the Christianity.

The Gnostic knowledge had to be transposed to a practical knowledge which could be transformed in action. The new mutant felt a necessity of being more integrated. It had the perception that there was something beyond its physical existence, which was the impulse of life.

This oneiric and ludic perception forced it to find harmony (happiness) and to act in a different way which he did not understand himself. In this situation of mental confusion the new mutants would throw themselves to the beasts in the roman circus, giving a demonstration of the existence of the impulse of life, to which they were only passing manifestations and which was integrated in such a way to every man that they forgave their executors.

This new Gnostic knowledge was decoded by Saint Paul, joining pieces of the principal religions of the time. That is how the Christianity was born. If we substitute the word Christ for the Impulse of Life we will notice that Jesus' words are not metaphors in the bible. His words are crystal clear.

Whatever you ask for in the name of the Impulse of life will be done – "I am the truth, the way and the life!"

Adaptation

The Christianity did not appear because of Jesus. Jesus symbolized the Christ in the Pieces Era or the manifestation of the impulse of life which at that moment gave man a new Gnostic knowledge – the Compassion.

There was a change in the interior of men's head in which the impulse of life transferred a Gnostic knowledge. The new mutants had to somehow decode the transformation to find their own equilibrium.

Man was modified inside by the impulse of life which transferred to him a new evolutionary knowledge. There was, therefore, the necessity of the new mutant to adapt to this knowledge. At the beginning there is great suffering until he finds a way of expressing in a harmonious way the transformation.

Man does not have in his free arbitration the adaptation to the new Gnostic knowledge which the impulse of life transfers to its neo cortex. He is forced to adapt to the new knowledge through a mental constraint and will only find harmony (happiness) when he begins to manifest his new knowledge.

The same way as Noah saw himself forced to construct an ark and Abraham saw himself constraint to kill his son the compassion mutant saw itself constricted in having compassion (love) to his close.

In these passages of Eras man did not change because he wanted. The change happened because man was constrained by the Impulse of life. He was forced to change because of the necessity of manifesting a new knowledge which

the impulse of life imposed to the neo cortex for judging it necessary for the evolutionary process.

Part 3

15

New Mutants

The first living creatures were assembled to react in a certain way to the environment without any option or decision. Its reaction was determined through the form in which the gnosis assembled its molecules.

With the evolutionary progress the gnosis, progressively, gives knowledge so that the living creature can adapt to each new situation. The evolutionary process is the Gnosis Philosophy (Gnoseogenia) – the transference of evolutionary knowledge to the living creature.

Climbing on the evolutionary chain we find even more living creatures with decision capacity or will to determined and specific situations. The impulse of life gradually passes the gnosis to the living creature. The way in which the gnosis transfers the knowledge is rigidly controlled by the form in which the molecules organize themselves. Each living creature has a limited capacity to exert its will.

The first living creatures had their molecule organizations defined during their assembly. This was done by a code book that was being written in their cells by the nucleic acids. This code represents effectively a Gnostic knowledge (evolutionary) to which the living creature has no access. It only serves for the assembling of its structure. These molecules keep the secret of the impulse of life which cannot yet be revealed to man.

Free-will

All the anterior living creatures, when the neo cortex appeared (man), operated inside an enclosure in which its species is included because the impulse of life limits its free-will to the species condition in its environment.

The free-will of one mammal is much reduced. It is forced to behave according to its species which is also ordered inside the exterior environment according to the impulse of life's membranes.

The vegetable kingdom sustains the animal kingdom. The predator feeds from the prey. The prey reproduces according to the equilibrium among the species. Even man was controlled by the manifestation of life in this equilibrium with the external environment.

The Power to Think

When the impulse of life gave man the possibility of using the evolutionary knowledge it permitted a broadening of the free-will which has been trying to limit. Since man's appearance the impulse of life continued acting in the activation of the neo cortex parts with the intention of exerting the control of its free-will.

Limiting Action

In the beginning man was limited by the law of the one that was the strongest. When the law of the strongest started to damage the appearance of new

evolutionary knowledge the impulse of life activated certain areas of the neo cortex giving him the concept of justice.

The concept of justice needed a repair further on, especially because it generated the great empires that went back to exerting the law of the strongest.

In face of this situation the impulse of life activated areas of the cerebral neo cortex which gave man understanding and compassion.

The understanding of compassion, when associated to justice and strength, generated a potentiality of destruction of the own planet.

New Millennium

In face of this situation the impulse of life is again activating areas in the cerebral neo cortex which give man an understanding that he and all the other men, to be able to perpetuate it, have to live in harmony and equilibrium with the environment.

That's why it has activated new neo cortex areas making individuals with a certain concept of globalization to appear at this end of millennium. They dedicate to ecology. They understand that the planet, like a living creature that only through equilibrium is capable of maintaining and perpetuating the impulse of life.

These new mutants have a new perception and do not understand it very well. They protest against the killing of seals and worry about the psychological health of their pet dogs. However they forget that children die of hunger in Africa and do not notice the psychological disruption which people sitting beside them on the subway are going through.

Repetition

They go through today a similar moment which happened when the compassion mutants appeared and had no comprehension of the compassion and throw themselves to the beasts as martyr in the roman circus.

Saint Francis

The new mutants usually have as a patron Saint Francis of Assisi. He was an extremely advanced man for his time because he presented this actual mutation.

He understood that every manifestation of the impulse of life was like his brother – the brother bear and the brother woof!

Aquarian Mutants

Since the end of the last century when the Era of Pisces changed to the Aquarian Era men have been born with a different activation of the neo cortex which gives them a universality concept in which all the living creatures are a same impulse of life manifestation.

It is very difficult for a man, anterior to this mutation, to understand that he should not destroy the forest to gain better profits. Even if the earlier men understood rationally that destroying the forest would jeopardize him in the long run his rationality transferred the problem to the other. Nowadays he needs to profit more because money commands the world.

The man of Taurus presented a similar difficulty to understand Justice and the roman to understand Compassion.

New Generation

I can see in the new generation the appearance of these mutants - Aquarians. They are youngsters of a remarkable good faith who have no ambition that goes beyond their necessity. They have an understanding of the manifestation of life to which they are part of; different from my generation when we believed that we were the owners of the manifestation of life.

These new mutants are everywhere.

Mechanism

The Aquarian mutants had their neo cortex activated in a selective way by the impulse of life. Such a neo cortex difference made them different from the others. They have difficulty in finding harmony with themselves and because of this they engage in ecologic and religious movements in search of harmony (happiness). They are not like this because they want. They are because they were forced to be this way. A great number of them are unknown by society. Many are not understood by their parents and family. They are going through a big suffering and will only find peace when they are able to manifest the universality which became part of their Gnostic knowledge. They suffer like Noah, Abraham and the Christianity martyrs, who also had to seek harmony (happiness) structured in their new paradigms or evolutionary knowledge.

Old Man

The aquarium mutants are still a few and are not yet capable of influencing the destiny of the world which seems to be heading to destruction. Although there seems to be sufficient indications that the planet heads to an irreversible situation of sustainability, the men of today are not capable of taking concrete measures to defend the planet.

Instead of using the wonders of science and technology to create a world of peace and harmony the ones who hold the power spend fortunes in space research. The man of the past wants to perpetuate his power creating artifacts which will take him to another area in space where he can go on with his domain. This man of the past is not capable of sharing what he has. Even when he rationally knows that his fellow African man dies of hunger and is part of the same humanity as he.

There is something missing in the old man's neo cortex which is activated in the young man; the aquarium mutant.

Critical Mass

Experience has shown that the intervention of the impulse of life to correct the bend that occurs inside the evolutionary process in the healthy man are initially forced in some mutants who suffer so that they can manifest a new gnosis order.

These mutants go on imposing their new knowledge with great effort and even with the social exclusion until the moment in which the mutation reaches a same critical mass and then the process inverts.

During the passage from the Aries to the Pisces Era we could see, clearly, the moment in which the compassion mutants reached a critical mass and took over Rome's power (justice). This moment occurred during the kingdom of Constantine who though the Milan Editorial made Christianity the official religion of the Empire.

Constantine, himself, was not catholic. But he felt forced to giving in to the critical mass of the mutants.

Point of Mutation

We see our world heading to a chaotic situation and we have difficulty to imagine in which way the impulse of life can save man. The understanding that there is a critical mass is very comforting and gives us hope.

The quantic theory and the mutation point were two concepts greatly spread and commented by the intellectuals at the end of the last century. At the moment in which one electron reaches a determined energy quantum he leaps to another orbit. The new orbit is unpredictable (uncertain). The moment of the quantic leap is more or less the moment of mutation.

Inside the Gnostic knowledge this is exactly the moment in which we are going through. The same way as in the Gnostic writings this is also, in an oneiric or allegoric way, in the gospels.

“You also must be ready; for the Son of man is coming at an hour you do not expect.”

Saint John

The Gnostic writings, specially the gospel, transmit the knowledge in which man will be saved at the end of times by a lord who will come when from where he is not expected.

Saint John's books are good examples of Gnostic knowledge accessed through ecstasy. In the first book the path which the impulse of life made to come from divinity and the manifestation in man is described. It begins with the verb and ends with man's manifest and crucified by the five wounds of the five senses.

The second book of Saint John is the Apocalypse (revelation) where he, John (man), describes in which way, oriented by the impulse of life (Christ), will be saved to a new dimension which is identified as the New Jerusalem. The course of the book is full of Gnostic knowledge expressed in an oneiric way.

The beginning of the book is explanatory – “I, John, prisoner on the island Patmos have a revelation and write this book”. John (man), isolated from the world (island), in a pathetic state (patmos) writes the Gnostic knowledge and perceives in ecstasy.

Christ of Pisces

In the Era of Pisces passage there was a similar transformation which occurs in this new millennium. This change has to somehow be transformed into a myth so that it can be understood by men.

So that the transformation can be assimilated by all it needs to be decoded into a story as was done in the Pisces Era passage when the passage of the Era was assembled over the life of Jesus. As I said before, if it had not been for Jesus, it would have been taken as a paradigm the story of Apollonian from Tyana.

Jesus was the Christ from the Pisces Era. He was the impulse of life which manifested among us to save us. It is interesting to notice that Jesus' passage was very little documented. There was a time in which there was doubt that he had even existed. What can be concluded is that all the Gnostic knowledge, flawlessly expressed in Jesus' history, was assembled later through the access of men in a state of ecstasy (static) to the Gnostic knowledge.

The staging of the Era passage is not done by an only person. It is the result of the knowledge of all the Era mutants. As myth assembles itself corrections occur which represent the mutant's acceptance or rejection in a way in which the final gospel holds the new knowledge.

Aquarium Christ

After the foreseen transformation by the neo cortex alteration of the aquarium mutants and the transformation spread when the critic mass will be affected, man will begin to live in a new world (kingdom). For the Gnostic knowledge to be accessed by the new man it will be necessary to assemble a story which will perform the passage of the Era, as was done in Pisces.

This staging will also have an oneiric and ludic form, which is proper of the emotional brain. The bases of the story should be assembled over an existing character who represents, in a clear way, the aquarium mutant and who has distinguished himself for some impact deed in the transference of a new knowledge.

The staging of the Aquarium Era should be similar to the one that occurred with Jesus. The Aquarium Christ will go unnoticed among men because most of them do not have the new Gnostic knowledge to identify it.

The Sanhedrim Jews were extremely religious. But they did not have the neo cortex mutation to understand the Compassion. They did not have the capacity to perceive the impulse of life manifest. They did not have "eyes to see".

Probably the Aquarium Christ has already manifested himself among us and we didn't have "eyes to see". The fact that we still do not have the neo cortex alteration, which perceives the generalization, prevents us of recognizing Christ. The same way as it did not allow the Pharisees to recognize Jesus.

The gospel of the third millennium will be written over a story that represents the passage of the Era, in which man was destroying the planet. It will be assembled about the life of an aquarium mutant who has had a deed strongly represented of the new Gnostic knowledge.

Chico Mendes

I'm not proposing exactly that Chico Mendes (rubber tapper, unionist, and environmentalist) was the Aquarium Christ. But his story can be well represented. To begin with his life was very obscure for the Aquarium mutants to later adapt it to the new gospel. He was a mutant who defended the Amazon Forest. The Amazon Forest is, undoubtedly, the most representative of the new generalized Gnostic knowledge on the planet. Chico Mendes was murdered by men of the past who disagreed to the manifestation of the new knowledge.

As can be seen, with some changes in this story, the new gospel can be born in the third millennium; where the sacrifice of the aquarium Christ will be expressed, manifested among men and which was not recognized by them and killed him in an ambush with a shot.

Repercussion

The gospel is not the result of a story or of a strong emotion. It is being built according to the acceptance or rejection of the new men.

The Chico Mendes' case shows the formation ingredients of a myth. His tomb, in a small town in the interior of Brazil, is covered with sayings which praise him as a divine hero and it is a pilgrimage point of every mutant involved with the

ecology. In Germany, where there is a great quantity of aquarium mutants, more than one hundred books have been written giving account of and acclaiming his story.

There is a certain consensus in putting Chico Mendes as the great aquarium mutant because his story has the necessary ingredients for turning him into Christ. Only time will tell if he will be transformed as Jesus or as Apollonian of Tyana.